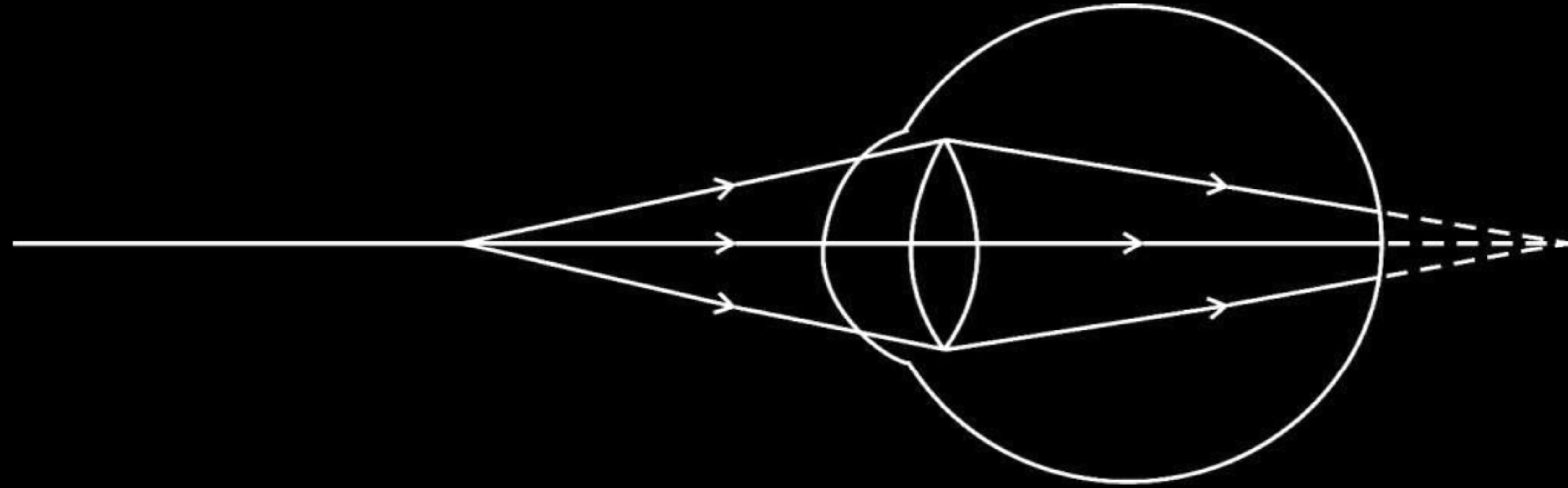
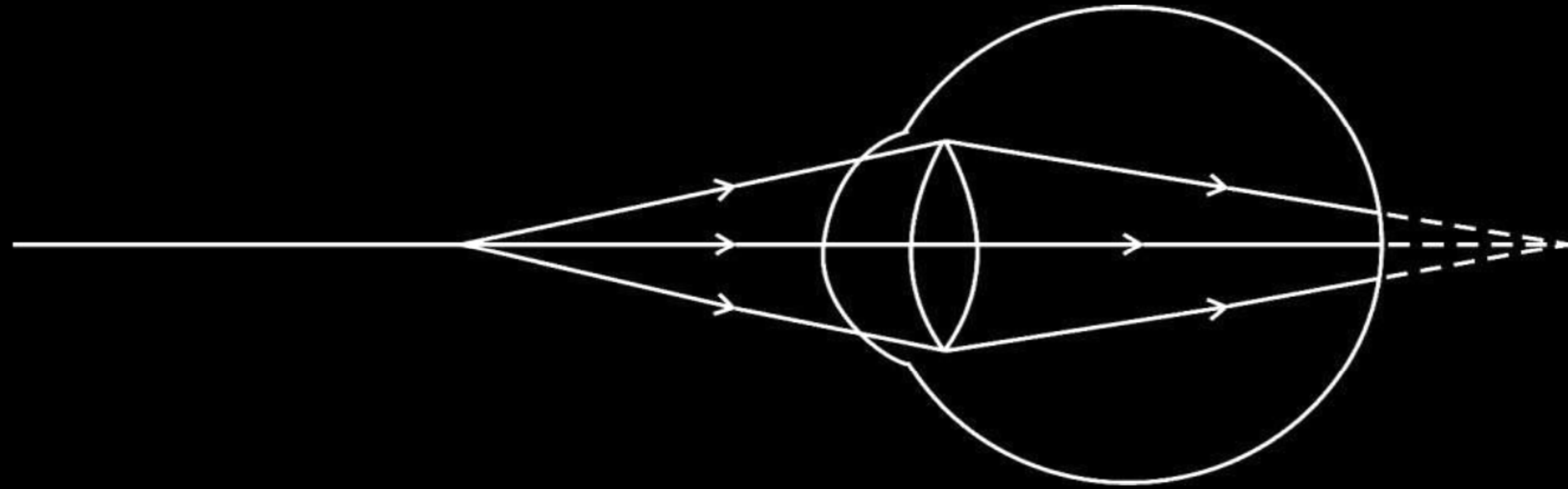


26. Observe the following diagram showing an image formation in an eye :



- (a) Identify the defect of vision shown in the figure.
- (b) List its two causes and suggest a suitable corrective lens to overcome this defect.

26. Observe the following diagram showing an image formation in an eye :



- (a) Identify the defect of vision shown in the figure.
- (b) List its two causes and suggest a suitable corrective lens to overcome this defect.

2

(a) Hypermetropia/Far sightedness	$\frac{1}{2}$
(b) •Focal length is too long	$\frac{1}{2}$
•Size of eyeball is small.	$\frac{1}{2}$
•Convex lens/Converging lens	$\frac{1}{2}$

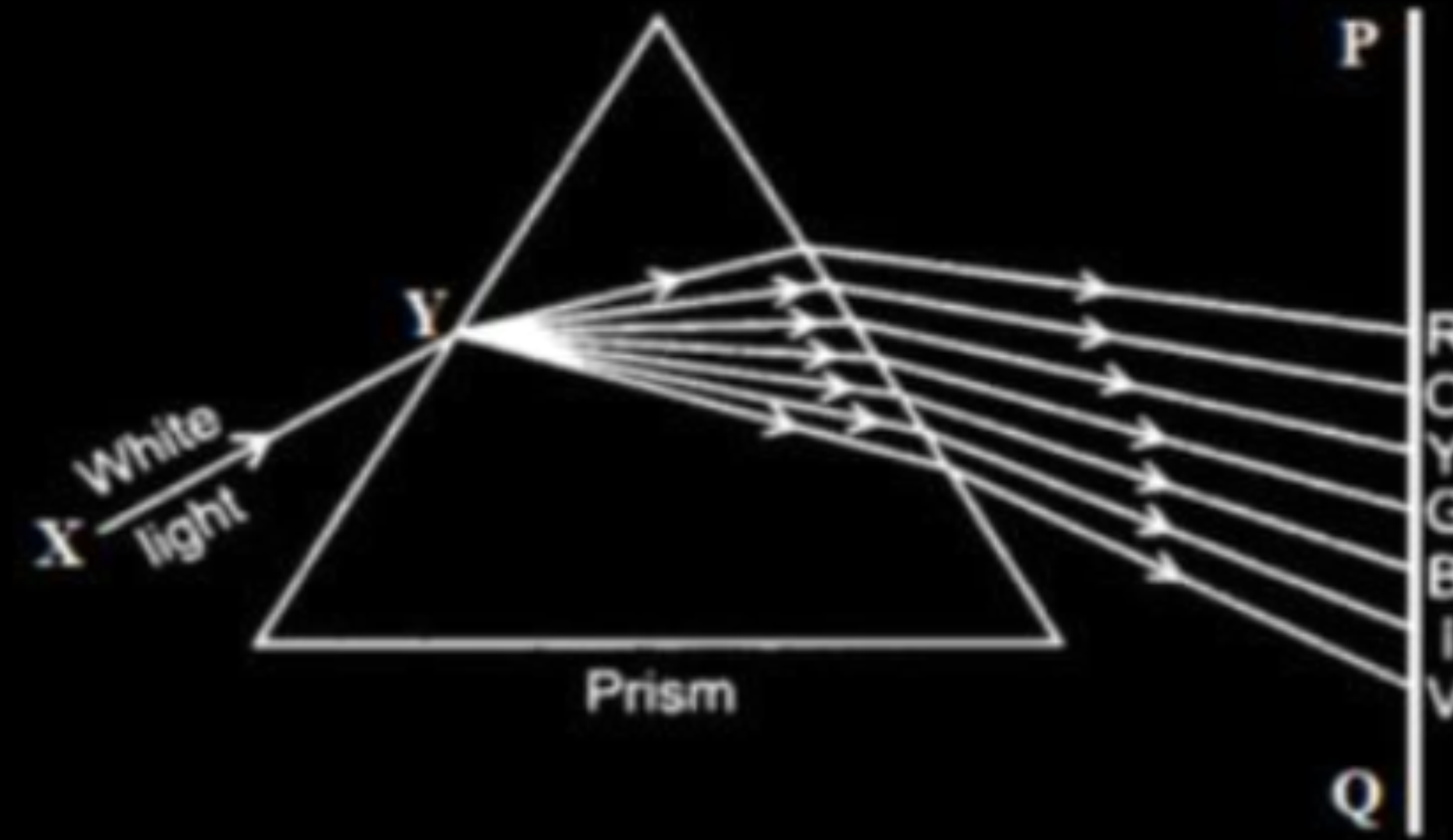
33. Draw a labelled diagram to show (i) dispersion of a beam of white light, and (ii) formation of a rainbow.

3

- 33.** Draw a labelled diagram to show (i) dispersion of a beam of white light, and (ii) formation of a rainbow.

3

(i)



$1\frac{1}{2}$

(ii)

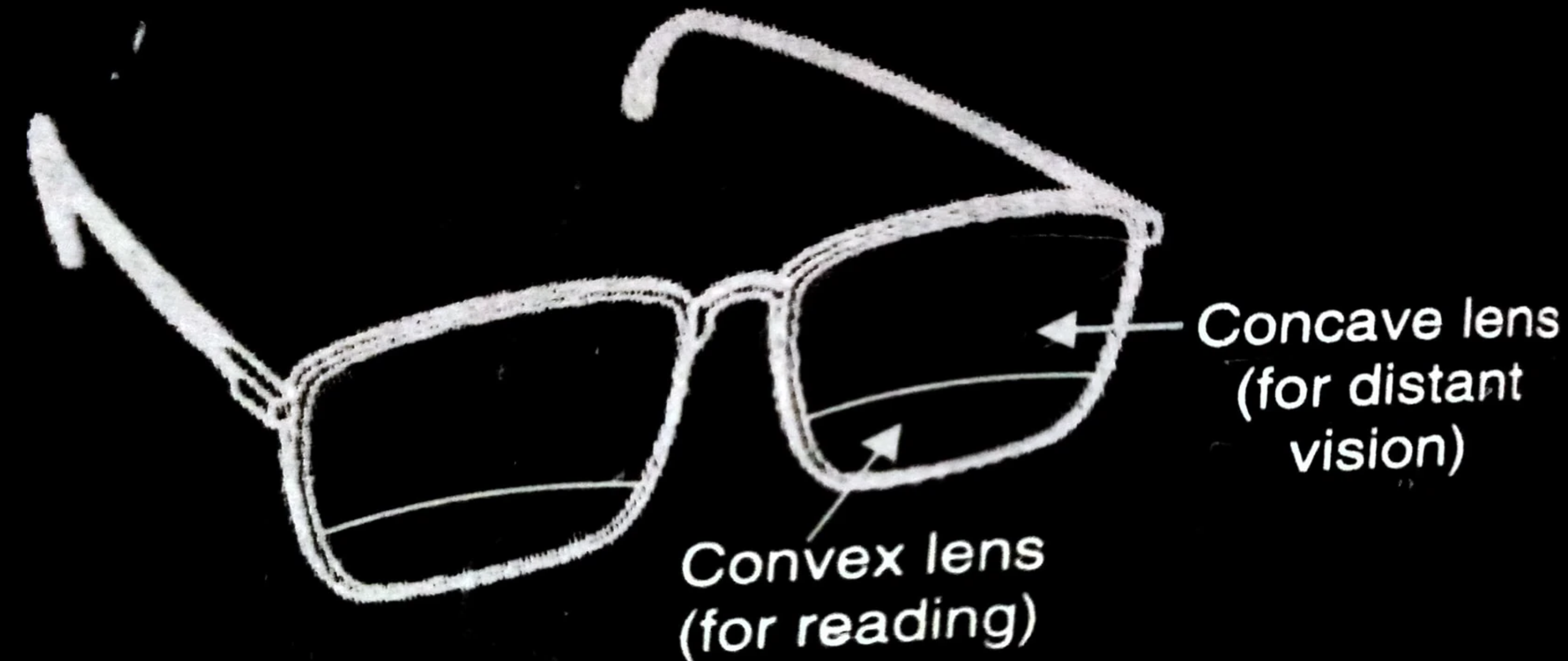


$1\frac{1}{2}$

- 29.** Draw a labelled diagram of a common type of bifocal lens. State the function of each of its parts. Name the defect for the correction of which it is used and state its main cause.

29. Draw a labelled diagram of a common type of bifocal lens. State the function of each of its parts. Name the defect for the correction of which it is used and state its main cause.

3



- The concave lens (upper portion of bifocal lens) facilitates distant vision.
- The convex lens (lower portion of bifocal lens) facilitates near vision.
- Presbyopia
 - Diminishing flexibility of the eye lens.
 - Gradual weakening of ciliary muscles

(any one cause)

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 31.** (a) State one important function of the following parts of the human eye :
- (i) Retina
 - (ii) Pupil
- (b) State the role of ciliary muscles in focussing objects at varying distances from the eye.

- 31.** (a) State one important function of the following parts of the human eye :
- (i) Retina
 - (ii) Pupil
- (b) State the role of ciliary muscles in focussing objects at varying distances from the eye.

3

(a) (i) Retina behaves like a light sensitive screen on which image of an object is formed.	1
(ii) Pupil : regulates and controls the amount of light entering the eye.	1
(b) When ciliary muscles relax or contract they change the curvature of eye lens and hence the focal length to focus objects at varying distances from the eye.	1
[Alternate answer of (b): It regulates the focal length or thickness of the lens.]	

19. *Assertion (A) :* The angle through which a ray of light bends on passing through a prism is called the angle of deviation.

Reason (R) : The peculiar shape of a prism makes the emergent ray bend at an angle to the direction of the incident ray.

19. *Assertion (A) :* The angle through which a ray of light bends on passing through a prism is called the angle of deviation.

Reason (R) : The peculiar shape of a prism makes the emergent ray bend at an angle to the direction of the incident ray.

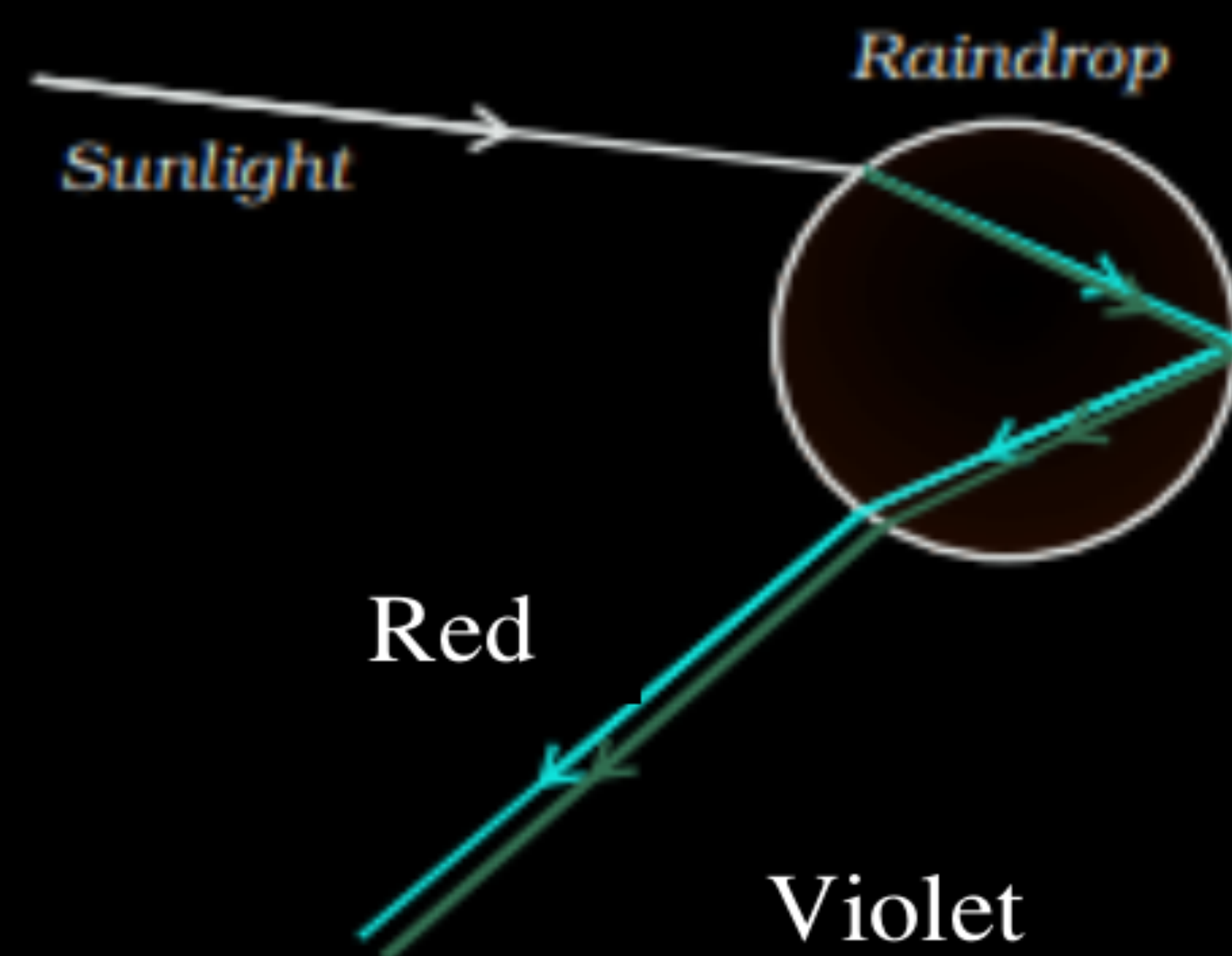
(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

- 25.** (a) When and where does a rainbow appear in the sky ? Draw a labelled ray diagram to show its formation. 2

OR

- (b) What is scattering of light ? Why does the clear sky appear blue ? 2

- (a)
- It is formed after a rain shower.
 - It is always formed in a direction opposite to that of the Sun.



OR

- (b) (i) The phenomenon of the change in direction of propagation of light caused by large sized molecules/ caused by colloidal particles.

When sunlight passes through the atmosphere, fine particles in the air scatter blue light (shorter wavelengths) more than the red colour (longer wavelengths).

20. *Assertion (A) :* A person suffering from myopia cannot see the distant objects clearly.

Reason (R) : A converging lens is used for the correction of myopic eye as it can form real as well as virtual images of the objects placed in front of it.

20. *Assertion (A) :* A person suffering from myopia cannot see the distant objects clearly.

Reason (R) : A converging lens is used for the correction of myopic eye as it can form real as well as virtual images of the objects placed in front of it.

(c) Assertion (A) is true, but Reason (R) is false.

- 25.** (a) (i) What is the (I) least distance of distinct vision, and (II) far point of a normal human eye ?
- (ii) A student is unable to see distinctly a chart hanging on the wall of his classroom. Name the defect of vision he is suffering from and the type of corrective lens he must use in his spectacles.

- 25.** (a) (i) What is the (I) least distance of distinct vision, and (II) far point of a normal human eye ?
- (ii) A student is unable to see distinctly a chart hanging on the wall of his classroom. Name the defect of vision he is suffering from and the type of corrective lens he must use in his spectacles.

2

(a)	
(i) (I) 25 cm (II) infinity / ∞	$\frac{1}{2}$, $\frac{1}{2}$
(ii) • myopia/near sightedness • concave lens/diverging lens	$\frac{1}{2}$, $\frac{1}{2}$

(b) Write the main function of each of the following in the human eye :

(i) Cornea

(ii) Iris

(iii) Retina

(iv) Ciliary muscles

(b) Write the main function of each of the following in the human eye :

(i) Cornea

(ii) Iris

(iii) Retina

(iv) Ciliary muscles

2

(b)	
(i) Cornea – Most of the refraction of light rays falling on the eye.	$\frac{1}{2}$
(ii) Iris – controls the size of the pupil.	$\frac{1}{2}$
(iii) Retina – image is formed on it	$\frac{1}{2}$
(iv) Ciliary muscles – To modify the curvature / focal length of the eye lens.	$\frac{1}{2}$

31. A person is unable to see clearly a poster fixed on a distant wall. He however sees it clearly when standing at a distance of about 2 m from the wall.
- (a) Draw ray diagram to show the formation of image by his eye lens when he is far away from the wall.
 - (b) List two possible reasons of this defect of vision.
 - (c) Draw ray diagram to show the correction of this defect using appropriate lens.

(a)



1

(b)

- Excessive curvature of the eye lens
- Elongation of the eyeball

$\frac{1}{2}$

$\frac{1}{2}$

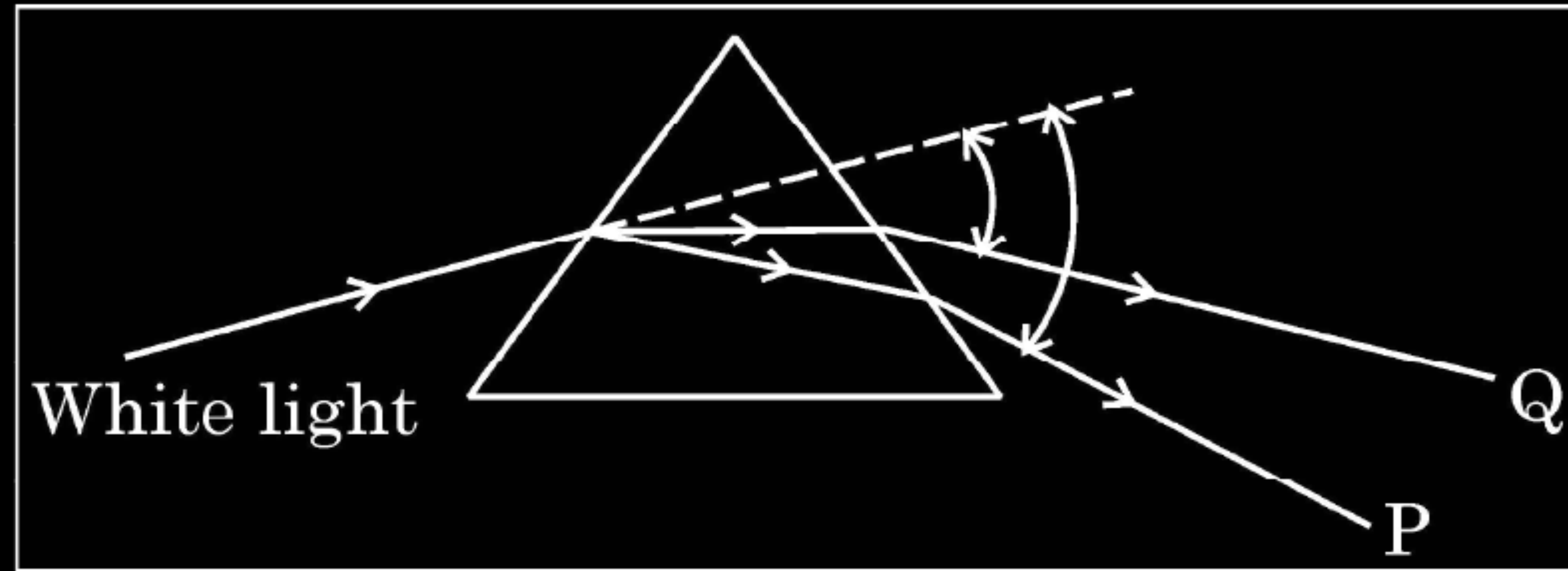
(c)



1

14. In the following diagram showing dispersion of white light by a glass prism, the colours 'P' and 'Q' respectively are –

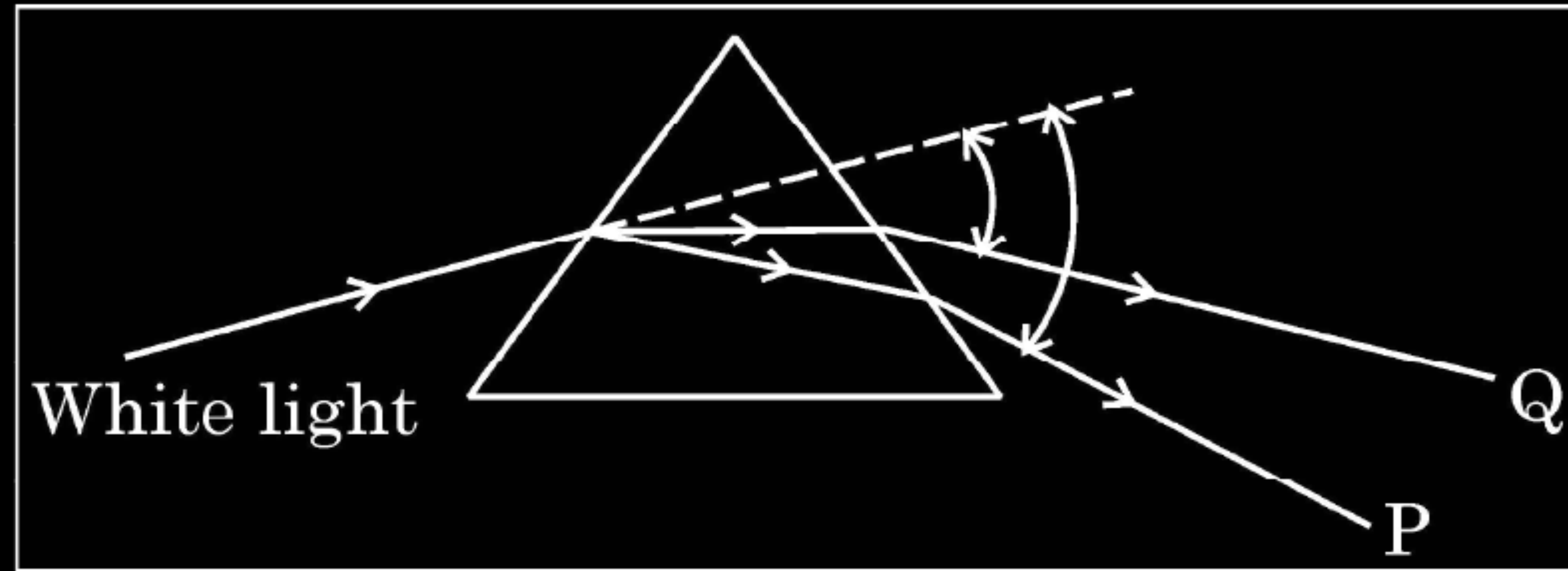
1



- | | |
|--------------------|----------------------|
| (a) Red and Violet | (b) Violet and Red |
| (c) Red and Blue | (d) Orange and Green |

14. In the following diagram showing dispersion of white light by a glass prism, the colours 'P' and 'Q' respectively are –

1



- | | |
|--------------------|----------------------|
| (a) Red and Violet | (b) Violet and Red |
| (c) Red and Blue | (d) Orange and Green |

(b) Violet and Red

12. The change in the focal length of an eye lens in human beings is caused by the action of

1

(a) optic nerves

(b) ciliary muscles

(c) retina

(d) cornea

12. The change in the focal length of an eye lens in human beings is caused by the action of

1

(a) optic nerves

(b) ciliary muscles

(c) retina

(d) cornea

(b) ciliary muscles

11. In human eye the part which allows light to enter into the eye is –

1

(a) Retina

(b) Pupil

(c) Eye lens

(d) Cornea

11. In human eye the part which allows light to enter into the eye is –

1

(a) Retina

(b) Pupil

(c) Eye lens

(d) Cornea

(d) Cornea

13. The phenomena of light involved in the formation of rainbow are

1

- (a) Refraction, dispersion and scattering.
- (b) Refraction, reflection and dispersion.
- (c) Refraction, dispersion and internal reflection.
- (d) Reflection, dispersion and total internal reflection.

13. The phenomena of light involved in the formation of rainbow are

1

- (a) Refraction, dispersion and scattering.
- (b) Refraction, reflection and dispersion.
- (c) Refraction, dispersion and internal reflection.
- (d) Reflection, dispersion and total internal reflection.

(c) Refraction, dispersion and internal reflection.

28. State reasons for Myopia. With the help of ray diagrams, show the

(a) image formation by a myopic eye, and

(b) correction of myopia using an appropriate lens.

3

28. State reasons for Myopia. With the help of ray diagrams, show the

(a) image formation by a myopic eye, and

(b) correction of myopia using an appropriate lens.

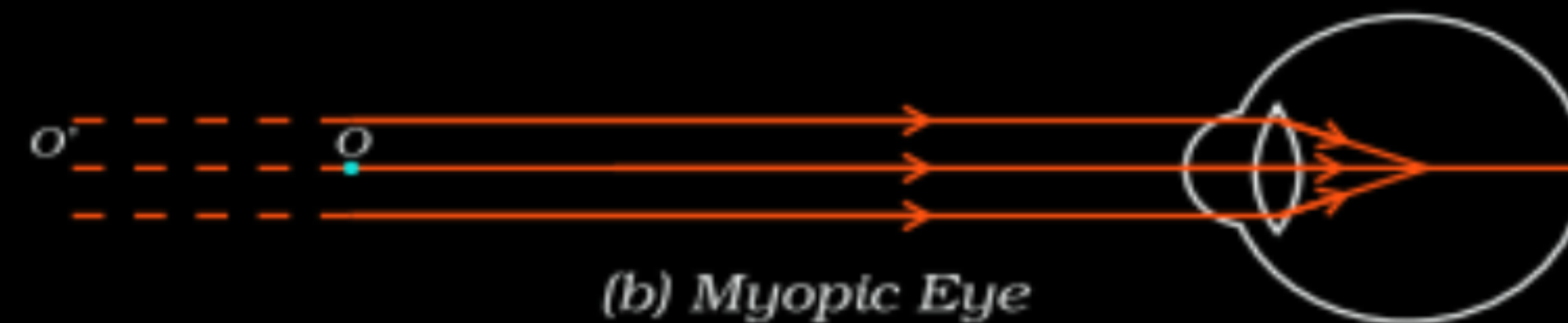
3

- excessive curvature of the eye lens
- elongation of the eyeball

$\frac{1}{2}$

$\frac{1}{2}$

(a)



1

(b)



1

33. A person cannot see distinctly the object placed beyond 5 m from his eyes. Name the defect of vision the person is suffering from. Draw a ray diagram to illustrate this defect. List its two possible causes. Name the lens used for the correction of this defect.

(b) It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age.

2

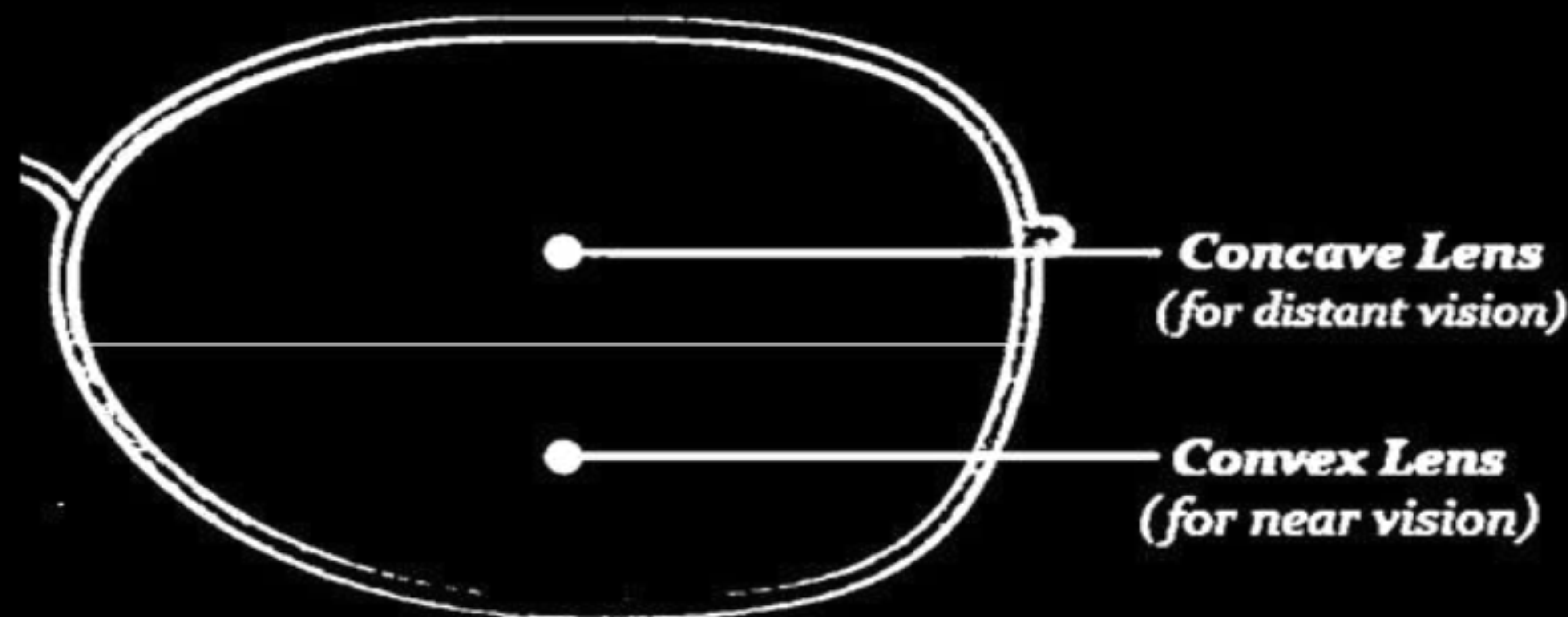
(i) Give reason for the above statement.

(ii) Name the defect that is likely to arise in the eyes in such a condition.

(iii) Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.

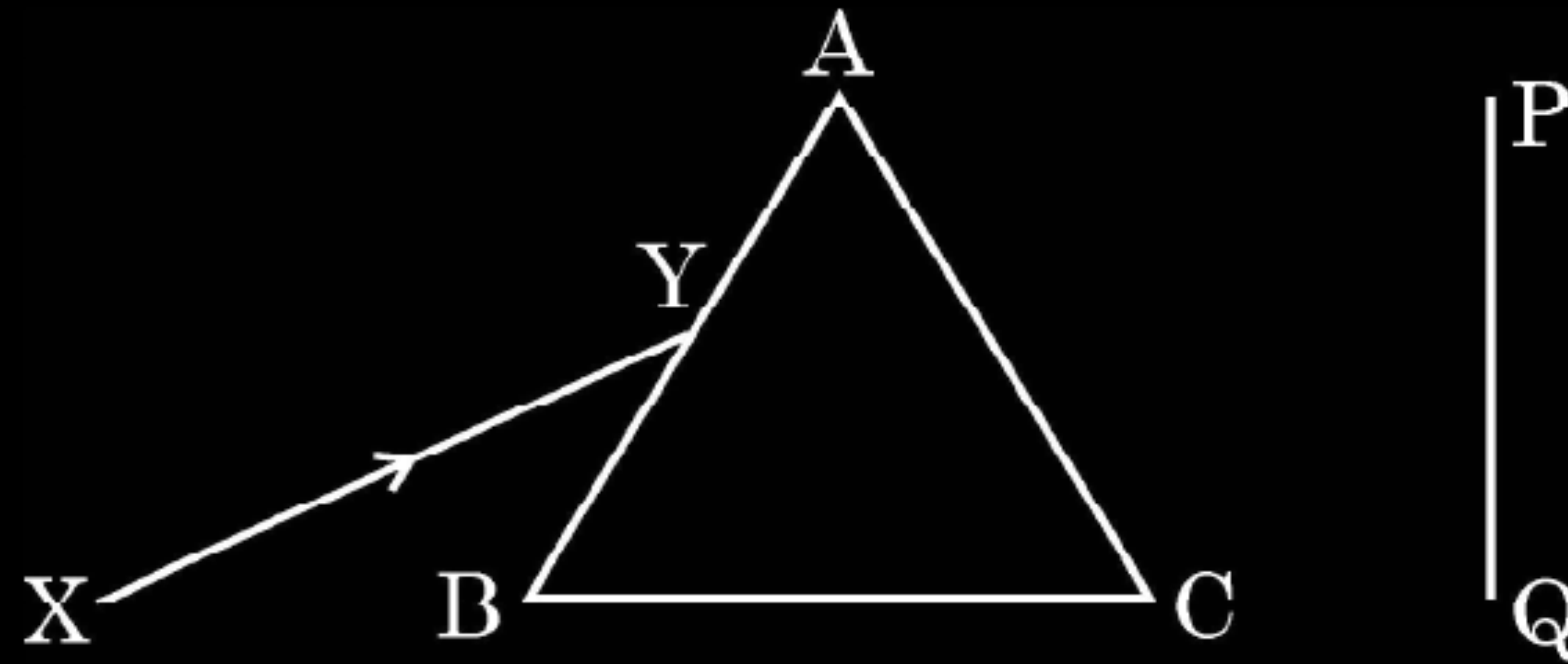
- (b) It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age.
- Give reason for the above statement.
 - Name the defect that is likely to arise in the eyes in such a condition.
 - Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.

- (b) (i) It is due to gradual weakening of the ciliary muscles and diminishing flexibility of the eye lens.
- (ii) Presbyopia/ Presbyopia + Myopia
- (iii) Bifocal /Concave + Convex lens/ Diagram

 $\frac{1}{2}, \frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$


26. (a) A narrow beam XY of white light is passing through a glass prism ABC as shown in the diagram :

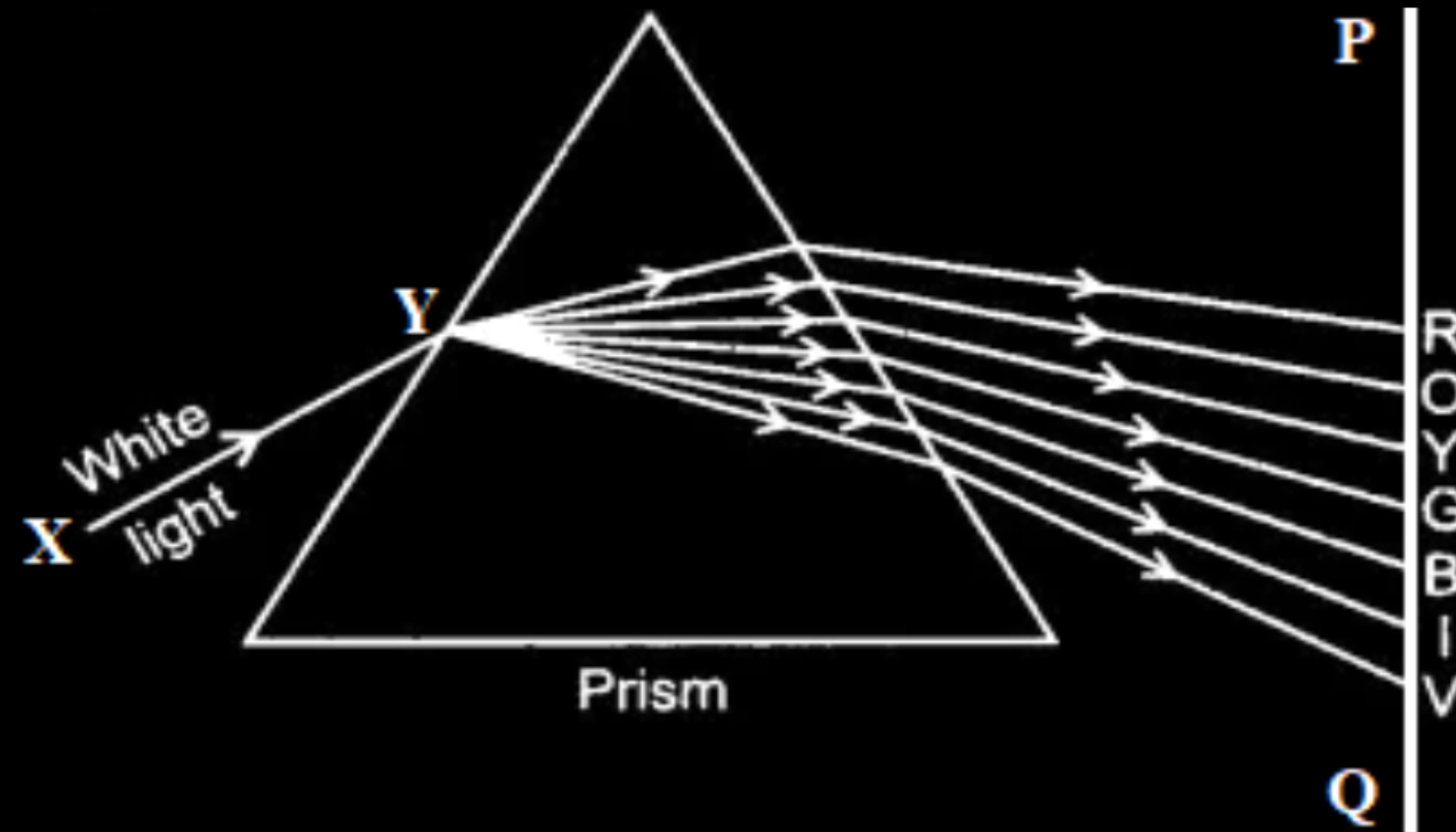
2



Trace it on your answer sheet and show the path of the emergent beam as observed on the screen PQ.

Name the phenomenon observed and state its cause.

(a)



- Dispersion of white light
- Cause: Different colours of light bend through different angles w.r.t. the incident ray. / Different colours have different wavelengths.

1

$\frac{1}{2}$

$\frac{1}{2}$

29. A person is suffering from an eye defect in which the far point of the eye is nearer than infinity. Identify the defect. List two main causes of this defect.

3

Draw a ray diagram to show how this defect is corrected by using a suitable lens.

29. A person is suffering from an eye defect in which the far point of the eye is nearer than infinity. Identify the defect. List two main causes of this defect.

3

Draw a ray diagram to show how this defect is corrected by using a suitable lens.

•Defect : Myopia/Short sightedness

Two Causes :

- Excessive curvature of the eye lens
- Elongation of the eyeball



1

$\frac{1}{2}$

$\frac{1}{2}$

1

31. Give reasons for the following :

3

- (a) Danger signals installed at airports and at the top of tall buildings are of red colour.
- (b) The sky appears dark to the passengers flying at very high altitudes.
- (c) The path of a beam of light passing through a colloidal solution is visible.

31. Give reasons for the following :

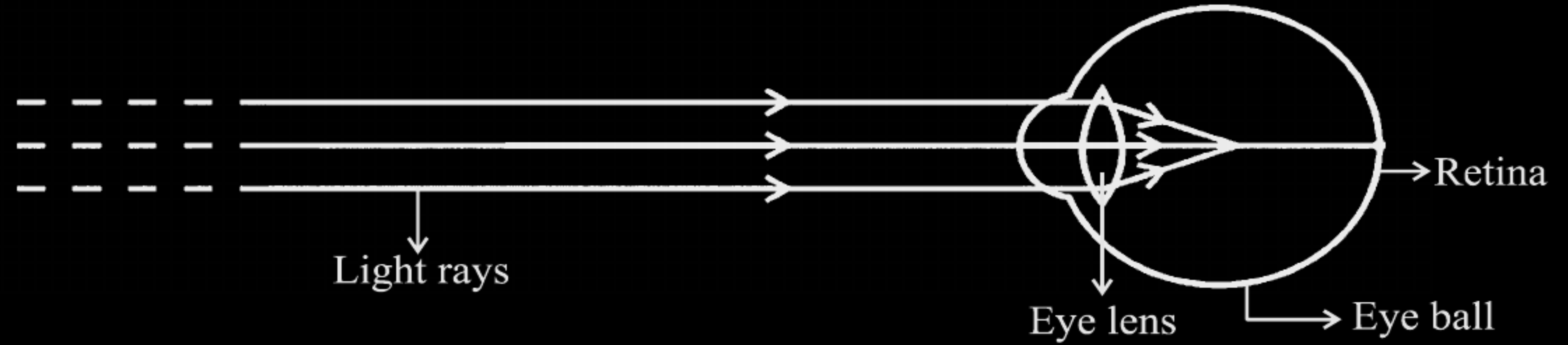
3

- (a) Danger signals installed at airports and at the top of tall buildings are of red colour.
- (b) The sky appears dark to the passengers flying at very high altitudes.
- (c) The path of a beam of light passing through a colloidal solution is visible.

(a) Red Coloured light is least scattered by fog or smoke and can be easily seen from a distance. / Red colour has high wavelength thus scattering is less.	1
(b) There is no scattering of light due to lack of atmosphere.	1
(c) Particles of colloid are big enough to scatter the beam of light.	1

26. (A) Observe the following diagram and answer the questions following it :

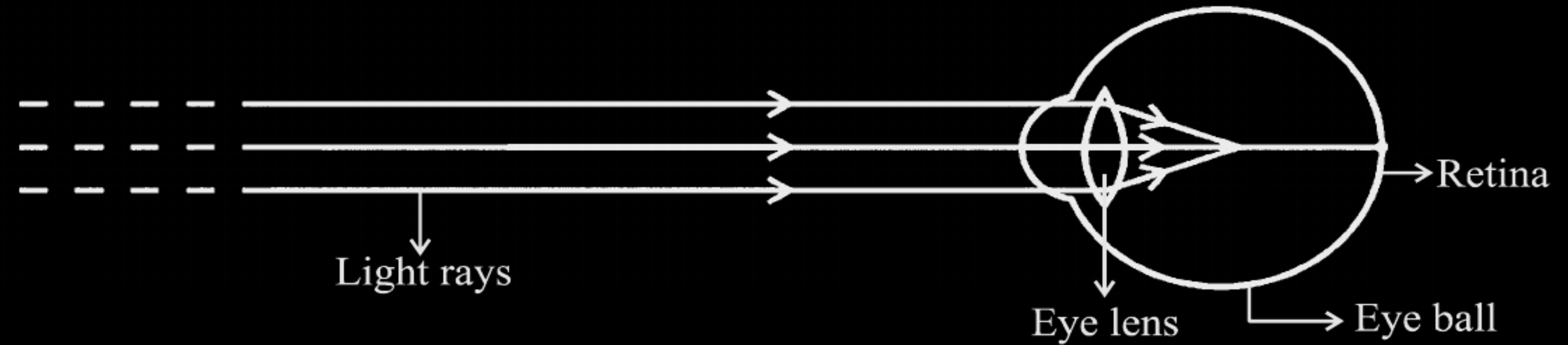
2



- (i) Identify the defect of vision shown.
- (ii) List its two causes.
- (iii) Name the type of lens used for the correction of this defect.

26. (A) Observe the following diagram and answer the questions following it :

2



- (i) Identify the defect of vision shown.
- (ii) List its two causes.
- (iii) Name the type of lens used for the correction of this defect.

(A) (i) Myopia / Short Sightedness
(ii) • Excessive curvature of eye lens
• Elongation of eye ball
(iii) Concave lens /Diverging Lens

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(B) The colour of clear sky from the earth appears blue but from the space it appears black. Why ?

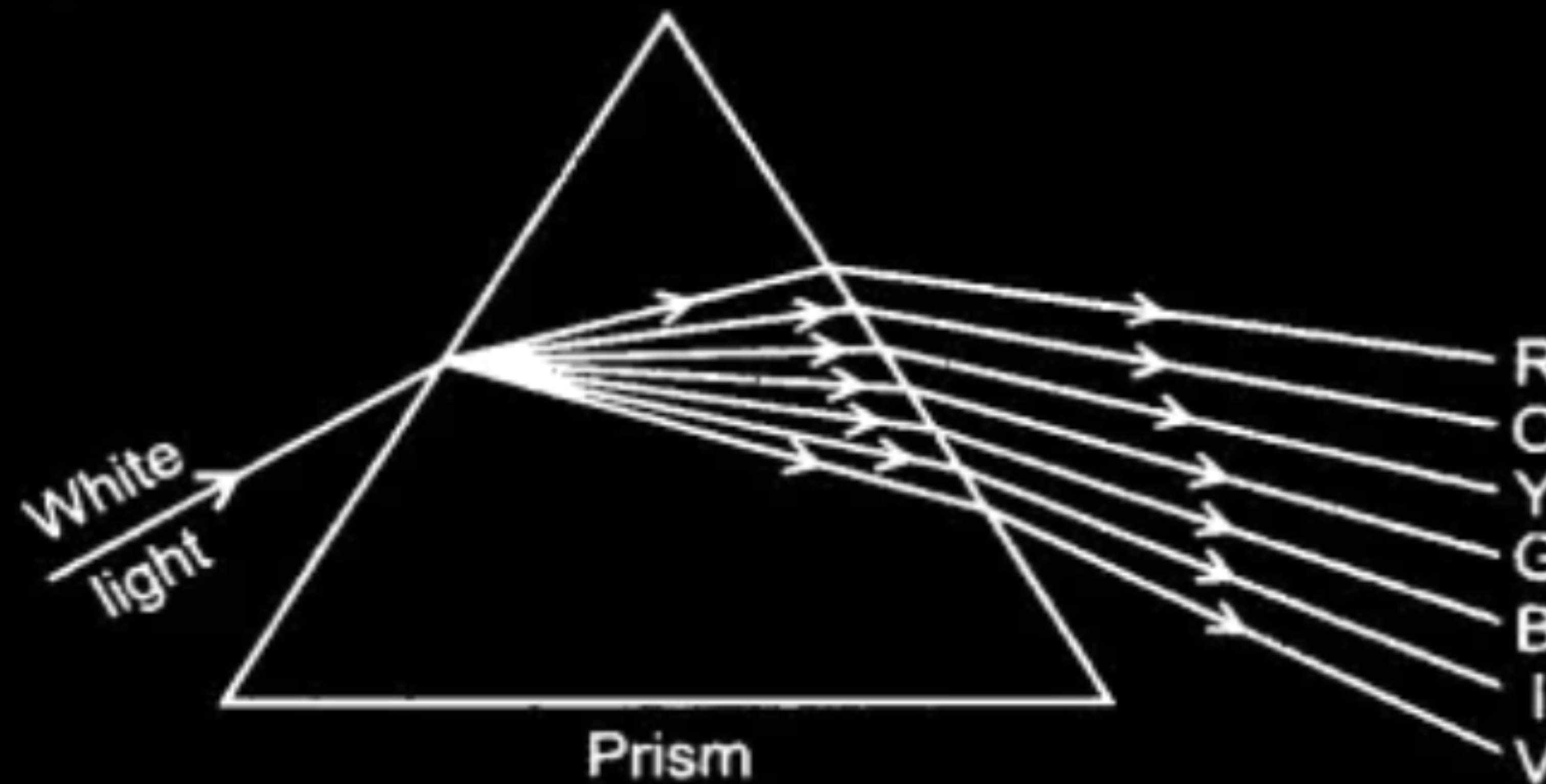
- 33. (A)** Define the term dispersion of white light. State the colour which bends (i) the most, (ii) the least while passing through a glass prism. Draw a diagram to show the dispersion of white light.

33. (A) Define the term dispersion of white light. State the colour which bends (i) the most, (ii) the least while passing through a glass prism. Draw a diagram to show the dispersion of white light.

3

(A)

- The splitting up of white light into its constituent/seven different colours while passing through a glass prism is called dispersion of white light.
- (i) Violet bends the most
- (ii) Red bends the least



1

$\frac{1}{2}$

$\frac{1}{2}$

1

(B) What is a rainbow ? Draw a labelled diagram to show its formation.

3

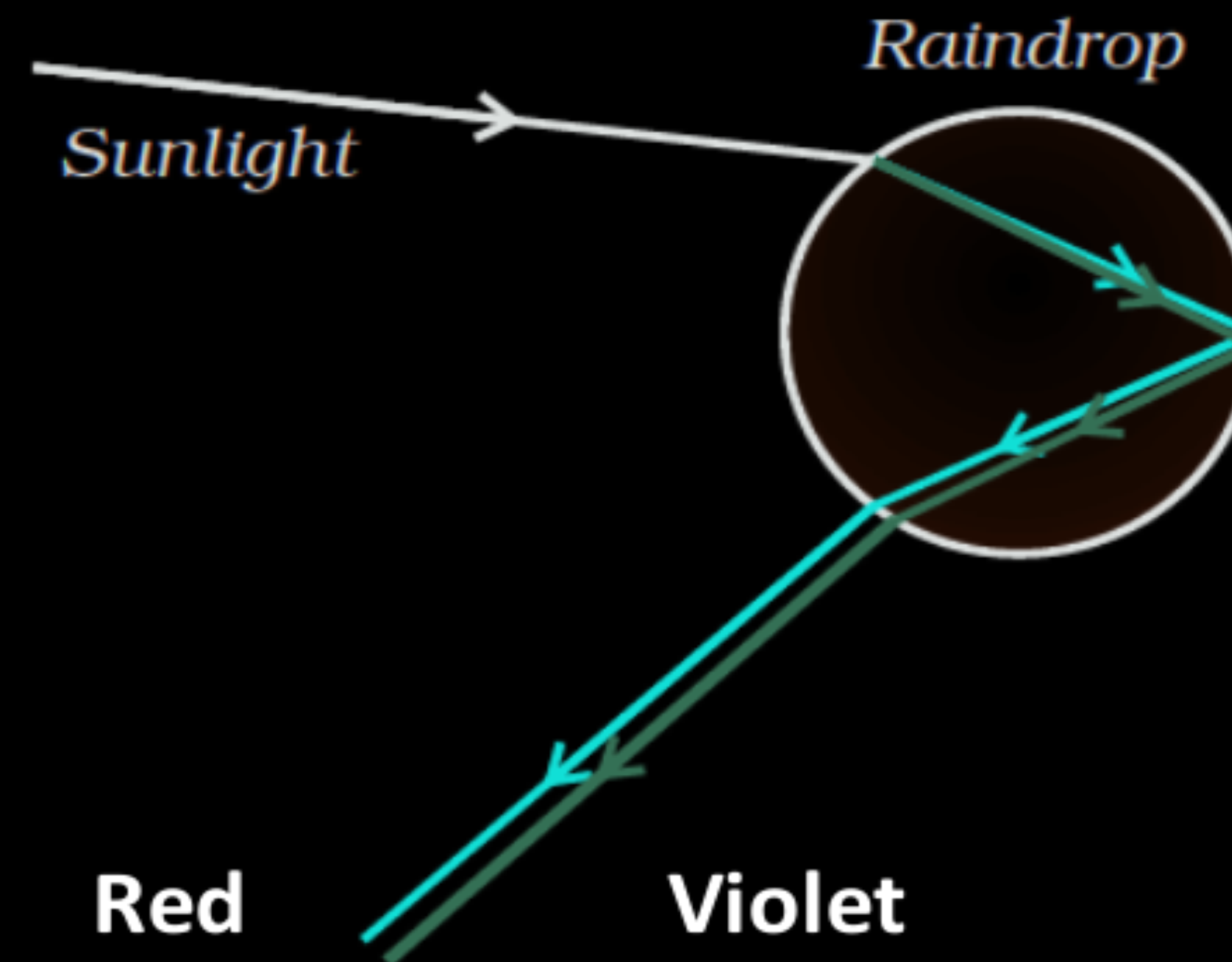
(B) What is a rainbow ? Draw a labelled diagram to show its formation.

3

(B)

- A rainbow is a natural spectrum of sunlight appearing in the sky after the rainfall.

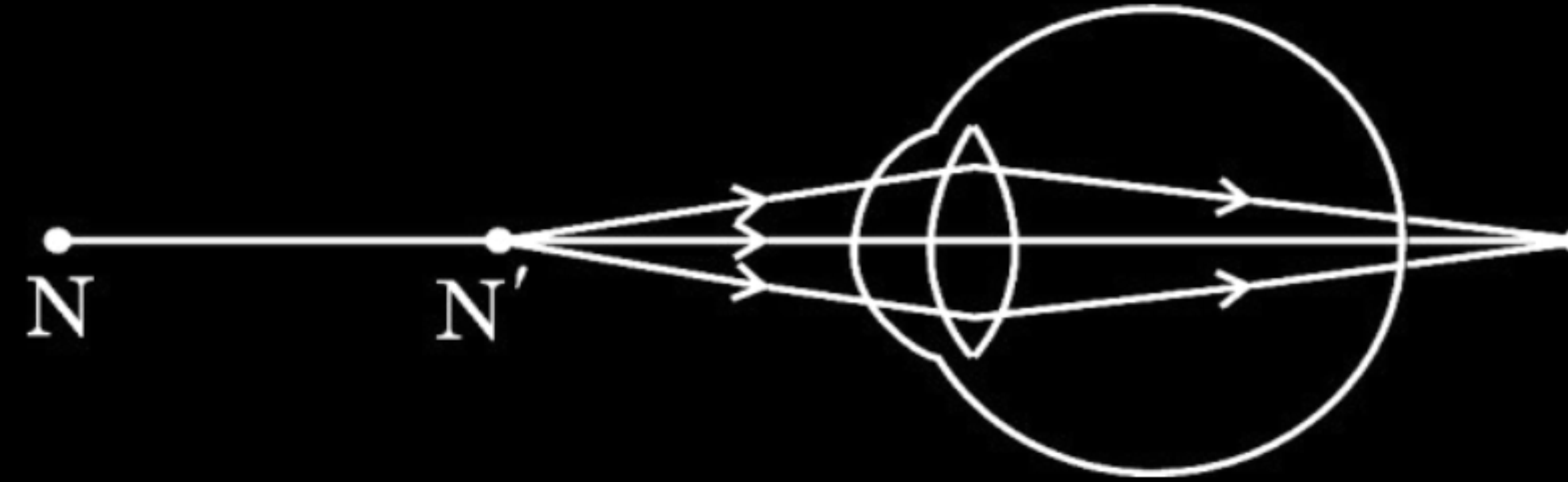
-



1

2

- 33.** (a) Study the diagram given below and answer the questions that follow :



- (i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition.
- (ii) List two causes of this defect.
- (iii) Name the type of lens used to correct this defect and state its role in this case.

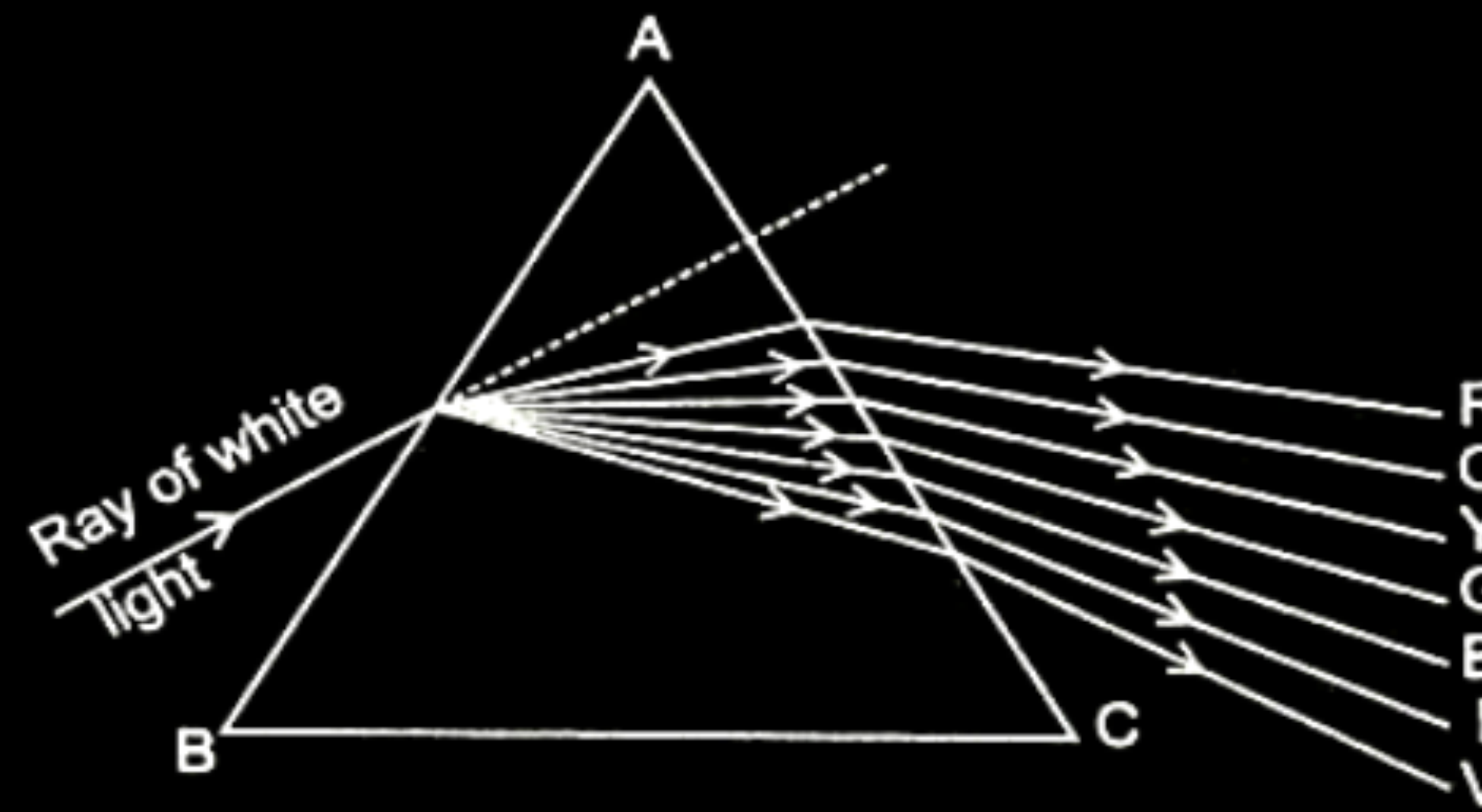
(a)	
(i) • Hypermetropia	$\frac{1}{2}$
• Ciliary muscles/ eye lens	$\frac{1}{2}$
(ii) • Focal length of the eye lens is too long.	$\frac{1}{2}$
• Eyeball becomes too small.	$\frac{1}{2}$
(iii) Converging lenses/ convex lens	$\frac{1}{2}$
They provide the additional focussing power required for forming the image on the retina./ Decrease the focal length of the eye lens	$\frac{1}{2}$

(b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

3

- (b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism. 3

(b)
The splitting of white light into its constituent colours is called dispersion.
Cause: Different colours of white light bend through different angles with respect to incident ray.



15. The colour of light for which the refractive index of glass is minimum, is :

(A) Red

(B) Yellow

(C) Green

(D) Violet

15. The colour of light for which the refractive index of glass is minimum, is :

(A) Red

(B) Yellow

(C) Green

(D) Violet

(A) / Red

- 14.** The phenomena of light involved in the formation of rainbow are :
- (A) Refraction, reflection and dispersion
 - (B) Refraction, dispersion and internal reflection
 - (C) Reflection, dispersion and internal reflection
 - (D) Refraction, dispersion, scattering and total internal reflection

14. The phenomena of light involved in the formation of rainbow are :
- (A) Refraction, reflection and dispersion
 - (B) Refraction, dispersion and internal reflection
 - (C) Reflection, dispersion and internal reflection
 - (D) Refraction, dispersion, scattering and total internal reflection

(B) / Refraction, Dispersion and internal reflection

- 25.** (a) A student cannot see distinctly the charts hanging in the science laboratory. Name the defect of vision he/she is suffering from. List two causes of this defect. Suggest the type of lenses for the modification of this defect.

- 25.** (a) A student cannot see distinctly the charts hanging in the science laboratory. Name the defect of vision he/she is suffering from. List two causes of this defect. Suggest the type of lenses for the modification of this defect.

2

(a) • Myopia / Near sightedness	$\frac{1}{2}$
Causes: i. Elongation of eyeball	$\frac{1}{2}$
ii. Excessive curvature of eye lens	$\frac{1}{2}$
• Modification – use of concave lens / diverging lens	$\frac{1}{2}$

- (b) A person cannot see distinctly the objects placed at the least distance of distinct vision for a normal eye. However, he can easily read a newspaper by placing it at a distance of 40 cm from his eyes. Name the defect of vision in this case. List its two causes. Suggest the type of lenses for his spectacles.

- (b) A person cannot see distinctly the objects placed at the least distance of distinct vision for a normal eye. However, he can easily read a newspaper by placing it at a distance of 40 cm from his eyes. Name the defect of vision in this case. List its two causes. Suggest the type of lenses for his spectacles.

2

b. • Hypermetropia / Far sightedness	$\frac{1}{2}$
Causes: i. Focal length of the eye lens increases	$\frac{1}{2}$
ii. Eyeball becomes too small	$\frac{1}{2}$
• convex lens / converging lens	$\frac{1}{2}$

16. When a narrow beam of white light passes through a glass prism it splits into its component colours ? This phenomenon is called :

- (a) Diffusion of light
- (b) Total reflection of light
- (c) Scattering of light
- (d) Dispersion of light

16. When a narrow beam of white light passes through a glass prism it splits into its component colours ? This phenomenon is called :

- (a) Diffusion of light
- (b) Total reflection of light
- (c) Scattering of light
- (d) Dispersion of light

(d) Dispersion of light

17. *Assertion (A) :* A rainbow is an artificial spectrum of white light appearing in the sky after a rain shower.

Reason (R) : The water droplets act like small prisms.

17. *Assertion (A) :* A rainbow is an artificial spectrum of white light appearing in the sky after a rain shower.

Reason (R) : The water droplets act like small prisms.

(d) Assertion (A) is false, but Reason (R) is true.

- 31.** A person keeps the reading material much beyond 25 cm from the eye for comfortable reading. Name the defect of vision he is suffering from. How can it be corrected ? Draw ray diagrams for (i) the defective eye and (ii) its correction.

- 31.** A person keeps the reading material much beyond 25 cm from the eye for comfortable reading. Name the defect of vision he is suffering from. How can it be corrected ? Draw ray diagrams for (i) the defective eye and (ii) its correction.

3

- Hypermetropia / far sightedness.
- By using a convex lens of appropriate power

$\frac{1}{2}$

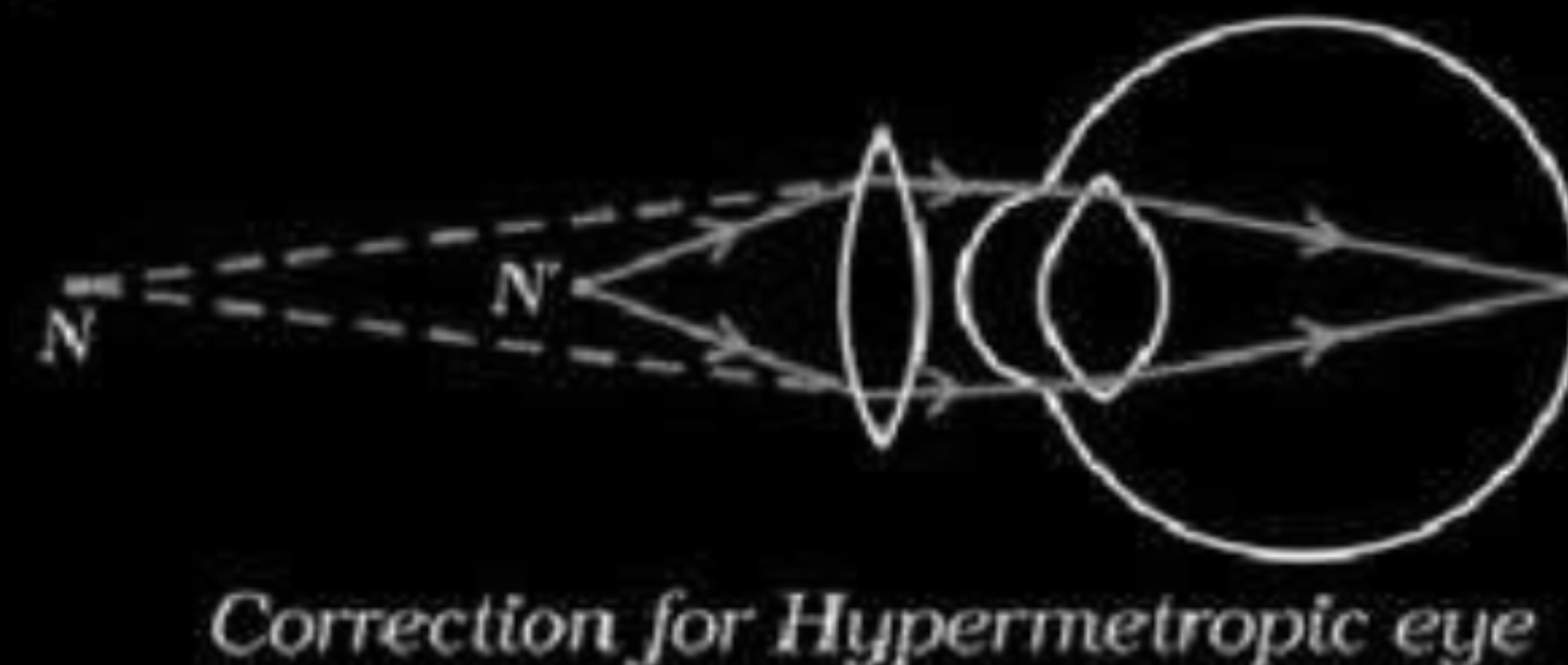
$\frac{1}{2}$

(i)



1

(ii)

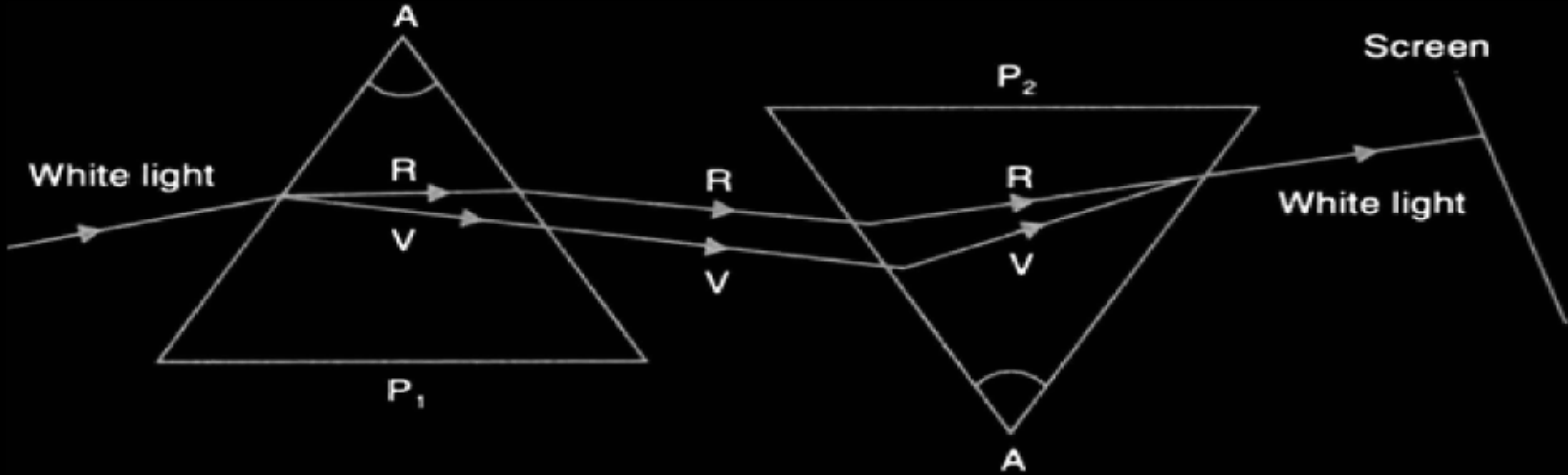


1

- 31.** (a) White light is dispersed into seven visible coloured components by a glass prism. Name the colour which bends (i) the most and (ii) the least.
- (b) How can the coloured components of white light be recombined after a prism has separated them ? Draw a labelled diagram to justify your answer.

- 31.** (a) White light is dispersed into seven visible coloured components by a glass prism. Name the colour which bends (i) the most and (ii) the least.
- (b) How can the coloured components of white light be recombined after a prism has separated them ? Draw a labelled diagram to justify your answer.

3

(a) (i) Violet (ii) Red (b) By placing another identical prism in an inverted position with respect to the first prism.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
	$1\frac{1}{2}$

31. What is Tyndall effect ? Give two examples where this phenomenon is observed. Give reason to explain this phenomenon.

31. What is Tyndall effect ? Give two examples where this phenomenon is observed. Give reason to explain this phenomenon.

3

• It is the phenomenon of scattering of light by the colloidal particles due to which the path of the light becomes visible. • Examples: 1. When sunlight passes through a canopy of dense forest. 2. A fine beam of sunlight enters a smoke/ dust filled dark room through a small hole. • Explanation: When a beam of light passes through a heterogenous mixture of minute particles (dust, smoke etc) it is reflected diffusely by the particles and gets scattered. This scattering of light makes the beam of light visible.	1 $\frac{1}{2}$ $\frac{1}{2}$ 1
---	---

13. The lens system of human eye forms an image on a light sensitive screen,
which is called as :

1

- (a) Cornea
- (b) Ciliary muscles
- (c) Optic nerves
- (d) Retina

13. The lens system of human eye forms an image on a light sensitive screen, which is called as :

1

- (a) Cornea
- (b) Ciliary muscles
- (c) Optic nerves
- (d) Retina

(d) /Retina

- 20. Assertion (A) :** The rainbow is a natural spectrum of sunlight in the sky.
- Reason (R) :** Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air.

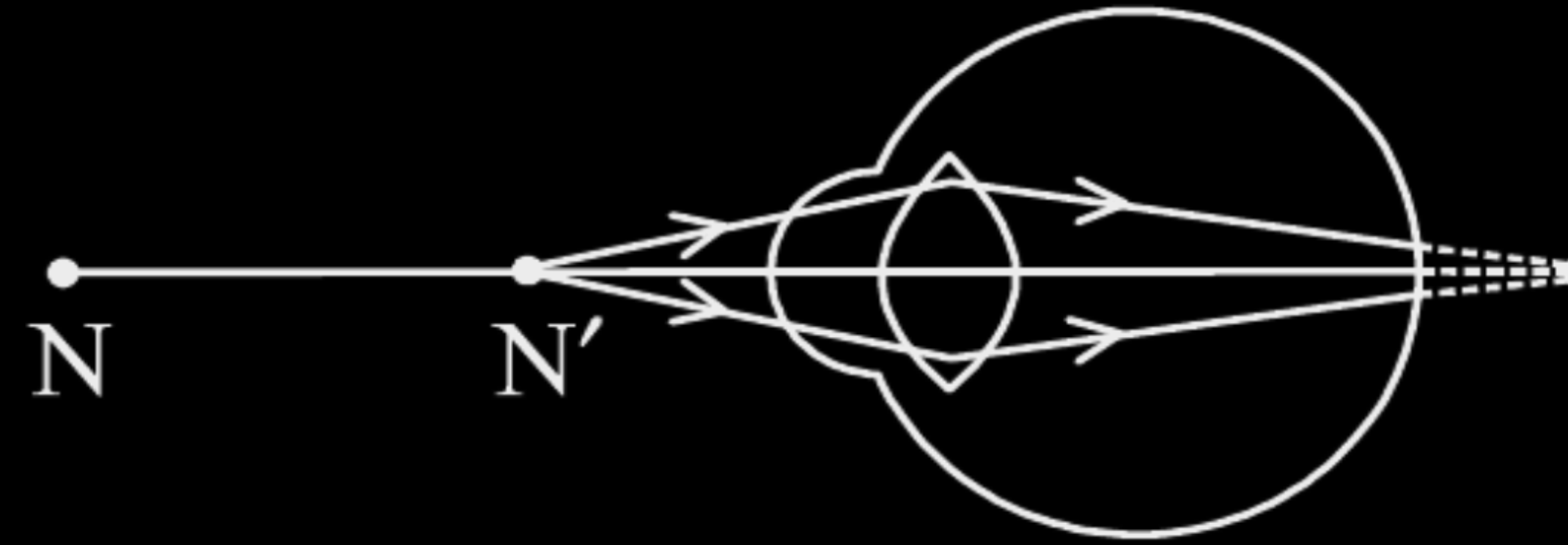
- 20. Assertion (A) :** The rainbow is a natural spectrum of sunlight in the sky.
Reason (R) : Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air.

1

(c) /Assertion (A) is true, but Reason (R) is false.

31. Study the diagram given below and answer the questions that follow :

3



- (i) Name the defect of vision represented in the diagram. Give reason for your answer.
- (ii) List two causes of this defect.
- (iii) With the help of a diagram show how this defect of vision is corrected.

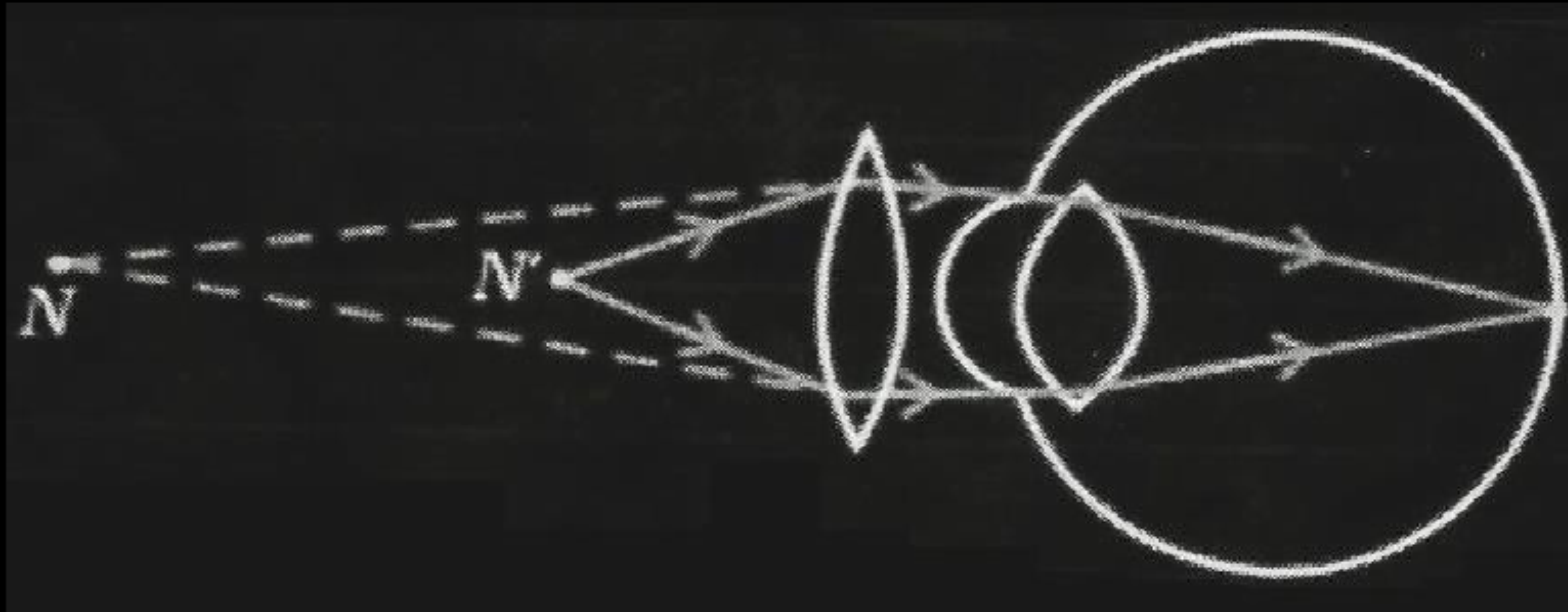
(i) Hypermetropia or Far-sightedness.

Reason – Image is formed behind the retina. / Near point for the person is farther away from the normal near point (25 cm)

(ii)

- Focal length of the eye lens is too long.
- The eyeball has become too small.

(iii)



N = Near point of a hypermetropic eye

N' = Near point of a normal eye

$\frac{1}{2}$

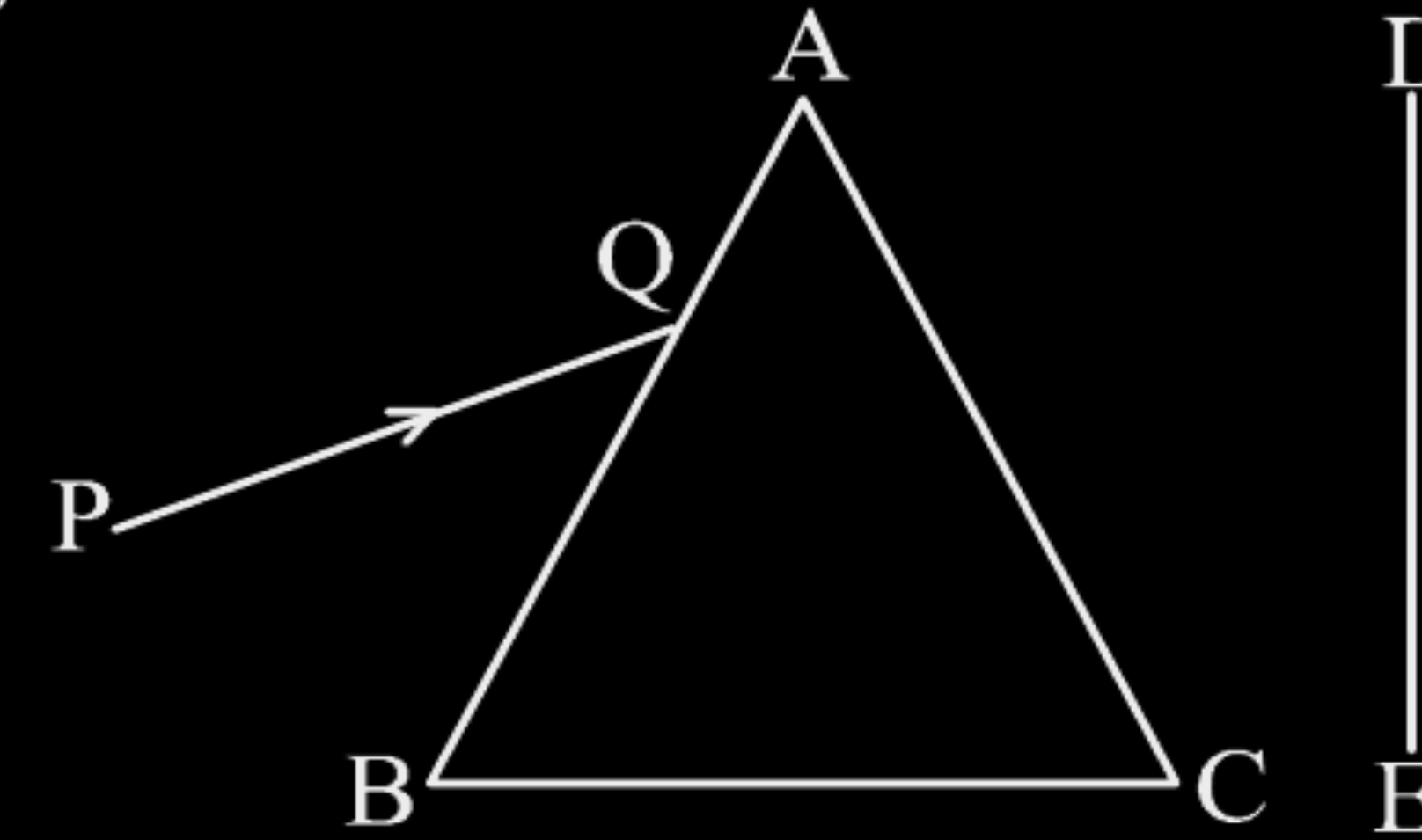
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$\frac{1}{2}$

$\frac{1}{2}$

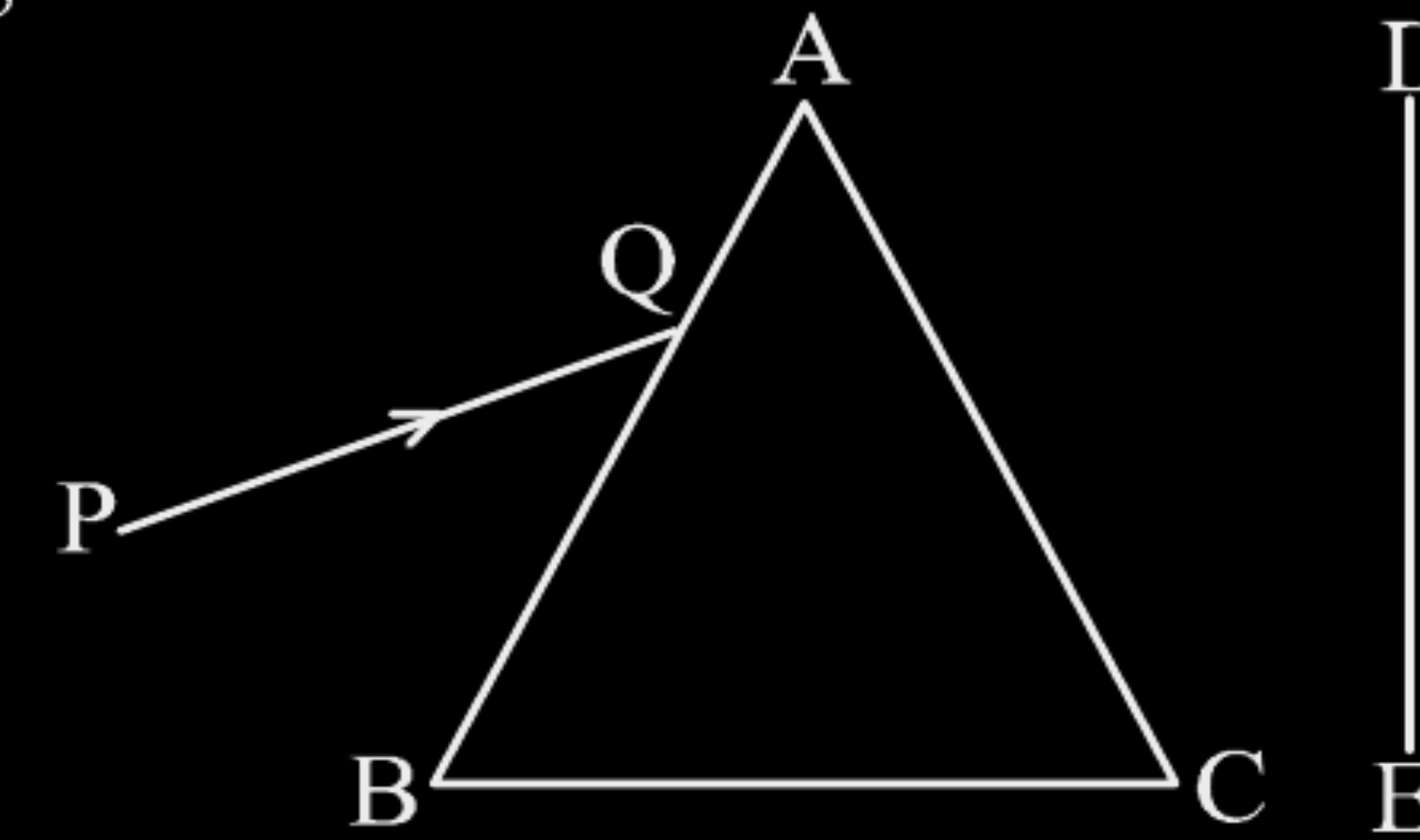
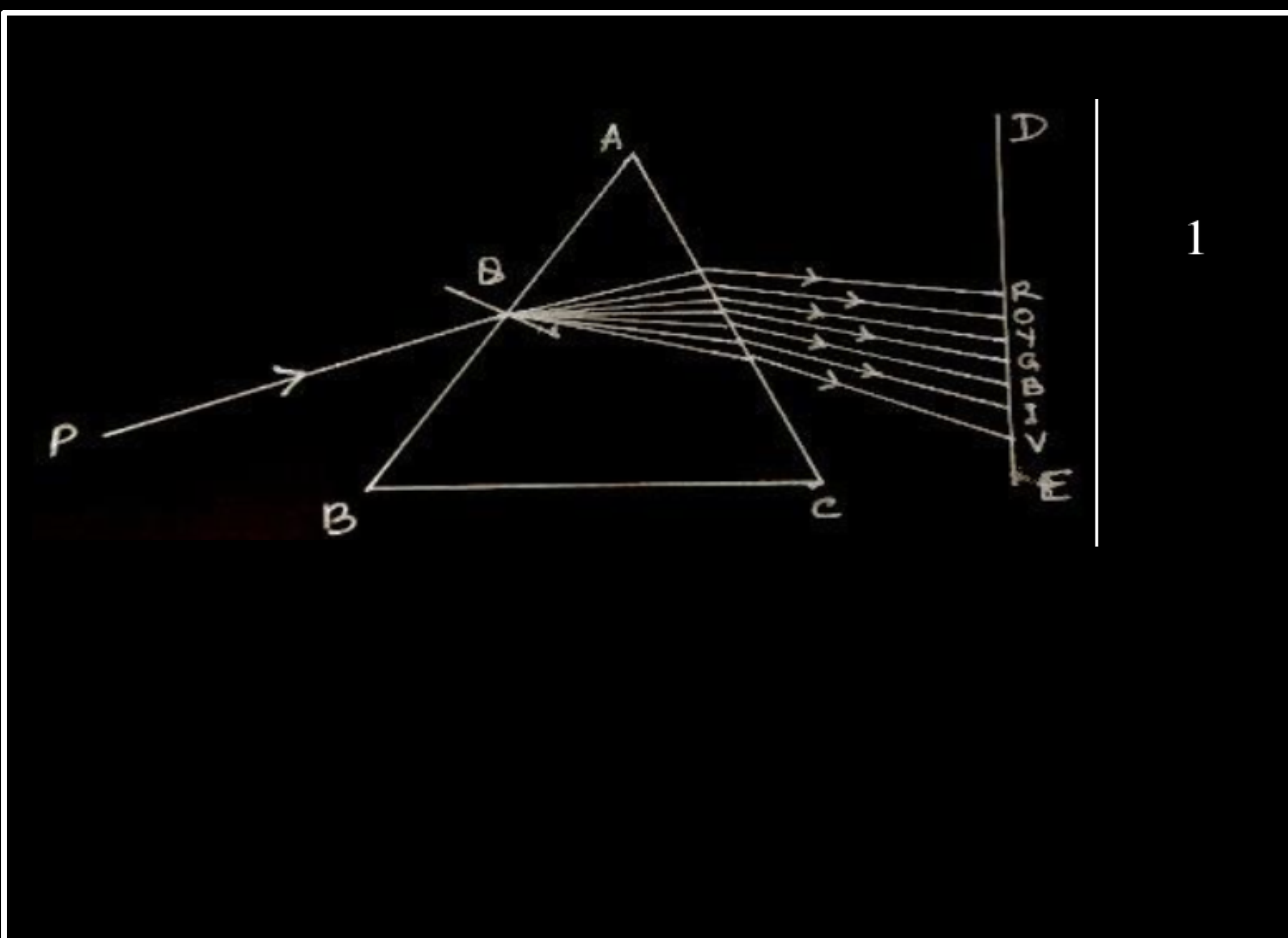
1

32. A narrow beam, PQ of white light is passing through a glass prism ABC as shown in the diagram.



Draw a ray diagram to show the emergent beam as it falls on the screen DE. Also write the phenomenon involved and its cause. Using the second law of refraction state which colour of light must have the highest value of refractive index amongst seven visible colours of light. Justify your answer.

32. A narrow beam, PQ of white light is passing through a glass prism ABC as shown in the diagram.



Draw a ray diagram to show the emergent beam as it falls on the screen DE. Also write the phenomenon involved and its cause. Using the second law of refraction state which colour of light must have the highest value of refractive index amongst seven visible colours of light. Justify your answer.

3

• Phenomenon: Dispersion of light	$\frac{1}{2}$
• Cause : Different colours of white light bend through different angles with respect to incident ray./ Different colours of white light have different wavelength therefore bend by different angles.	$\frac{1}{2}$
• Refractive index of glass is highest for violet colour.	$\frac{1}{2}$
• Justification :For same $\angle i$ the $\angle r$ is minimum for the violet light. $\left(\frac{\sin i}{\sin r} \right)$ or refractive index is highest	$\frac{1}{2}$

14. In bifocal lenses used for the correction of presbyopia :

1

- (a) the upper portion is of convex lens for the near vision and lower part is of concave lens for the distant vision.
- (b) the upper portion is of convex lens for the distant vision and lower part is of concave lens for the near vision.
- (c) the upper portion is of concave lens is for the near vision and lower part is of convex lens for the distant vision.
- (d) the upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.

14. In bifocal lenses used for the correction of presbyopia :

1

- (a) the upper portion is of convex lens for the near vision and lower part is of concave lens for the distant vision.
- (b) the upper portion is of convex lens for the distant vision and lower part is of concave lens for the near vision.
- (c) the upper portion is of concave lens is for the near vision and lower part is of convex lens for the distant vision.
- (d) the upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.

(d)/ The upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.

19. **Assertion (A)** : The colour of clear sky appears blue. **1**
- Reason (R)** : Light of blue colour has longer wavelength as compared to the light of red colour so it is scattered more in the upper atmosphere.

19. **Assertion (A)** : The colour of clear sky appears blue. **1**

Reason (R) : Light of blue colour has longer wavelength as compared to the light of red colour so it is scattered more in the upper atmosphere.

(C) / (A) is true, but (R) is false

25. (a) A person suffering from an eye defect uses lenses of power -1 D. Name the defect of vision and list its two causes. State the nature (converging/diverging) of the corrective lens.

25. (a) A person suffering from an eye defect uses lenses of power -1 D. Name the defect of vision and list its two causes. State the nature (converging/diverging) of the corrective lens.

2

(a)	
• Myopia	$\frac{1}{2}$
• Two causes :	
Excessive curvature of eye lens	$\frac{1}{2}$
Elongation of eye ball	$\frac{1}{2}$
• Diverging lens	$\frac{1}{2}$

- (b) What is presbyopia ? Name the type of lenses used for the correction of this defect. State the nature (converging/diverging) of the upper part of such lenses.

(b) What is presbyopia ? Name the type of lenses used for the correction of this defect. State the nature (converging/diverging) of the upper part of such lenses.

2

(b) • The power of accommodation of eye lens usually decreases with ageing and the person finds it difficult to see nearby objects comfortably and distinctly. • Convex lens (Bifocal lens if the person has myopia also.) • The upper part of bifocal lens will be diverging.	1 $\frac{1}{2}$ $\frac{1}{2}$
---	--

31. (a) Define the term power of accommodation of human eye. Write the name of the part of eye which plays a major role in the process of accommodation and explain what happens when human eye focuses (i) nearby objects and (ii) distant objects.

31. (a) Define the term power of accommodation of human eye. Write the name of the part of eye which plays a major role in the process of accommodation and explain what happens when human eye focuses (i) nearby objects and (ii) distant objects.

3

(a)	
• Ability of the eye lens to adjust its focal length.	1
• Ciliary muscles	1
• (i) While focusing on nearby objects ciliary muscles contract, eye lens becomes thick and its focal length decreases.	$\frac{1}{2}$
(ii) While focusing on distant objects ciliary muscles relax, eye lens becomes thin and its focal length increases.	$\frac{1}{2}$

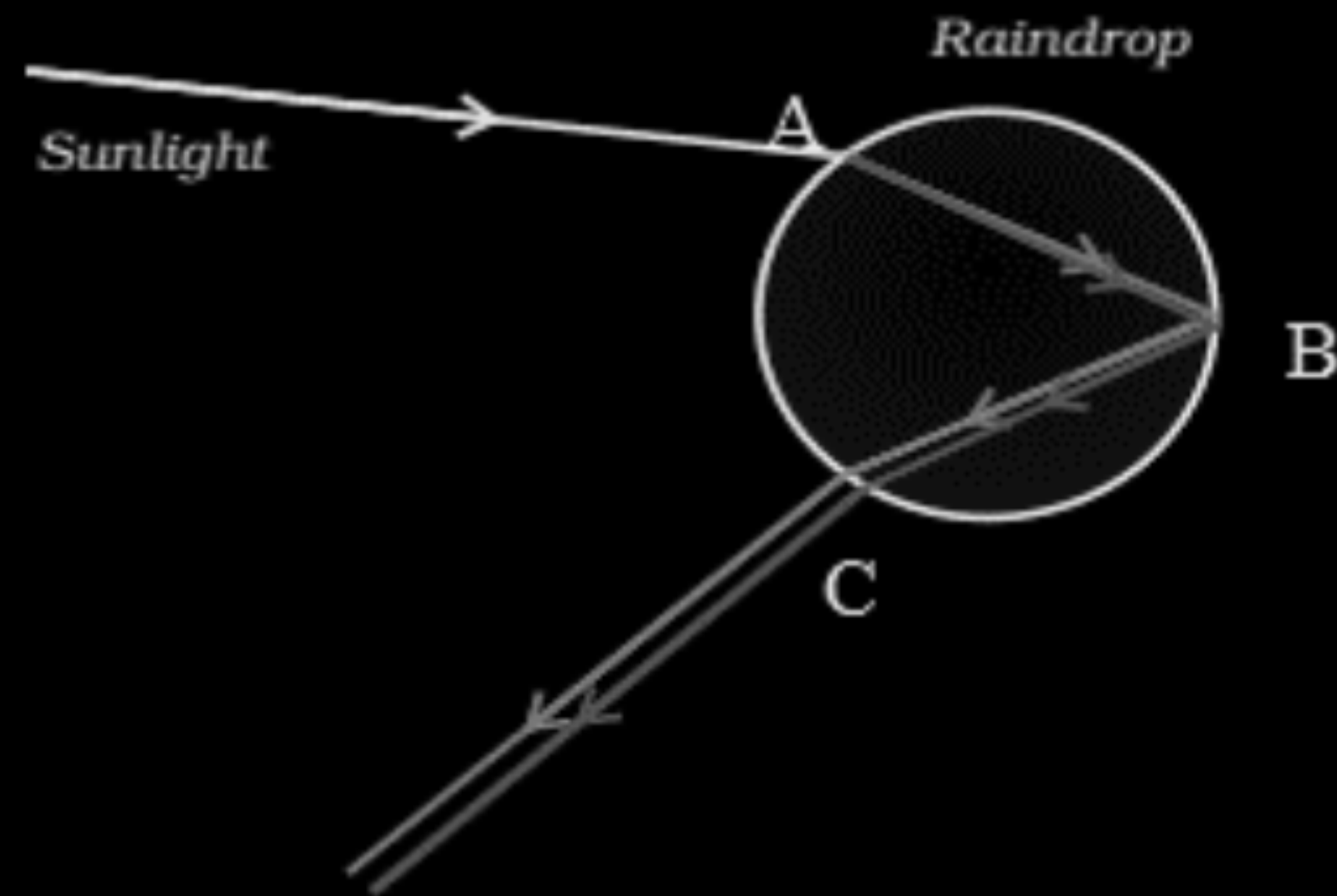
- (b) Draw a ray diagram to show the formation of a rainbow in the sky. On this diagram mark A – where dispersion of light occurs, B – where internal reflection of light occurs and C – where refraction of light occurs. List two necessary conditions to observe a rainbow.

3

- (b) Draw a ray diagram to show the formation of a rainbow in the sky. On this diagram mark A – where dispersion of light occurs, B – where internal reflection of light occurs and C – where refraction of light occurs. List two necessary conditions to observe a rainbow.

3

(b)



3 Labellings (A, B, C)

- Two conditions :
- (i) Presence of tiny water droplets in the atmosphere.
- (ii) Position of Sun at the back of/behind the observer.

$\frac{1}{2}$

$\frac{1}{2} \times 3$

$\frac{1}{2}$

$\frac{1}{2}$

19. **Assertion (A)** : Myopic eye cannot see distant objects distinctly. **1**
Reason (R) : For the correction of myopia converging lenses of appropriate power are prescribed by eye-surgeons.

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Reason (R) : For the correction of myopia converging lenses of appropriate power are prescribed by eye-surgeons.

(C) / (A) is true, but (R) is false.

13. Consider the following statements in the context of human eye :

1

- (a) The diameter of the eye ball is about 2.3 cm.
- (b) Iris is a dark muscular diaphragm that controls the size of the pupil.
- (c) Most of the refraction for the light rays entering the eye occurs at the crystalline lens.
- (d) While focusing on the objects at different distances the distance between the crystalline lens and the retina is adjusted by ciliary muscles.

The correct statements are –

- | | |
|----------------------|----------------------|
| (A) (a) and (b) | (B) (a), (b) and (c) |
| (C) (b), (c) and (d) | (D) (a), (c) and (d) |

13. Consider the following statements in the context of human eye :

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- (a) The diameter of the eye ball is about 2.3 cm.
- (b) Iris is a dark muscular diaphragm that controls the size of the pupil.
- (c) Most of the refraction for the light rays entering the eye occurs at the crystalline lens.
- (d) While focusing on the objects at different distances the distance between the crystalline lens and the retina is adjusted by ciliary muscles.

The correct statements are –

- | | |
|----------------------|----------------------|
| (A) (a) and (b) | (B) (a), (b) and (c) |
| (C) (b), (c) and (d) | (D) (a), (c) and (d) |

(A)/ (a) and (b)

19. **Assertion (A)** : Red light signals are used to stop the vehicles on the road. **1**

Reason (R) : Red coloured light is scattered the most so as to be visible from a large distance.

19. **Assertion (A)** : Red light signals are used to stop the vehicles on the road. **1**

Reason (R) : Red coloured light is scattered the most so as to be visible from a large distance.

(C) / Assertion (A) is true, but Reason (R) is false

24. When do we say that a particular person is suffering from hypermetropia ?
List two causes of this defect. Name the type of lens used to correct this defect.

24. When do we say that a particular person is suffering from hypermetropia ?
List two causes of this defect. Name the type of lens used to correct this defect.

2

• When he cannot see nearby objects distinctly but can see far object clearly.	$\frac{1}{2}$
• 2 causes: Focal length of the eye lens is too long.	$\frac{1}{2}$
Eyeball becomes too small.	$\frac{1}{2}$
• Convex or Converging lens	$\frac{1}{2}$

13. The Phenomena of light involved in the formation of a rainbow in the sky are

1

- (A) Refraction, dispersion and reflection
- (B) Refraction, dispersion and total internal reflection
- (C) Dispersion, scattering and reflection
- (D) Dispersion, refraction and internal reflection

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- (D) Dispersion, refraction and internal reflection

(D) Dispersion, refraction and internal reflection

31. Name and explain the phenomenon of light due to which the path of a beam of light becomes visible when it enters a smoke filled room through a small hole. Also state the dependence of colour of the light we receive on the size of the particle of the medium through which the beam of light passes.

31. Name and explain the phenomenon of light due to which the path of a beam of light becomes visible when it enters a smoke filled room through a small hole. Also state the dependence of colour of the light we receive on the size of the particle of the medium through which the beam of light passes.

3

• Scattering of light / Tyndall effect	1
• When a beam of light strikes fine particles of smoke, it is reflected diffusely and the path of the light becomes visible.	1
• Very fine particles scatter mainly blue light/short wavelength colours while the particles of larger size scatter longer wavelength colours.	1

15. The colour of light for which the refractive index of glass is minimum, is :

(A) Red

(B) Yellow

(C) Green

(D) Violet

15. The colour of light for which the refractive index of glass is minimum, is :

(A) Red

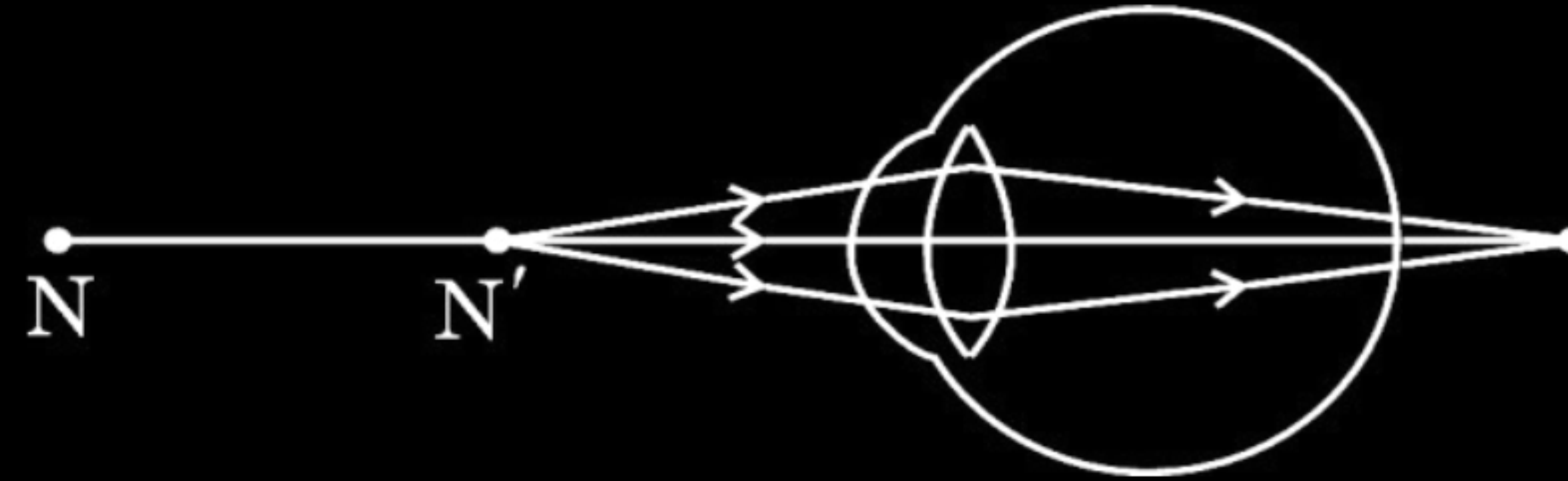
(B) Yellow

(C) Green

(D) Violet

(A) / Red

- 33.** (a) Study the diagram given below and answer the questions that follow :



- (i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition.
- (ii) List two causes of this defect.
- (iii) Name the type of lens used to correct this defect and state its role in this case.

(a)

(i) • Hypermetropia

• Ciliary muscles/ eye lens

$\frac{1}{2}$

$\frac{1}{2}$

(ii) • Focal length of the eye lens is too long.

$\frac{1}{2}$

• Eyeball becomes too small.

$\frac{1}{2}$

(iii) Converging lenses/ convex lens

$\frac{1}{2}$

They provide the additional focussing power required for forming the image on the retina./ Decrease the focal length of the eye lens

$\frac{1}{2}$

(b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

3

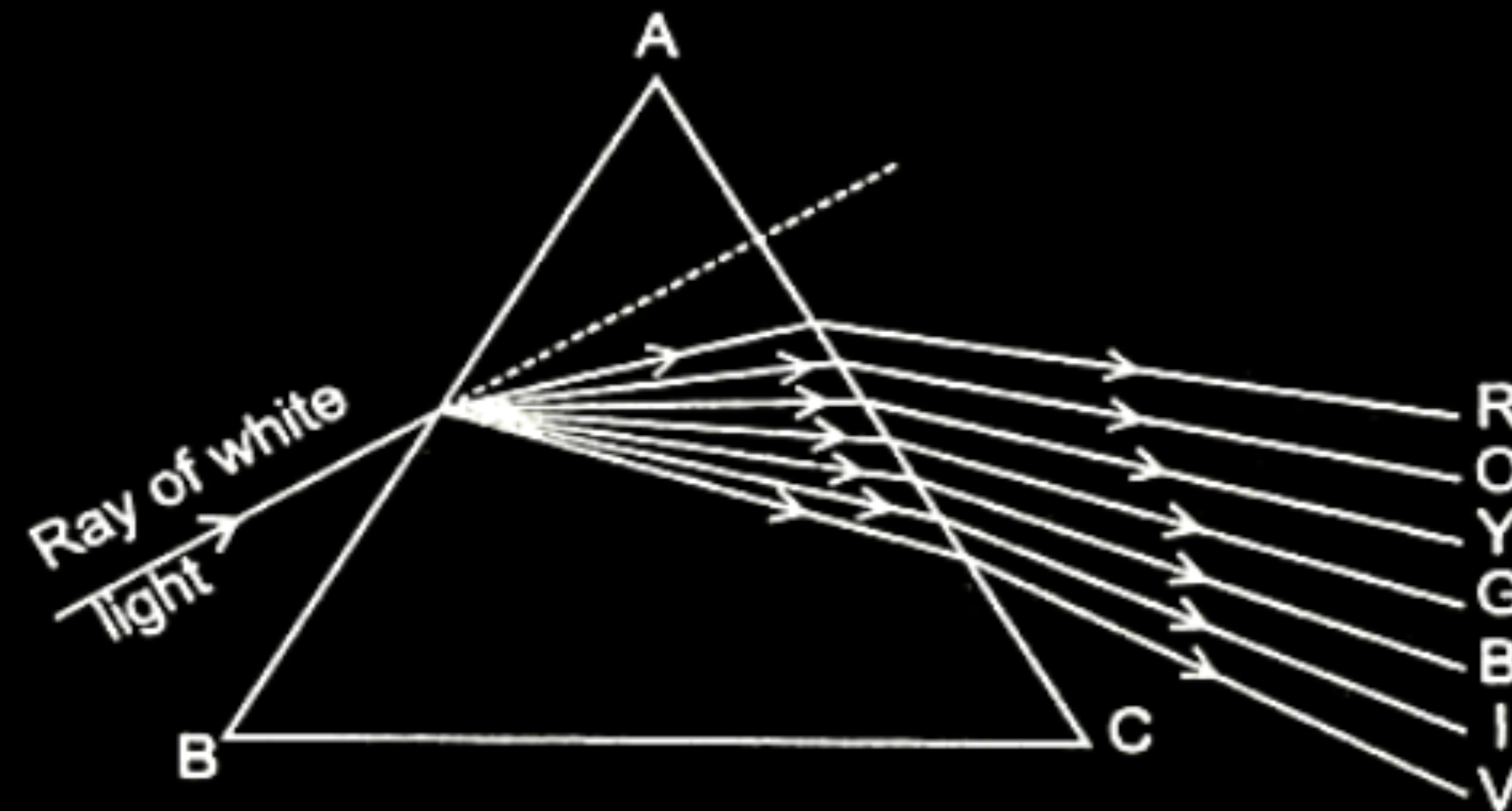
- (b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism.

3

(b)

The splitting of white light into its constituent colours is called dispersion.

Cause: Different colours of white light bend through different angles with respect to incident ray.



1

- 27.** Define the term power of accommodation of human eye. What happens to the image distance in the eye when we increase the distance of an object from the eye ? Name and explain the role of the part of human eye responsible for it in this case.

- 27.** Define the term power of accommodation of human eye. What happens to the image distance in the eye when we increase the distance of an object from the eye ? Name and explain the role of the part of human eye responsible for it in this case.

3

•Ability of the eye lens to adjust its focal length.	1
• Image distance remains unchanged	1
•Ciliary muscles –	$\frac{1}{2}$
While focusing on distant objects ciliary muscles relax, eye lens becomes thin and its focal length increases.	$\frac{1}{2}$

25. What would have been the colour of the sky, if the Earth had no atmosphere ? Give reason to justify your answer.

25. What would have been the colour of the sky, if the Earth had no atmosphere ? Give reason to justify your answer.

2

• Black colour	1
• As there would not be any particle to scatter light.	1

15. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is :

(A) Red

(B) Orange

(C) Blue

(D) Yellow

15. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is :

(A) Red

(B) Orange

(C) Blue

(D) Yellow

(C)/Blue

25. Name the phenomenon of light responsible for Tyndall effect. Write an event where this phenomenon can be observed.

25. Name the phenomenon of light responsible for Tyndall effect. Write an event where this phenomenon can be observed.

2

• Scattering of light.	1
• Example – When sunlight passes through a canopy of dense forest/ when a fine beam of sunlight enters a smoke filled room through a small hole. (or any other)	1

- 25.** A person suffering from presbyopia needs bifocal lens. If he needs two lens of power -4.0 dioptre and $+2.0$ dioptre, which one of these two lenses is for the correction of distant vision and what is its focal length ? 2

- 25.** A person suffering from presbyopia needs bifocal lens. If he needs two lens of power -4.0 dioptre and $+2.0$ dioptre, which one of these two lenses is for the correction of distant vision and what is its focal length ? 2

• For distant vision: lens of power -4.0 dioptre	1
• $f_{(meter)} = \frac{1}{P} = \frac{1}{-4.0D} = -0.25 \text{ m or } -25 \text{ cm}$	1

- (a) Name the type of lens presented by opticians for the correction of hypermetropia. State the role of such lenses in correcting the vision of the person suffering from this defect.

- (a) Name the type of lens presented by opticians for the correction of hypermetropia. State the role of such lenses in correcting the vision of the person suffering from this defect.

2

(a) • Convex lens/ converging lens • It provides the additional focussing power required for forming the image on the retina. / These lenses decrease the focal length of the eye lens, thereby forming the image on the retina.	1 1
---	-------------------

5. Consider the following statements :
- i. Ciliary muscles adjust for changing the intensity of light entering the eye.
 - ii. Myopic eye can be corrected by converging lenses of suitable power.
 - iii. The function of the pupil is to regulate the quantity of light entering the eye.
 - iv. When ciliary muscles are completely relaxed, the focal length of the eye lens is maximum.

The correct statements are :

- (A) i, ii and iv
- (B) i, ii and iii
- (C) iii and iv
- (D) ii and iv

5. Consider the following statements :
- i. Ciliary muscles adjust for changing the intensity of light entering the eye.
 - ii. Myopic eye can be corrected by converging lenses of suitable power.
 - iii. The function of the pupil is to regulate the quantity of light entering the eye.
 - iv. When ciliary muscles are completely relaxed, the focal length of the eye lens is maximum.

The correct statements are :

- (A) i, ii and iv
- (B) i, ii and iii
- (C) iii and iv
- (D) ii and iv

(C) /iii and iv

28. When and where does a rainbow appear in the sky ? List the phenomena of light in proper sequence which occurs during the formation of a rainbow. Why can a rainbow not be seen from the surface of the moon ?

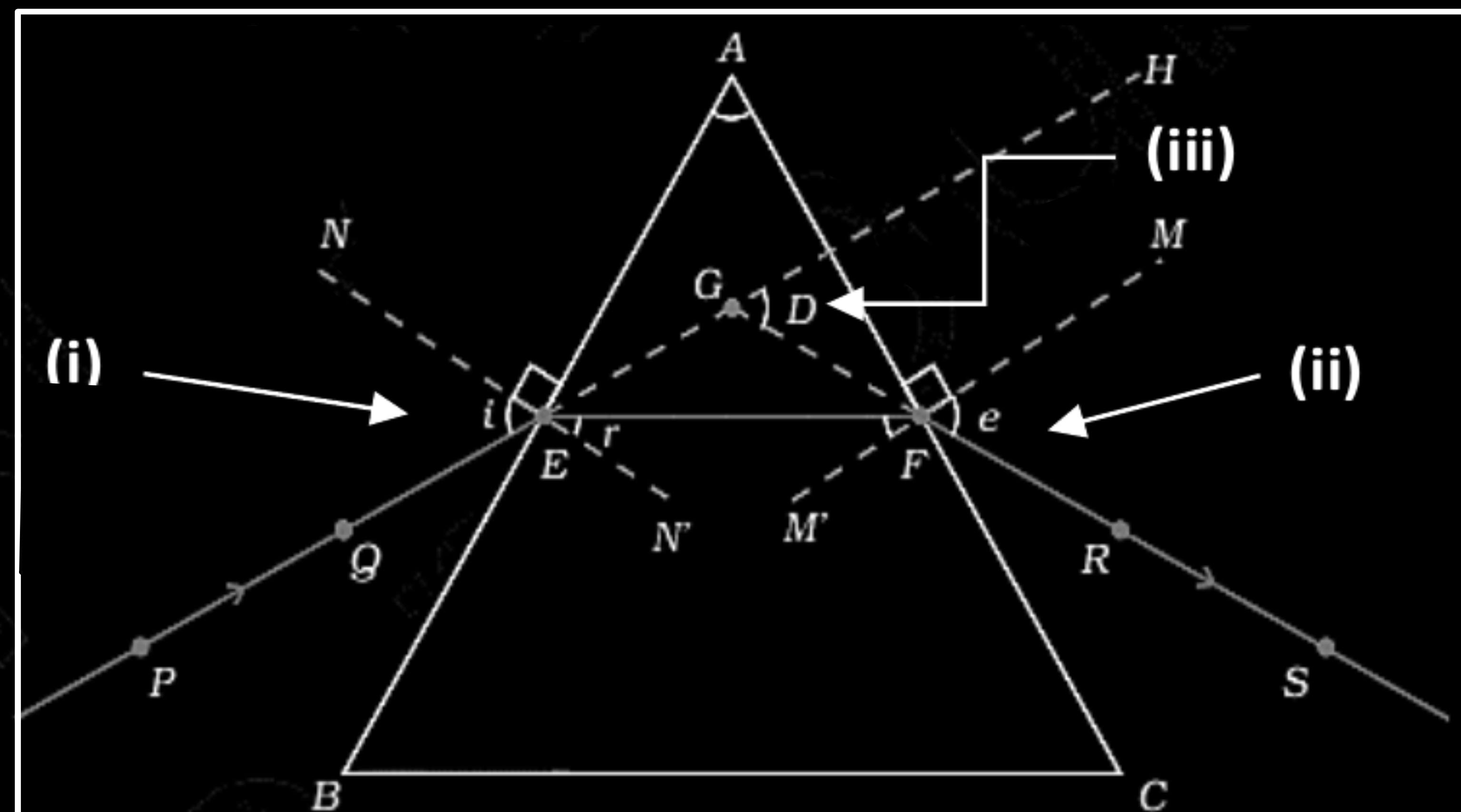
28. When and where does a rainbow appear in the sky ? List the phenomena of light in proper sequence which occurs during the formation of a rainbow. Why can a rainbow not be seen from the surface of the moon ?

3

• After the rain shower during day time	$\frac{1}{2}$
• In the sky opposite to the Sun	$\frac{1}{2}$
• Refraction; dispersion of light; Internal reflection	$\frac{1}{2} \times 3$
• Because there is no atmosphere (water droplets) for the refraction of light to occur.	$\frac{1}{2}$

33. Draw a ray diagram to show the refraction of a ray of light through a triangular glass prism and mark on it (i) angle of incidence (i), (ii) angle of emergence (e), and (iii) angle of deviation (D). What would happen, if instead of a ray, a beam of white light falls on the prism ? Give reason for the phenomenon that occurs in this case.

3



[Fig 10.4, page 166 of NCERT]

Labelled diagram

1 ½

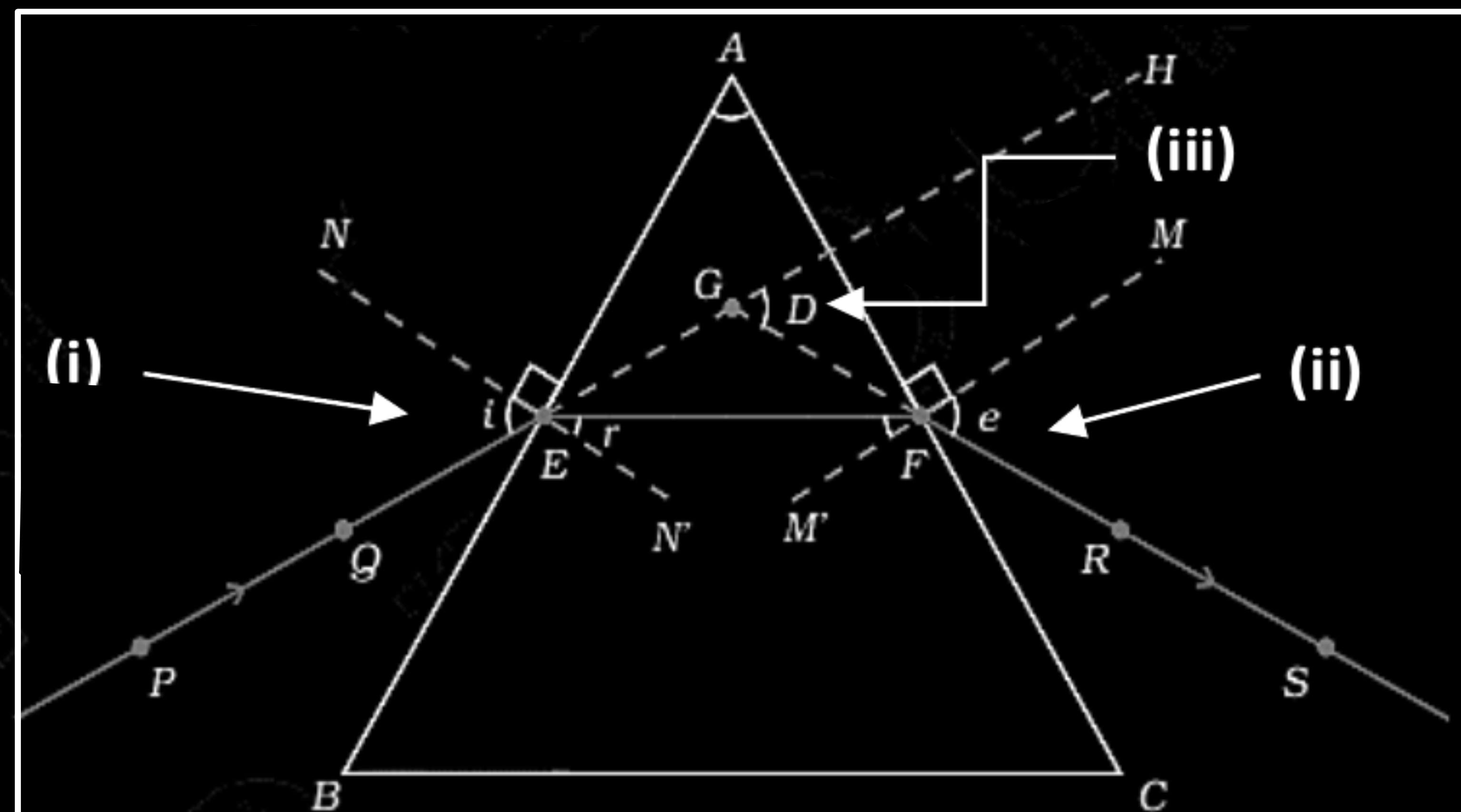
- White light will split into a band of seven colours of spectrum.
- Different colours of white light bend / refract through different angles with respect to incident ray as they pass through a prism.

½

1

33. Draw a ray diagram to show the refraction of a ray of light through a triangular glass prism and mark on it (i) angle of incidence (i), (ii) angle of emergence (e), and (iii) angle of deviation (D). What would happen, if instead of a ray, a beam of white light falls on the prism ? Give reason for the phenomenon that occurs in this case.

3



[Fig 10.4, page 166 of NCERT]

Labelled diagram

1 ½

- White light will split into a band of seven colours of spectrum.
- Different colours of white light bend / refract through different angles with respect to incident ray as they pass through a prism.

½

1

23. (a) Draw a ray diagram to show the formation of a rainbow in the sky. Mark on the diagram three points A, B and C as given below :

2

A – where dispersion of light occurs

B – where internal reflection of dispersed light occurs

C – where refraction of dispersed light occurs

23. (a) Draw a ray diagram to show the formation of a rainbow in the sky. Mark on the diagram three points A, B and C as given below :

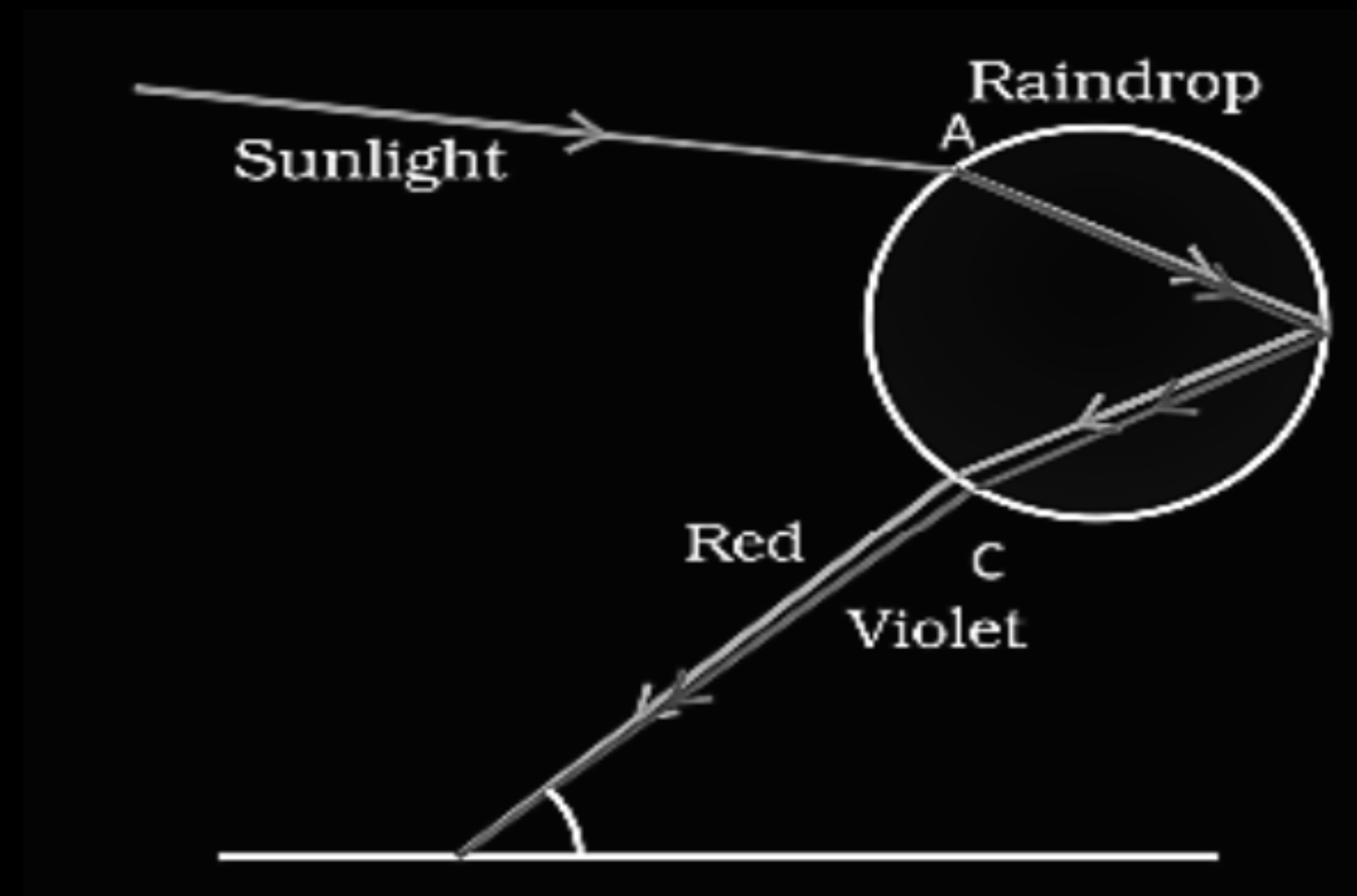
2

A – where dispersion of light occurs

B – where internal reflection of dispersed light occurs

C – where refraction of dispersed light occurs

(a)



[Fig 10.8 page 16]

Diagram

$\frac{1}{2}$

Correct labelling of A,B and C

$1\frac{1}{2}$

- (b) Name the muscles present in the human eye which enable it to focus on objects at varied distances (i.e., distant as well as nearby objects). Explain how it happens.

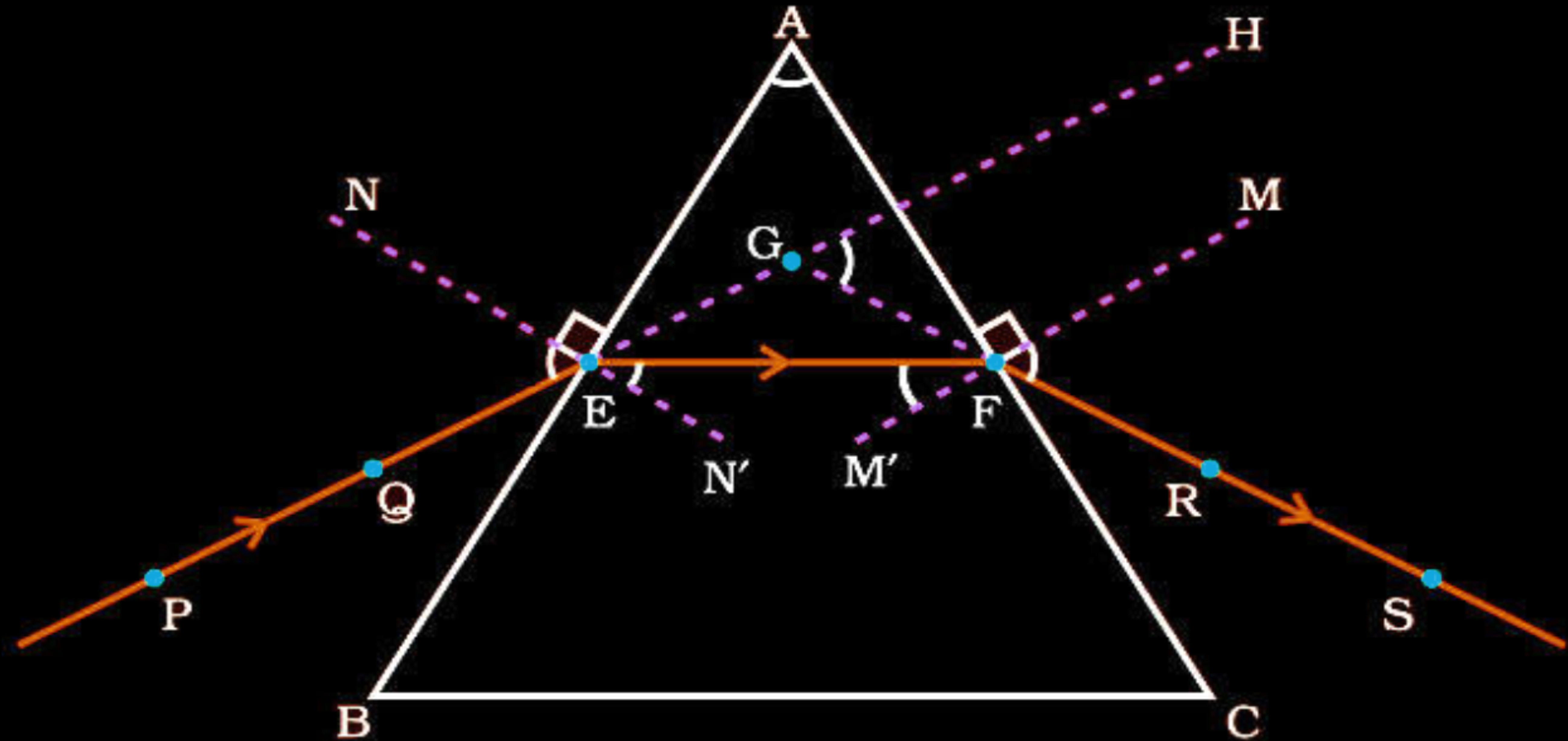
- (b) Name the muscles present in the human eye which enable it to focus on objects at varied distances (i.e., distant as well as nearby objects). Explain how it happens.

2

(b) • Ciliary muscles • Ciliary muscles adjust/change the focal length of eye lens by changing the curvature of eye lens.	1 1
---	------------

33. (a) Study the following ray diagram and name the angles given below :

$\angle PEN$, $\angle FEN'$, $\angle SFM$ and $\angle FGH$



- (b) In this diagram, if $\frac{\sin \angle PEN}{\sin \angle FEN'} = \frac{3}{2}$, what is the value of $\frac{\sin \angle SFM}{\sin \angle EFM}$?

Give reason to justify your answer.

(a)

$\angle PEN = \angle i$ / angle of incidence

$\angle FEN' = \angle r$ / angle of refraction

$\angle SFM = \angle e$ / angle of emergence

$\angle FGH = \angle D$ / angle of deviation

1/2

1/2

1/2

1/2

(b)

At face, AC of the prism

$$\frac{\sin i}{\sin r} = \frac{\sin \angle EFM'}{\sin \angle SFM} = \frac{2}{3}$$

$\therefore$

$$\frac{\sin \angle SFM}{\sin \angle EFM'} = \frac{3}{2}$$

Reason: Path of light is reversed at the face AC of the prism.

1

Note:- Give one mark for part (b) to all students as in question

given, $\frac{\sin \angle SFM}{\sin \angle EFM}$ is wrong. It should be $\frac{\sin \angle SFM}{\sin \angle EFM'}$

14. A person is advised to use lenses of power -0.25 D for the correction of his defect of vision. It implies that the person is suffering from :
- (A) hypermetropia and he has to use converging lens of focal length 25 cm in his spectacles.
 - (B) hypermetropia and he has to use diverging lens of focal length 4 m in his spectacles.
 - (C) myopia and he has to use diverging lens of focal length 4 m in his spectacles.
 - (D) myopia and he has to use diverging lens of focal length 25 cm in his spectacles.

14. A person is advised to use lenses of power -0.25 D for the correction of his defect of vision. It implies that the person is suffering from :

- (A) hypermetropia and he has to use converging lens of focal length 25 cm in his spectacles.
- (B) hypermetropia and he has to use diverging lens of focal length 4 m in his spectacles.
- (C) myopia and he has to use diverging lens of focal length 4 m in his spectacles.
- (D) myopia and he has to use diverging lens of focal length 25 cm in his spectacles.

(C) myopia and he has to use diverging lens of focal length 4 m in his spectacles.
--

- 32.** (a) Define the term dispersion of white light. State the colour of light which bends (i) the least, (ii) the most, while passing through a glass prism. Also, state the cause of dispersion of a narrow beam of white light when it passes through a prism.

32. (a) Define the term dispersion of white light. State the colour of light which bends (i) the least, (ii) the most, while passing through a glass prism. Also, state the cause of dispersion of a narrow beam of white light when it passes through a prism.

3

- | | |
|---|---------------|
| • Splitting of white light into its component colours when it passes through a prism/rain drop/oil bubble. | 1 |
| • Red colour bends the least | $\frac{1}{2}$ |
| • Violet colour bends the most. | $\frac{1}{2}$ |
| • Different colours of white light bend/refract through different angles with respect to the incident ray as they pass through a prism. | 1 |

- (b) When and where does a rainbow appear in the sky ? Write sequence-wise the phenomena of light responsible for the formation of a rainbow. List two essential conditions for observing a rainbow.

(b) When and where does a rainbow appear in the sky ? Write sequence-wise the phenomena of light responsible for the formation of a rainbow. List two essential conditions for observing a rainbow.

3

(b) • ➤ After the rain shower. ➤ In a direction opposite to that of the sun. • The tiny water droplets ➤ refract and disperse sunlight then ➤ reflect it internally and finally ➤ refract it again when it comes out of the rain drop. • Two necessary condition – (i) Presence of tiny rain droplets in air. (ii) Sun behind the observer.	$\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2} + \frac{1}{2}$
---	--

- 25.** (a) The far point of a defective eye is nearer than infinity.
- (i) Identify the defect of vision.
 - (ii) List two possible causes responsible for this defect.
 - (iii) Name the type of lens (converging/diverging) used for the correction of this defect.

25. (a) The far point of a defective eye is nearer than infinity.
- (i) Identify the defect of vision.
- (ii) List two possible causes responsible for this defect.
- (iii) Name the type of lens (converging/diverging) used for the correction of this defect.

2

(a)	
(i)	Myopia
(ii)	
•	Excessive curvature of eye lens / Converging power of eye lens increases
•	Elongation of the eye ball.
(ii)	Diverging lens

- (b) “The colour of the clear sky from the earth appears blue but the sky appears dark to the passengers flying at very high altitudes.”
Explain.

- (b) “The colour of the clear sky from the earth appears blue but the sky appears dark to the passengers flying at very high altitudes.”
Explain.

2

(b) • Very fine particles of the atmosphere scatter short wavelength colour, mainly blue light, but scattering is not prominent at very high altitudes. So, sky appears black.	1+1
--	-----

37. The students in a class took a thick sheet of cardboard and made a small hole in its centre. Sunlight was allowed to fall on this small hole and they obtained a narrow beam of white light. A glass prism was taken and this white light was allowed to fall on one of its faces. The prism was turned slowly until the light that comes out of the opposite face of the prism appeared on the nearby screen. They studied this beautiful band of light and concluded that it is a spectrum of white light.

- (i) Give any one more instance in which this type of spectrum is observed. **1**
- (ii) What happens to white light in the above case? **1**
- (iii) (A) List two conditions necessary to observe a rainbow. **2**

OR

- (iii) (B) Draw a ray diagram to show the formation of a rainbow. Mark on it, points (a), (b) and (c) as given below : **2**
 - (a) Where dispersion of light occurs.
 - (b) Where light gets reflected internally.
 - (c) Where final refraction occurs.

(i) A rainbow (or any other)

(ii) Dispersion of white light takes place.

(iii) (A)

- The presence of water droplets in the atmosphere.
- The sun must be at the back of the observer.

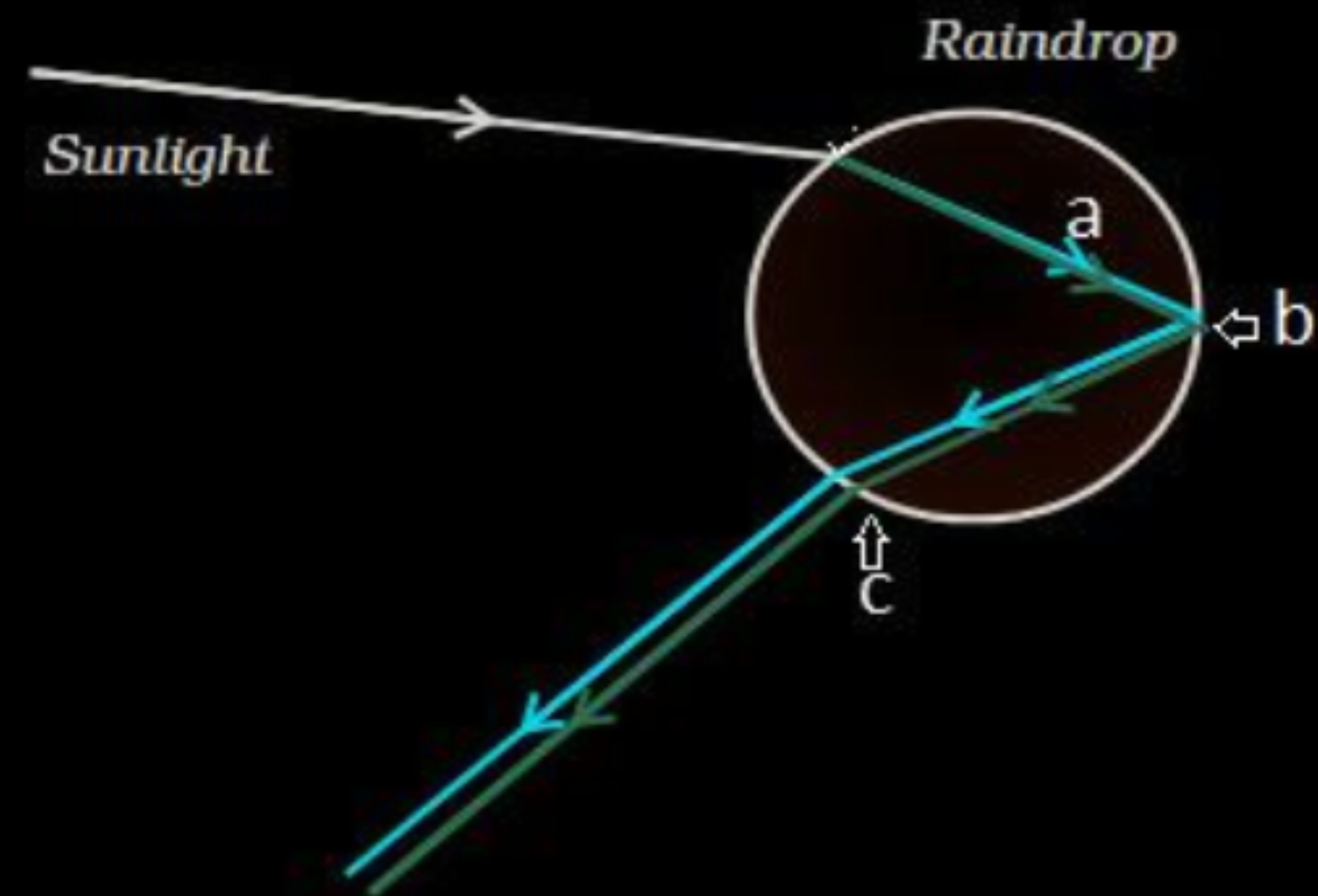
1

1

1+1

OR

(iii) (B)



$\frac{1}{2} \times 4$

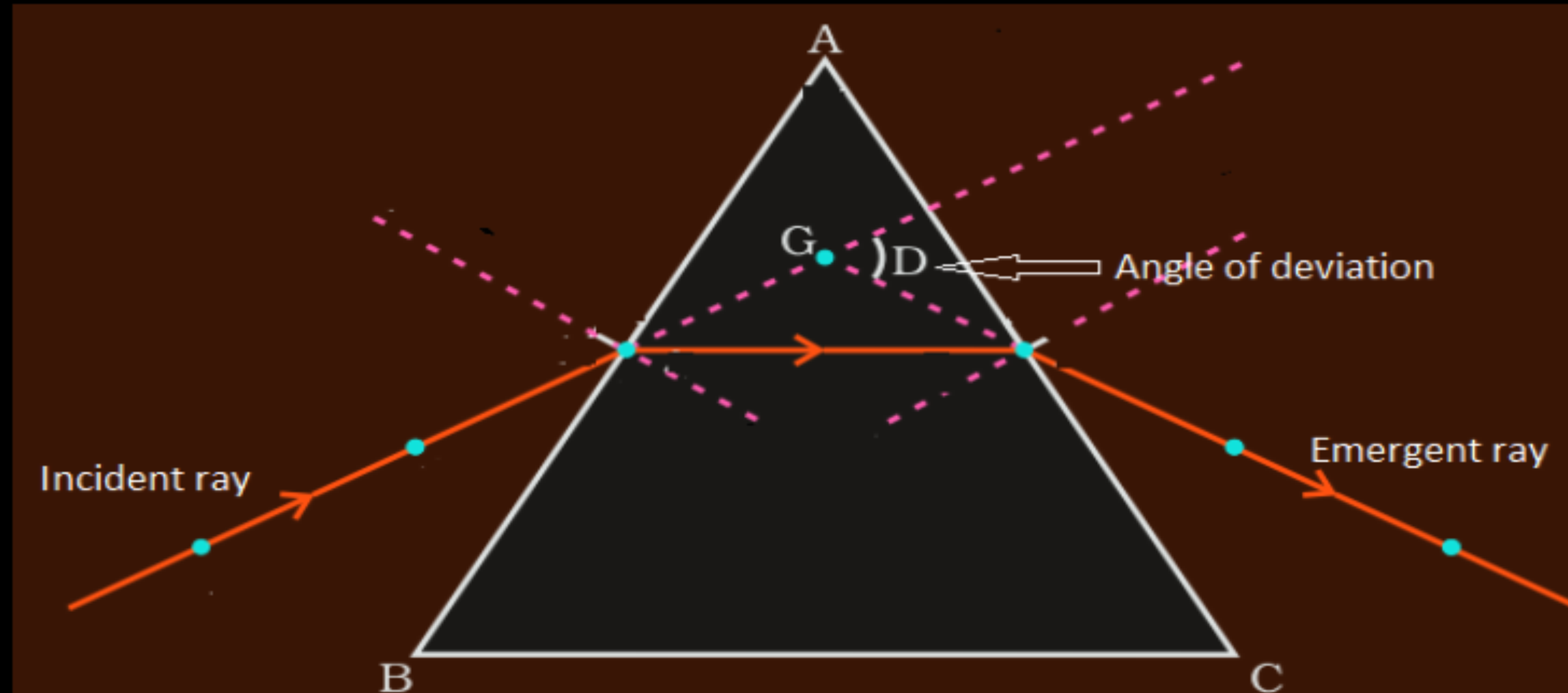
($\frac{1}{2}$ mark for diagram and $\frac{1}{2}$ for labelling a, b, c)

- 25. (A)** Draw a ray diagram to show the refraction of a ray of light passing through an equilateral glass prism. Mark the angle through which the emergent ray bends from the direction of the incident ray and also name it.

25. (A) Draw a ray diagram to show the refraction of a ray of light passing through an equilateral glass prism. Mark the angle through which the emergent ray bends from the direction of the incident ray and also name it.

2

(A)



(one mark for diagram and $\frac{1}{2}$ for labelling.)

- Angle of deviation

$1\frac{1}{2}$

$\frac{1}{2}$

(B) Name the type of lenses required by the persons for the correction of their defect of vision called presbyopia. Write the structure of the lenses commonly used for the correction of this defect giving reason for such designs.

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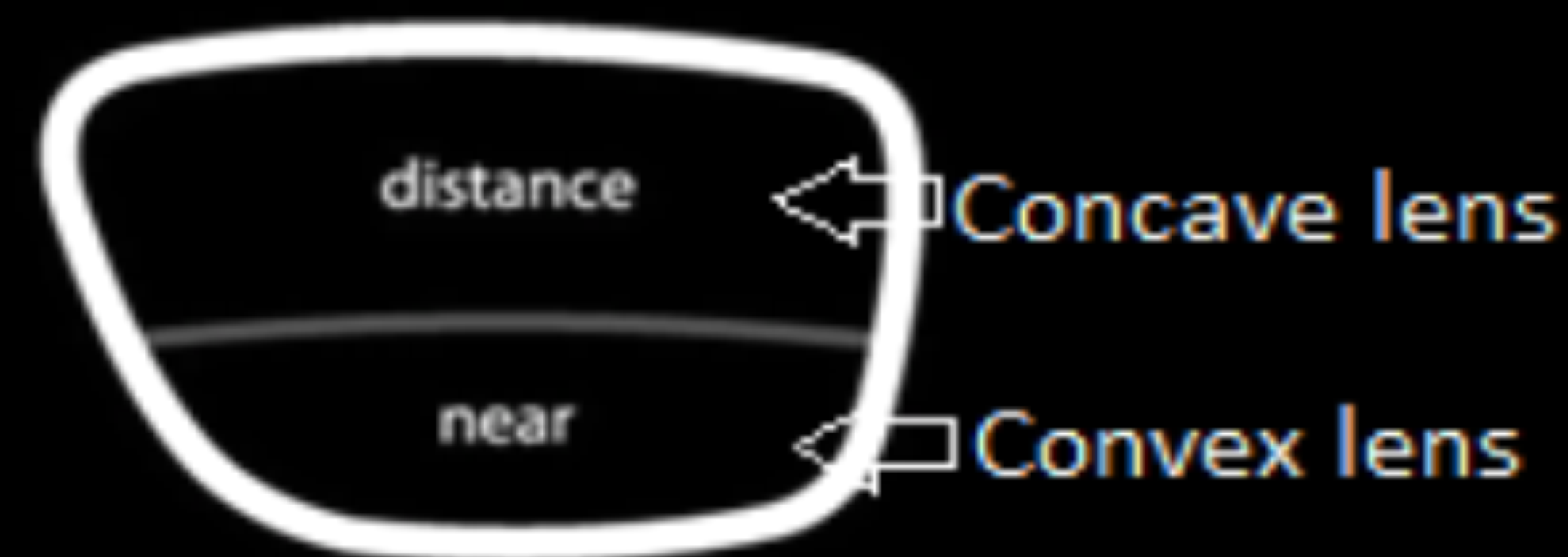
2

(B)

I.

Bi-focal lens.

Bi-focal lens having upper portion consists of a concave lens and lower portion consists convex lens. /



to facilitate the distant and near vision respectively.

$\frac{1}{2}$

1

$\frac{1}{2}$

20. Assertion (A) : White light is dispersed by a glass prism into seven colours.

Reason (R) : The red light bends the least while the violet the most when a beam of white light passes through a glass prism.

20. Assertion (A) : White light is dispersed by a glass prism into seven colours.

Reason (R) : The red light bends the least while the violet the most when a beam of white light passes through a glass prism.

1

(b) / Both Assertion and Reason are true and Reason (R) is not the correct explanation of Assertion (A).

18. Assertion (A) : When ciliary muscles contract, eye lens becomes thin.

Reason (R) : Ciliary muscles control the power of the eye lens.

1

18. Assertion (A) : When ciliary muscles contract, eye lens becomes thin.

Reason (R) : Ciliary muscles control the power of the eye lens.

1

(d) / Assertion (A) is false but Reason (R) is true.

32. A person uses lenses of power -0.5 D in his spectacles for the correction of his vision.

- (a) Name the defect of vision the person is suffering from.
- (b) List two causes of this defect.
- (c) Determine the focal length of the lenses used in the spectacles.

32. A person uses lenses of power -0.5 D in his spectacles for the correction of his vision.

- (a) Name the defect of vision the person is suffering from.
- (b) List two causes of this defect.
- (c) Determine the focal length of the lenses used in the spectacles.

3

(a) Myopia / Near Sightedness/ Short sightedness	1
(b) Two causes: • Excessive curvature of eye lens • Eye ball is elongated	$\frac{1}{2}$ $\frac{1}{2}$
(c) $f(\text{m}) = 1/P = 1/-0.5 = -2\text{ m}$	1

23. (a) Give reasons :

(i) The sky appears dark to passengers flying at very high altitude.

(ii) 'Danger' signal lights are red in colour.

2

OR

(b) What is a rainbow ? "We see a rainbow in the sky only after the rainfall." Why ?

2

23. (a) Give reasons :

- (i) The sky appears dark to passengers flying at very high altitude.
- (ii) 'Danger' signal lights are red in colour.

2

OR

(b) What is a rainbow ? "We see a rainbow in the sky only after the rainfall." Why ?

2

(a) (i) Scattering of light is not prominent at such heights.

1

(ii) The red colour is least scattered by smoke or fog. / Red colour has longer wavelength.

1

OR

(b)

- Rainbow is a natural spectrum appearing in the sky after a rain shower.

1

After rain, small water droplets act as a tiny prism, when light enters, it gets refracted and dispersed.

1

13. The curvature of eye lens of human eye

1

- (A) is fixed.
- (B) can be increased.
- (C) can be decreased.
- (D) increases or decreases as the case may be.

13. The curvature of eye lens of human eye

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- (A) is fixed.
- (B) can be increased.
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(D) / increases or decreases as the case may be

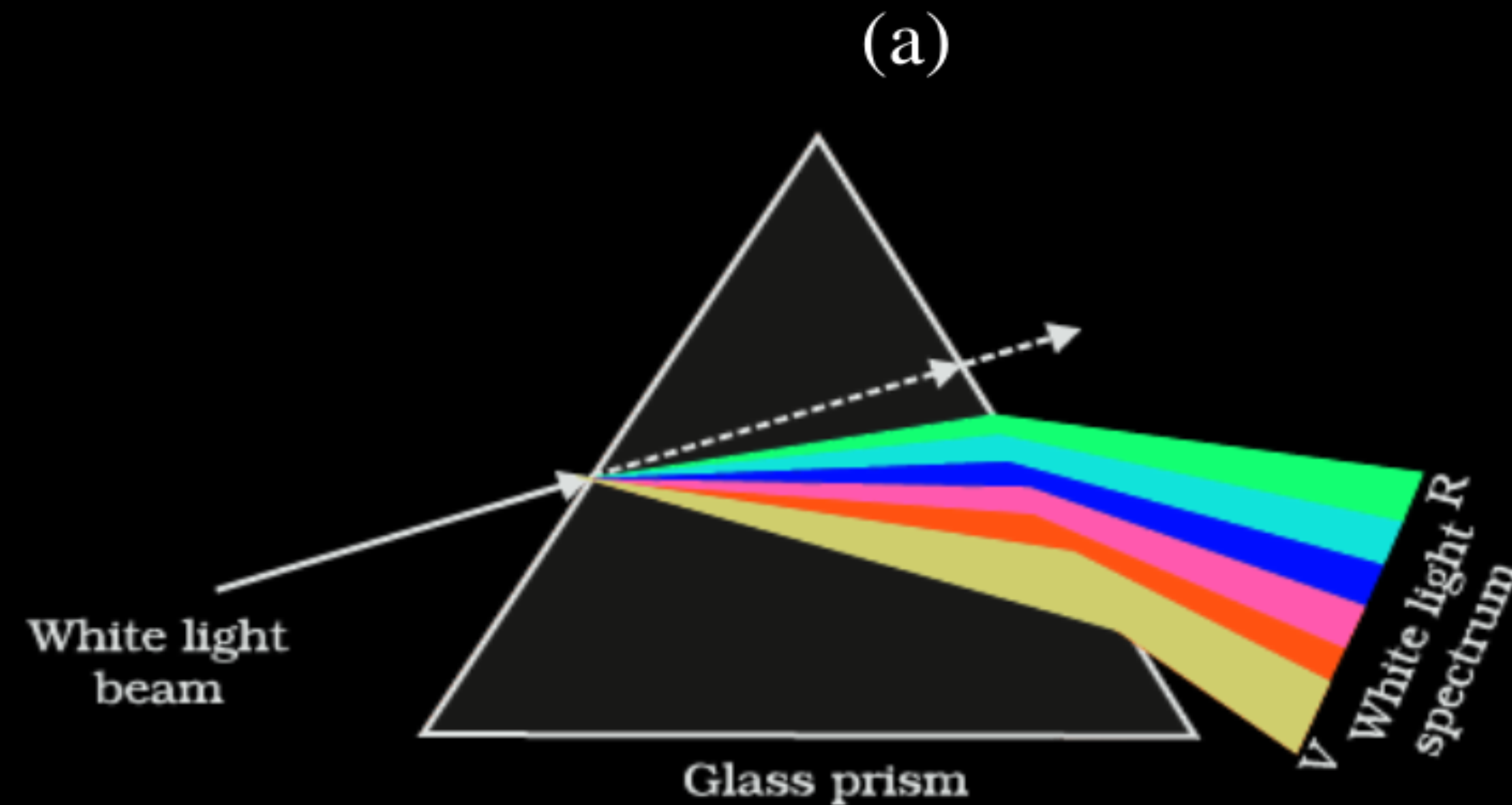
33. (a) Draw a diagram to show the dispersion of white light by a glass prism.

(b) What is spectrum ? Give reason for its formation.

33. (a) Draw a diagram to show the dispersion of white light by a glass prism.

(b) What is spectrum ? Give reason for its formation.

3



1

(b) The band of seven colours obtained on the screen after dispersion of white light

1

Reason: Different components of white light (colours) bend through different angles with respect to the incident ray as they pass through a prism

1

14. Most of the refraction for the light rays entering the eye occurs at

1

(A) Iris

(B) Pupil

(C) Crystalline lens

(D) Outer surface of Cornea

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(D) / Outer surface of cornea

32. A person uses lenses of +2.0 D power in his spectacles for the correction of his vision.

- (a) Name the defect of vision the person is suffering from.
- (b) List two causes of this defect.
- (c) Determine the focal length of the lenses used in the spectacles.

3

32. A person uses lenses of +2.0 D power in his spectacles for the correction of his vision.

- (a) Name the defect of vision the person is suffering from.
- (b) List two causes of this defect.
- (c) Determine the focal length of the lenses used in the spectacles.

3

(a)	Hypermetropia	1
(b)	Two causes:	
	• Eye ball has become too small /Eye ball is shortened	1/2 x2
	• The focal length of the eye Lens is long / eye lens becomes less convergent	
(c)	Focal length = $\frac{1}{P}$	1/2
	$= \frac{1}{2} = + 0.5 \text{ m}$	1/2

14. The part of human eye which controls the amount of light entering into it. **1**
- (A) Iris (B) Cornea
- (C) Ciliary muscles (D) Pupil

14. The part of human eye which controls the amount of light entering into it. **1**
- (A) Iris (B) Cornea
(C) Ciliary muscles (D) Pupil

(D) Pupil

39. The human eye is one of the most valuable and sensitive sense organs. Eyes enable us to see the wonderful world around us. We observe that some persons cannot see the objects distinctly and comfortably. The vision becomes blurred due to the refractive defects of the eye.

4

- (a) When do we say that a person is suffering from myopia ?
- (b) List two main causes of myopia.
- (c) (i) When do we say that a person is suffering from presbyopia. Write the structure of common type of bifocal lenses used by the persons suffering from presbyopia.

OR

- (ii) A person is using corrective lenses of power $+0.25$ D in his spectacles for seeing the objects clearly. Name the defect of vision he is suffering from. Also calculate the focal length of the lenses used in his spectacles.

(a) When a person cannot see distant objects distinctly but can see nearby objects clearly.	1
(b) Two main reasons of myopia:	
• Excessive curvature of the eye lens	$\frac{1}{2}$
• Elongation of the eye ball	$\frac{1}{2}$
(c) (i) • The power of accommodation of the eye usually decreases with ageing, the near point gradually recedes away and find it difficult to see nearby objects comfortably and distinctly without corrective eye glasses.	1
Structure of bifocal lens :	
• Upper portion consists of concave lens.	$\frac{1}{2}$
• Lower portion consists of convex lens	$\frac{1}{2}$
OR	
(ii) Power $P = + 0.25\text{D}$	
• Hypermetropia	1
• Focal length = $\frac{1}{\text{Power}}$	$\frac{1}{2}$
$\therefore$ Focal length = $\frac{1}{+0.25\text{D}} = + 4.0 \text{ m}$	$\frac{1}{2}$

19. **Assertion (A)** : The colour of the scattered light depends on the size of the scattering particles.

Reason (R) : Very fine particles scatter mainly red light while particles of larger size scatter light of shorter wavelengths.

19. **Assertion (A)** : The colour of the scattered light depends on the size of the scattering particles.

Reason (R) : Very fine particles scatter mainly red light while particles of larger size scatter light of shorter wavelengths.

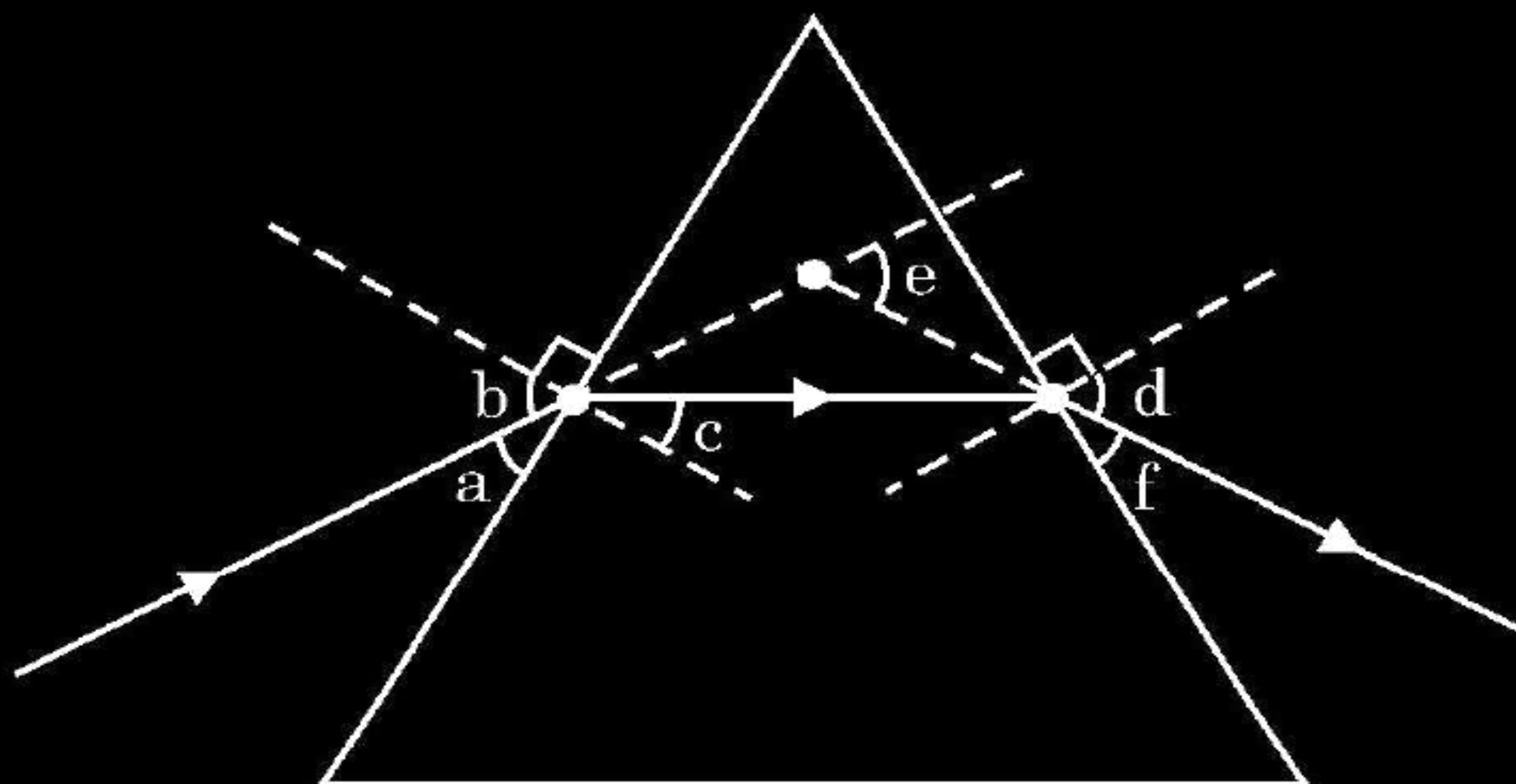
(C)/ Assertion (A) is true, but Reason (R) is false.

14. When we focus our eyes on a distance object, the ciliary muscles of our eyes are relaxed. In this case the eye lens becomes
- (A) thin and its curvature is maximum.
 - (B) thin and its curvature is minimum.
 - (C) thick and its curvature is minimum.
 - (D) thick and its curvature is maximum.

14. When we focus our eyes on a distance object, the ciliary muscles of our eyes are relaxed. In this case the eye lens becomes
- (A) thin and its curvature is maximum.
 - (B) thin and its curvature is minimum.
 - (C) thick and its curvature is minimum.
 - (D) thick and its curvature is maximum.

(B)/ thin and its curvature is minimum

15. Consider the following ray diagram :



Here, the angle of incidence and angle of deviation respectively are

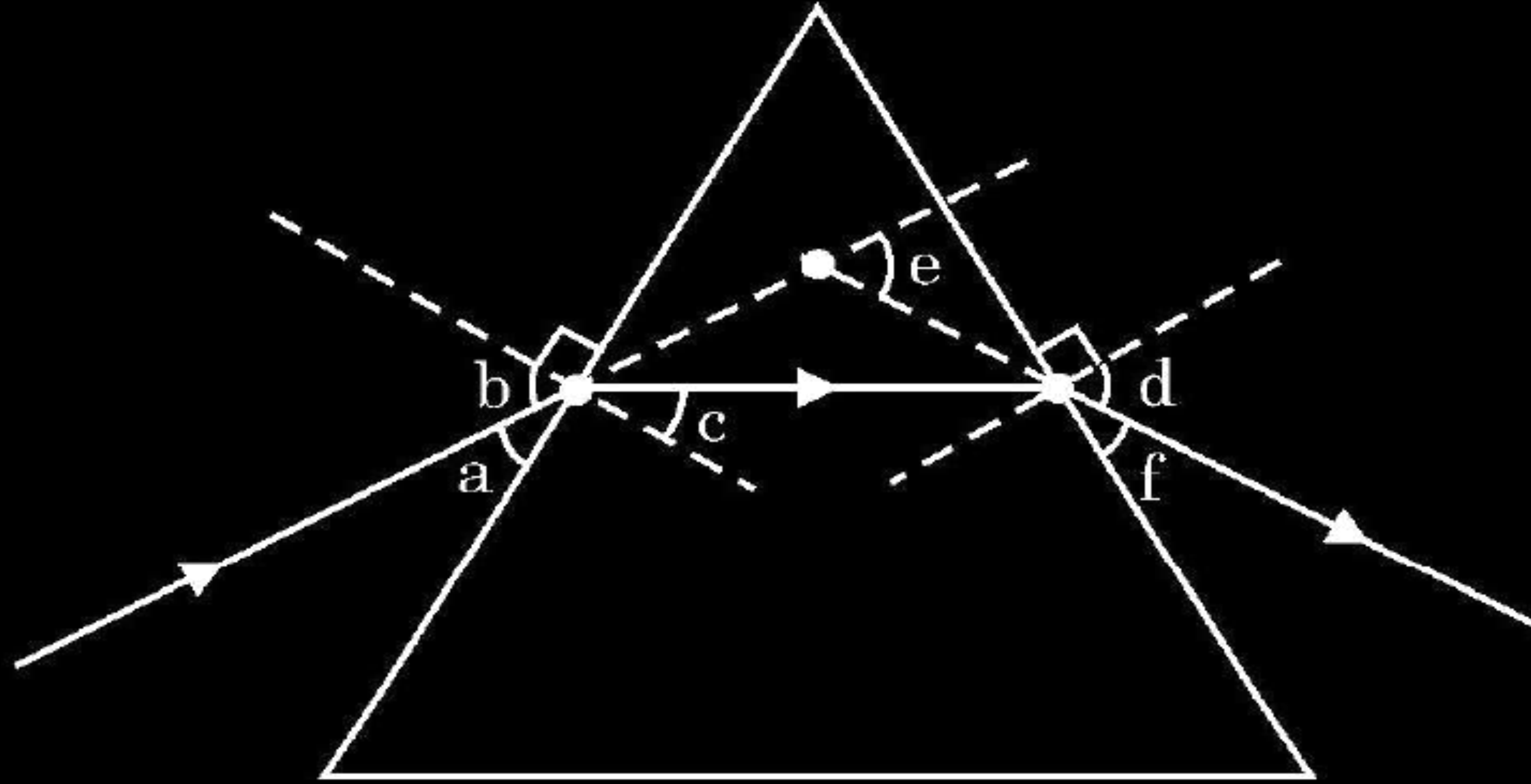
(A) a and e

(B) b and d

(C) b and e

(D) a and f

15. Consider the following ray diagram :



Here, the angle of incidence and angle of deviation respectively are

(A) a and e

(B) b and d

(C) b and e

(D) a and f

(C) / b and e

14. A boy while reading a book, keeps it much beyond 25 cm from his eyes. This defect of vision has arisen because of
- (A) excessive curvature of the eye lens
 - (B) the focal length of the eye lens has increased
 - (C) the eye ball has elongated
 - (D) the focal length of the eye lens has become too small

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- (A) excessive curvature of the eye lens
 - (B) the focal length of the eye lens has increased
 - (C) the eye ball has elongated
 - (D) the focal length of the eye lens has become too small

(B) / The focal length of the eye lens has increased

14. The possible way to restore clear vision of those people whose eyeball has elongated is the use of suitable

(A) bifocal lens

(B) concave lens

(C) converging lens

(D) convex lens

14. The possible way to restore clear vision of those people whose eyeball has elongated is the use of suitable

(A) bifocal lens

(B) concave lens

(C) converging lens

(D) convex lens

(B) / Concave lens

- 31.** A person is suffering from an eye defect in which the far point of the eye is much nearer than infinity. Name the defect of vision the person is suffering from. List two main causes of this defect. Write the type of the corrective lens and draw a ray diagram to show the function of the corrective lens.

- 31.** A person is suffering from an eye defect in which the far point of the eye is much nearer than infinity. Name the defect of vision the person is suffering from. List two main causes of this defect. Write the type of the corrective lens and draw a ray diagram to show the function of the corrective lens.

3

• Myopia /Shortsightedness	1/2
Causes :	
(i) Excessive curvature of the eye lens	1/2
(ii) Elongation of the eye ball	1/2
• Diverging lens/concave lens	1/2
The diagram illustrates the correction of myopia. On the left, an object 'O' is shown. Two parallel rays of light from 'O' travel horizontally to the right. They first pass through a concave lens (diverging lens), which causes the rays to diverge. These diverging rays then enter the eye, represented by a circle on the right. Inside the eye, the rays pass through the eye's lens and converge exactly on the retina at the back of the eye, forming a sharp image. This shows how the concave lens compensates for the eye's inability to focus light from distant objects properly.	1

- 25.** (a) A student has difficulty in reading his textbooks but can read the blackboard clearly while sitting in the last row. Name the defect of vision the student is suffering from. List two reasons due to which this defect arises. Write the nature of the lenses required to correct this defect.

- 25.** (a) A student has difficulty in reading his textbooks but can read the blackboard clearly while sitting in the last row. Name the defect of vision the student is suffering from. List two reasons due to which this defect arises. Write the nature of the lenses required to correct this defect.

2

(a) Hypermetropia /Farsightedness/Longsightedness.	$\frac{1}{2}$
Reasons:	
(i) Focal length of the eye lens is too long	$\frac{1}{2}$
(ii) Eyeball becomes too small.	$\frac{1}{2}$
Correction	
Convex lens /Converging lens	$\frac{1}{2}$

- (b) Draw a ray diagram to show the path of a ray of light falling obliquely on one of the refracting faces of a triangular glass prism and mark the angle of deviation on it.

- (b) Draw a ray diagram to show the path of a ray of light falling obliquely on one of the refracting faces of a triangular glass prism and mark the angle of deviation on it.

2

(b)

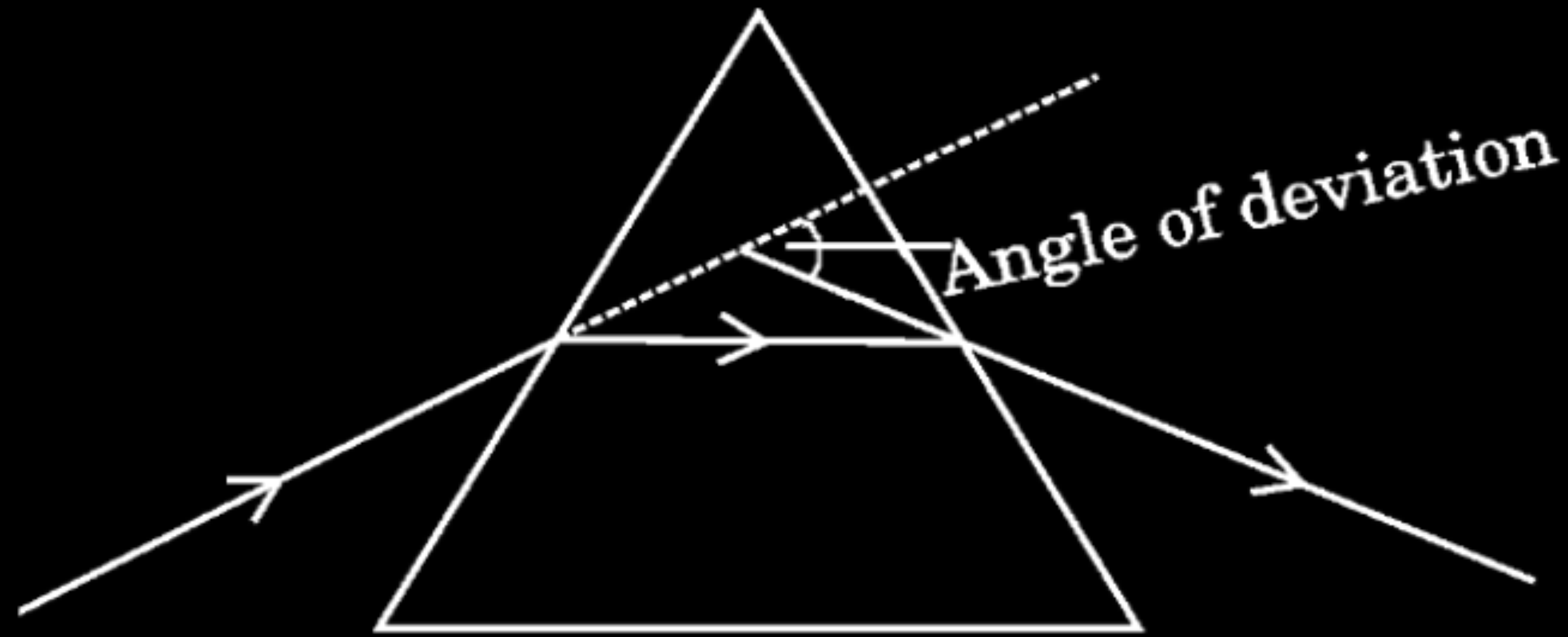


DIAGRAM
DIRECTION OF RAYS
MARKING OF ANGLE

1
 $\frac{1}{2}$
 $\frac{1}{2}$

12. Select the correct statement from the following :

- (A) The size of the molecules of air is larger than the wavelength of visible light.
- (B) The blue light has a wavelength about 1.8 times greater than that of red light.
- (C) When sunlight passes through the fine particles in air, they scatter the blue colour of visible light more strongly than red.
- (D) The light of red colour is scattered the most by fog or smoke.

12. Select the correct statement from the following :

- (A) The size of the molecules of air is larger than the wavelength of visible light.
- (B) The blue light has a wavelength about 1.8 times greater than that of red light.
- (C) When sunlight passes through the fine particles in air, they scatter the blue colour of visible light more strongly than red.
- (D) The light of red colour is scattered the most by fog or smoke.

(C) / When sunlight passes through the fine particles in air, they scatter the blue colour of visible light more strongly than red.

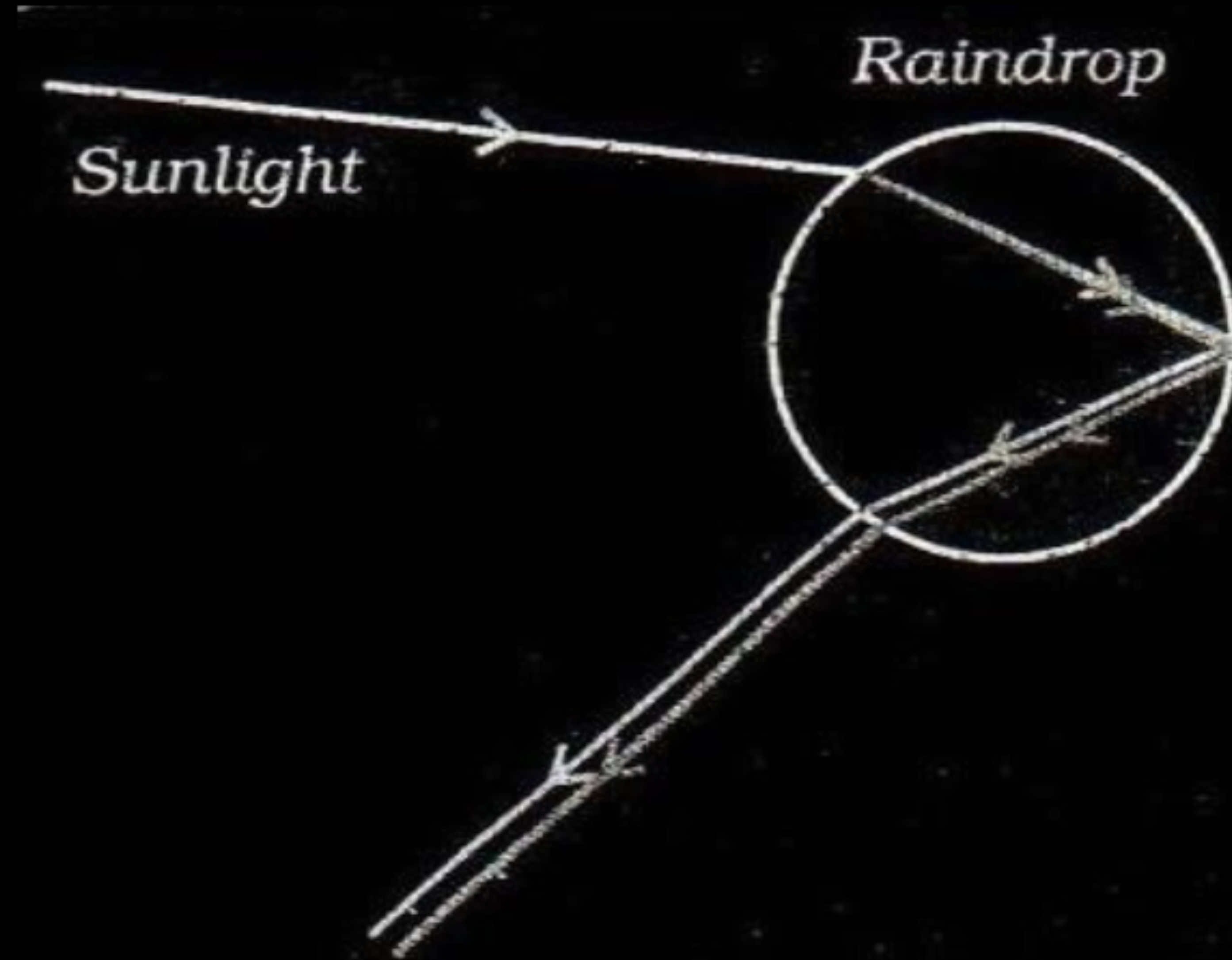
31. What is a rainbow ? Draw a labelled diagram to show its formation.

3

31. What is a rainbow ? Draw a labelled diagram to show its formation.

3

A rainbow is a natural spectrum appearing in the sky after a rain shower.



(Deduct half mark if arrows are not marked)

1

2

12. When a beam of white light passes through a region of very fine dust particles, the colour of light that scatters the most in that region is :

(A) red

(B) orange

(C) blue

(D) yellow

12. When a beam of white light passes through a region of very fine dust particles, the colour of light that scatters the most in that region is :

(A) red

(B) orange

(C) blue

(D) yellow

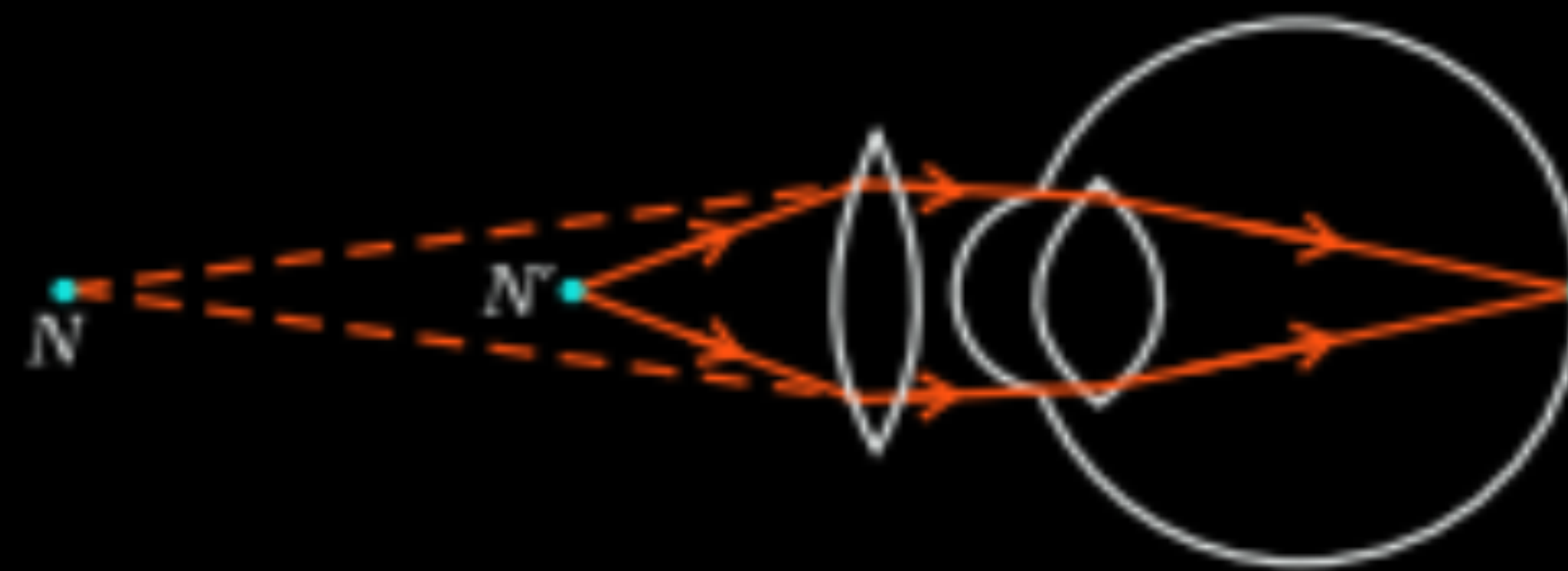
(C) /Blue

- 31.** A person has to keep reading material much beyond 25 cm (say at 50 cm) from the eye for comfortable reading. Name the defect of vision he is suffering from. List two causes responsible for arising of this defect. Draw a labelled diagram showing correction of this defect using eye-glasses. Are these glasses convergent or divergent of light ?

- 31.** A person has to keep reading material much beyond 25 cm (say at 50 cm) from the eye for comfortable reading. Name the defect of vision he is suffering from. List two causes responsible for arising of this defect. Draw a labelled diagram showing correction of this defect using eye-glasses. Are these glasses convergent or divergent of light ?

3

- Hypermetropia/ far-sightedness
- (i) Focal length of the eye lens is too long
- (ii) Eye ball has become too small/shortened



- Convergent of light

$\frac{1}{2}$

$\frac{1}{2} \times 2$

1

$\frac{1}{2}$

- 34.** (a) (i) The power of a lens 'X' is -2.5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens ?

- 34.** (a) (i) The power of a lens 'X' is -2.5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens ?

(a) (i)	
• Concave lens	$\frac{1}{2}$
• $P = \frac{1}{f(m)}$	$\frac{1}{2}$
$-2.5 = \frac{1}{f}$	
$f = \frac{10}{-2.5} = -0.4 \text{ m} = -40 \text{ cm}$	$\frac{1}{2}$
• Myopia	$\frac{1}{2}$

15. The phenomenon responsible for making the smoke particles visible when a beam of sunlight enters a smoke filled room through a narrow hole is :

- (A) scattering of light
- (B) dispersion of light
- (C) reflection of light
- (D) internal reflection of light

15. The phenomenon responsible for making the smoke particles visible when a beam of sunlight enters a smoke filled room through a narrow hole is :

- (A) scattering of light
- (B) dispersion of light
- (C) reflection of light
- (D) internal reflection of light

(A) scattering of light

13. The white light entering a glass prism, gets split into its constituent colours. It is observed that :

- (A) Red light deviates the most.
- (B) Violet light deviates the least.
- (C) Yellow light deviates more than the blue light.
- (D) Green light deviates more than the orange light.

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- (A) Red light deviates the most.
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- (D) Green light deviates more than the orange light.

D/Green light deviates more than the orange light

14. An old person is suffering from an eye defect caused by weakening of ciliary muscles and diminishing flexibility of the eye lens. If the defect of vision is 'a' which can be corrected by lens 'b', then 'a' and 'b' respectively are :

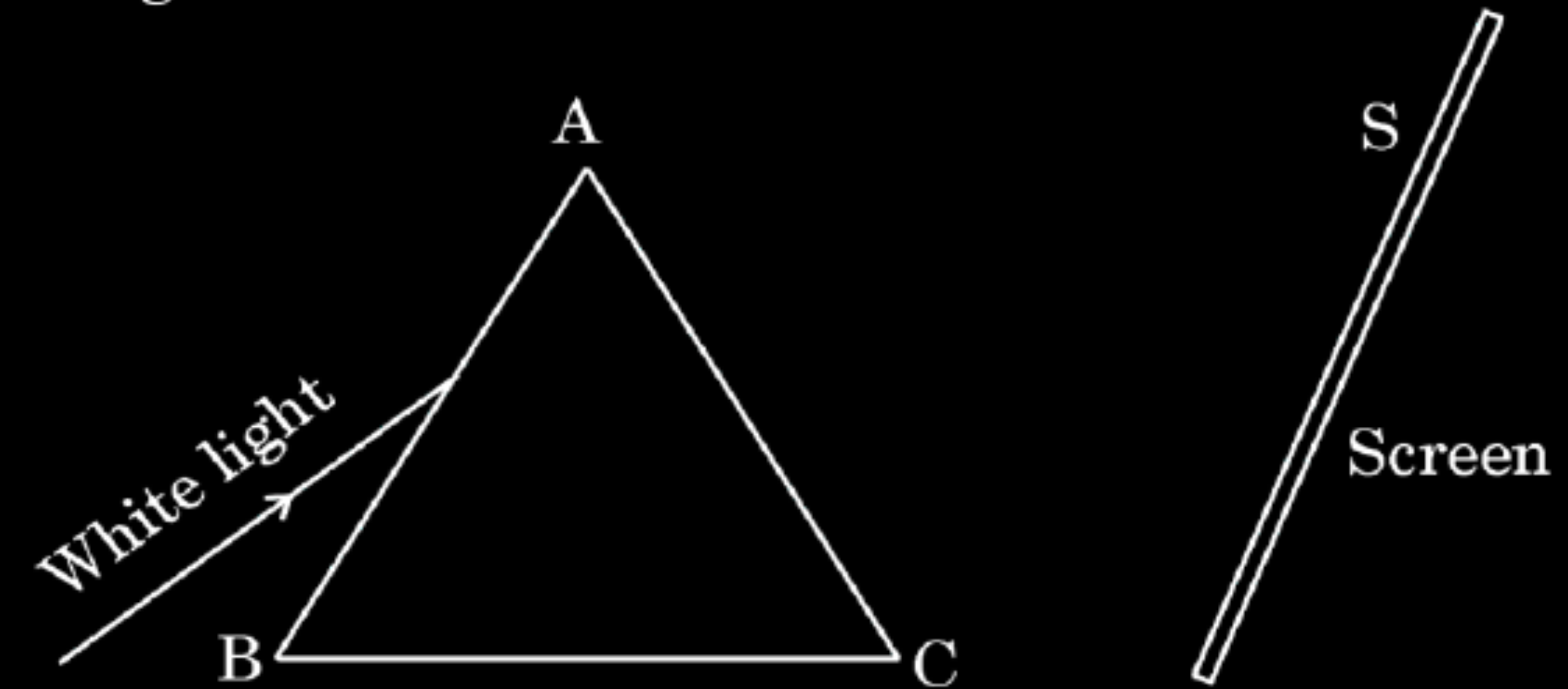
- (A) hypermetropia and convex lens
- (B) presbyopia and bifocal lens
- (C) myopia and concave lens
- (D) myopia and bifocal lens

14. An old person is suffering from an eye defect caused by weakening of ciliary muscles and diminishing flexibility of the eye lens. If the defect of vision is 'a' which can be corrected by lens 'b', then 'a' and 'b' respectively are :

- (A) hypermetropia and convex lens
- (B) presbyopia and bifocal lens
- (C) myopia and concave lens
- (D) myopia and bifocal lens

B / Presbyopia and bifocal lens

- 39.** A person allowed a narrow beam of white light from the sun to enter a dark room through a small aperture and placed a glass prism in its path in such a manner that the beam falls on the face AB of the prism as shown in the figure.



A screen S is placed on the other side of the prism, facing AC. On turning the prism slowly, a beautiful band of colours is obtained on the screen. It is the spectrum of sunlight.

- (a) Name the phenomenon due to which a prism splits the incident white light into a band of colours. 1
- (b) State the reason of getting a band of seven colours in the above case. 1
- (c) (i) Explain with the help of a labelled ray diagram, an experimental arrangement to show the recombination of the spectrum of white light. 2

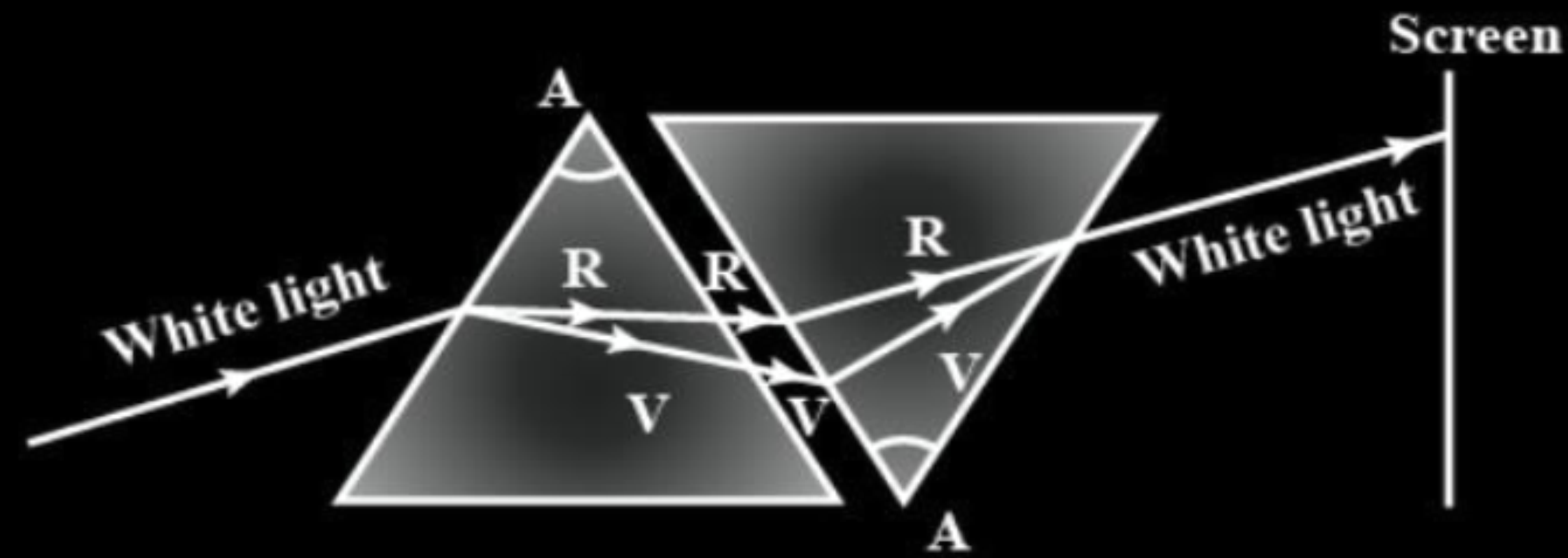
OR

- (c) (ii) Draw a labelled ray diagram to show the formation of a rainbow. 2

- (a) Dispersion of light
- (b) Different colours of light bend through different angles with respect to the incident ray as they pass through a prism.
- (c) (i) Two identical prisms are placed in inverted position with respect to each other as shown. When spectrum produced by prism A is passed through the prism B, a beam of white light emerges from the other side of the prism B.

1

1

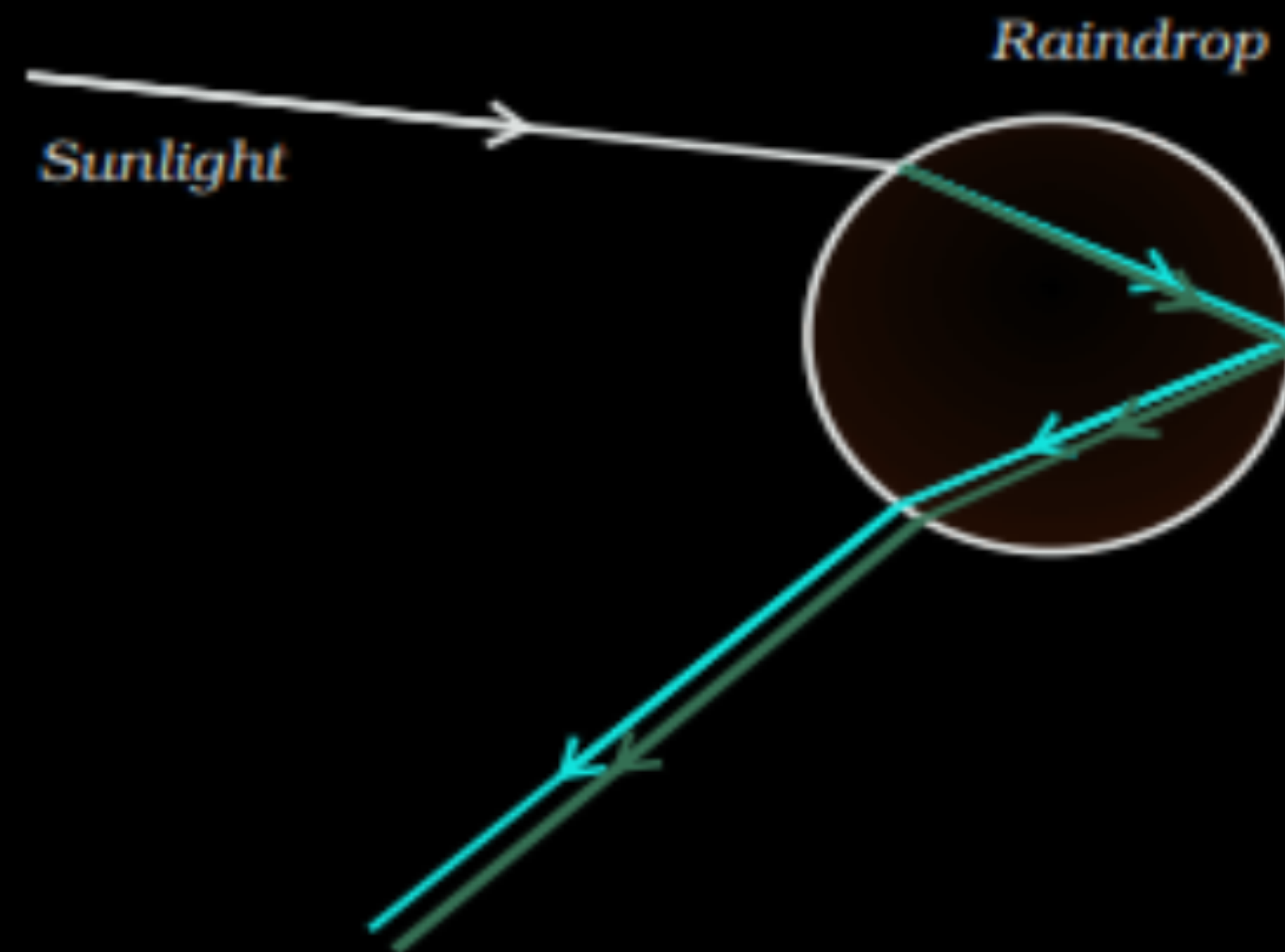


(award full marks even if only labelled ray diagram is given)

2

OR

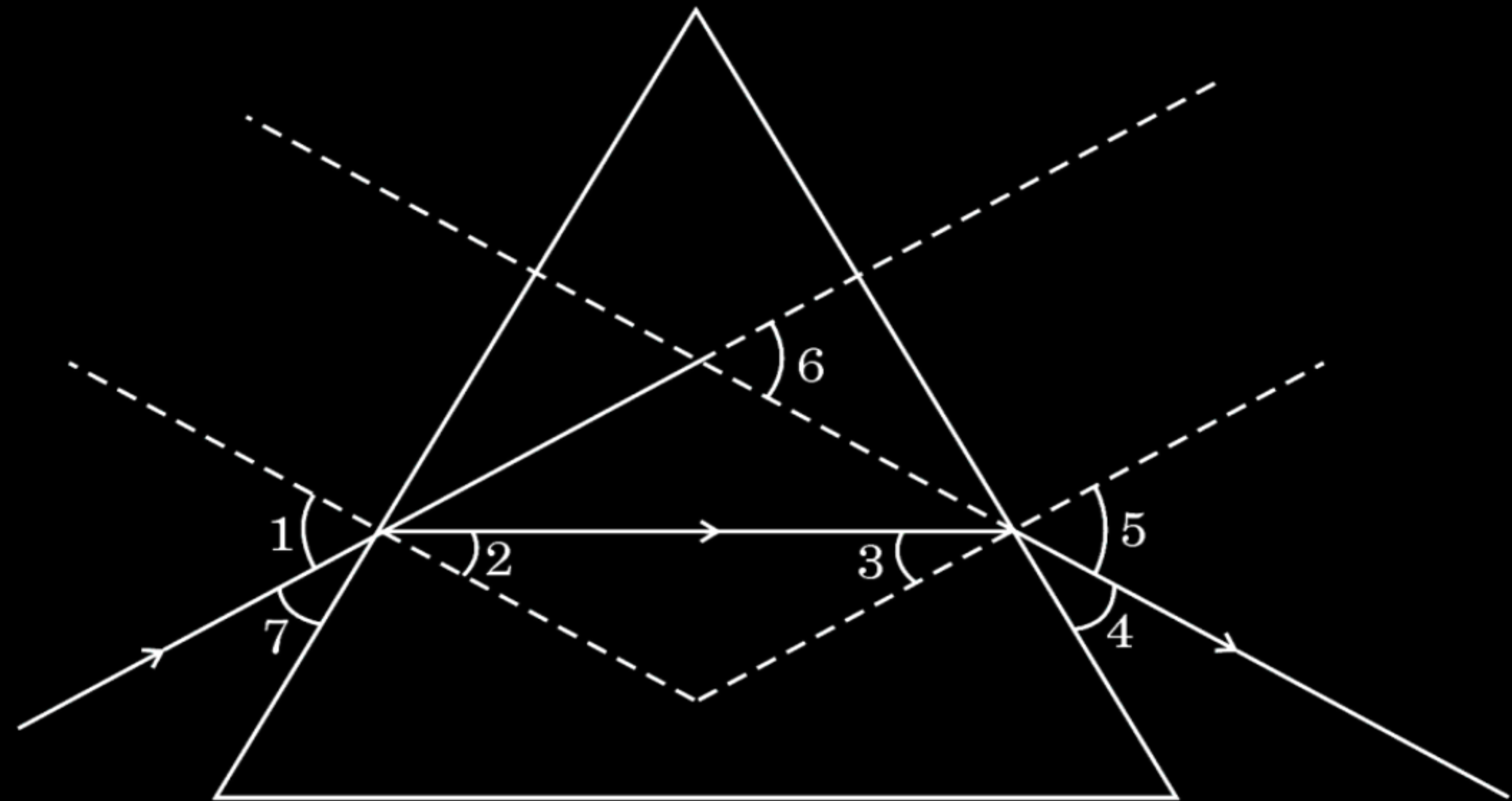
(c)(ii)



(deduct ½ marks if arrows are not marked)

2

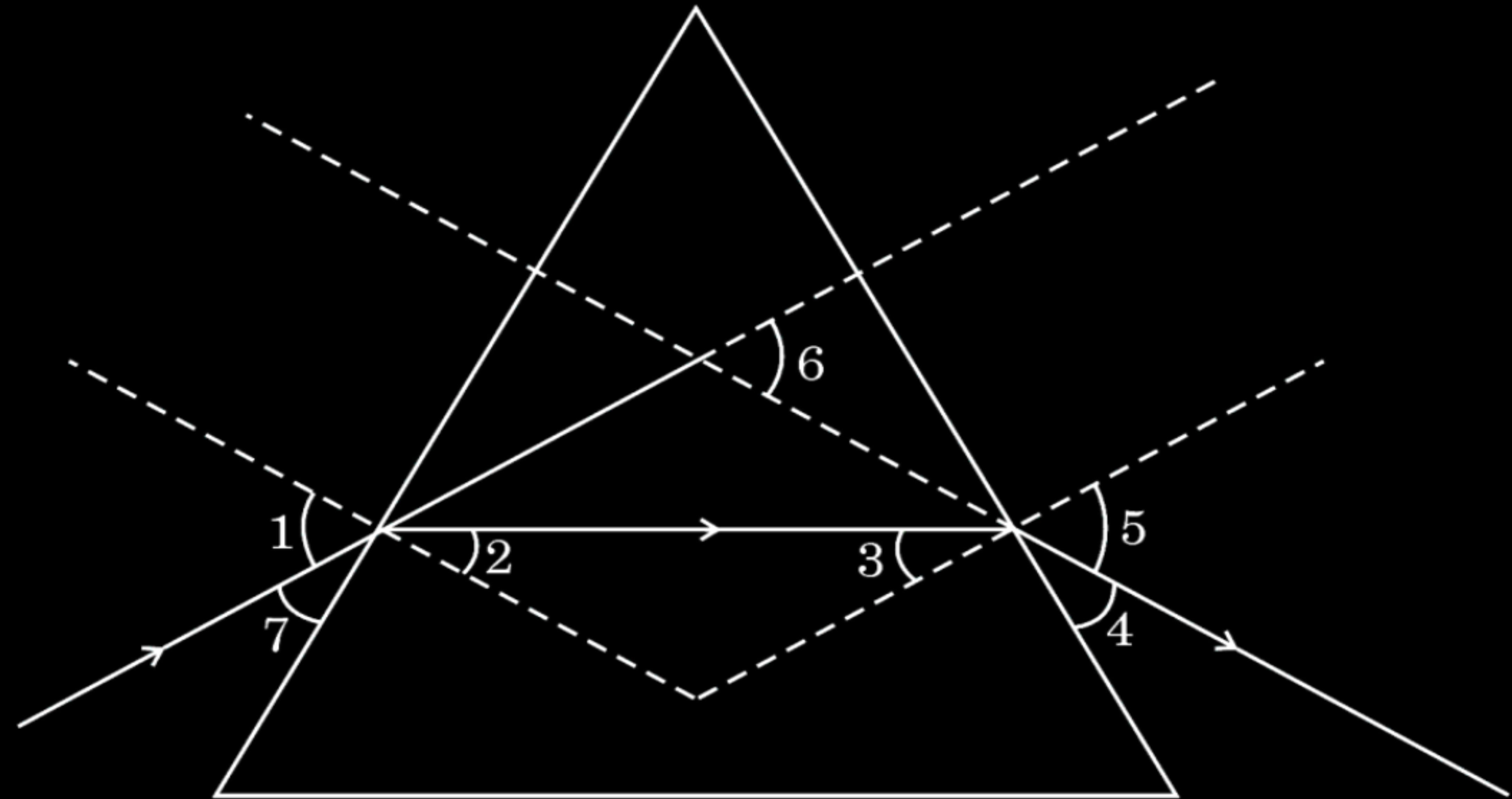
- 13.** In the given figure the angle of incidence and the angle of deviation respectively are :



- (A) 1 and 5
(C) 7 and 4

- (B) 7 and 6
(D) 1 and 6

13. In the given figure the angle of incidence and the angle of deviation respectively are :



- (A) 1 and 5
(C) 7 and 4

- (B) 7 and 6
(D) 1 and 6

(D) / 1 and 6

13. In the human eye, most of the refraction for the light rays entering the eye occurs at the surface of :

(A) Cornea

(B) Crystalline lens

(C) Pupil

(D) Aqueous humor

13. In the human eye, most of the refraction for the light rays entering the eye occurs at the surface of :

(A) Cornea

(B) Crystalline lens

(C) Pupil

(D) Aqueous humor

(A) / Cornea

13. The part of the human eye which can modify the curvature of the eye lens to some extent is :

- (A) Pupil
- (B) Cornea
- (C) Ciliary muscles
- (D) Aqueous humour

13. The part of the human eye which can modify the curvature of the eye lens to some extent is :

- (A) Pupil
- (B) Cornea
- (C) Ciliary muscles
- (D) Aqueous humour

(C)/ Ciliary muscles

28. When and where does a rainbow appear in the sky ? List the phenomena of light in proper sequence which occurs during the formation of a rainbow. Why can a rainbow not be seen from the surface of the moon ?

5. The image distance from the eye lens in the normal eye when we increase the distance of an object from the eye

1

- (A) increases.
- (B) decreases.
- (C) remains unchanged.
- (D) depends on the size of the eyeball.

5. The image distance from the eye lens in the normal eye when we increase the distance of an object from the eye

1

- (A) increases.
- (B) decreases.
- (C) remains unchanged.
- (D) depends on the size of the eyeball.

(C) / remains unchanged

- 24.** A student uses spectacles of focal length -2.5 m.
- (a) Name the defect of vision he is suffering from.
 - (b) Which lens is used for the correction of this defect ?
 - (c) List two main causes of developing this defect.
 - (d) Compute the power of this lens.

24. A student uses spectacles of focal length – 2.5 m.
- Name the defect of vision he is suffering from.
 - Which lens is used for the correction of this defect ?
 - List two main causes of developing this defect.
 - Compute the power of this lens.

3

(a) Myopia/Short sightedness	1/2
(b) Concave/Diverging lens.	1/2
(c) - Excessive curvature of eye lens - elongation of eye ball	1/2+ 1/2
(d) $P(D) = \frac{1}{f(m)}$ $P(D) = \frac{1}{-2.5(m)} = \frac{10}{-25} = \frac{2}{-5} = -0.4D$	1
(Deduct 1/2 mark if unit is not mentioned)	

Give reasons :

- (a) Red colour is selected for danger signals.
- (b) The sky appears dark in space.
- (c) The time difference between actual sunset and apparent sunset is about 2 minutes.

Give reasons :

- (a) Red colour is selected for danger signals.
- (b) The sky appears dark in space.
- (c) The time difference between actual sunset and apparent sunset is about 2 minutes.

3

(a) The Red colour is least scattered by fog or smoke, hence visible from a long distance.	1
(b) Because in the absence of atmosphere there is no scattering of light.	1
(c) Because of atmospheric refraction, the sun appears above the horizon even after actual sunset.	1

- (a) A person suffering from myopia (near-sightedness) was advised to wear corrective lens of power -2.5 D. A spherical lens of same focal length was taken in the laboratory. At what distance should a student place an object from this lens so that it forms an image at a distance of 10 cm from the lens ?
- (b) Draw a ray diagram to show the position and nature of the image formed in the above case.

$$(a) \quad f(m) = \frac{1}{P(D)}$$

$$f = \frac{1}{-2.5D} = \frac{-10}{25D} = -0.4m = -40cm$$

$$f = -40 \text{ cm}$$

$$v = -10 \text{ cm} \quad u = ?$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

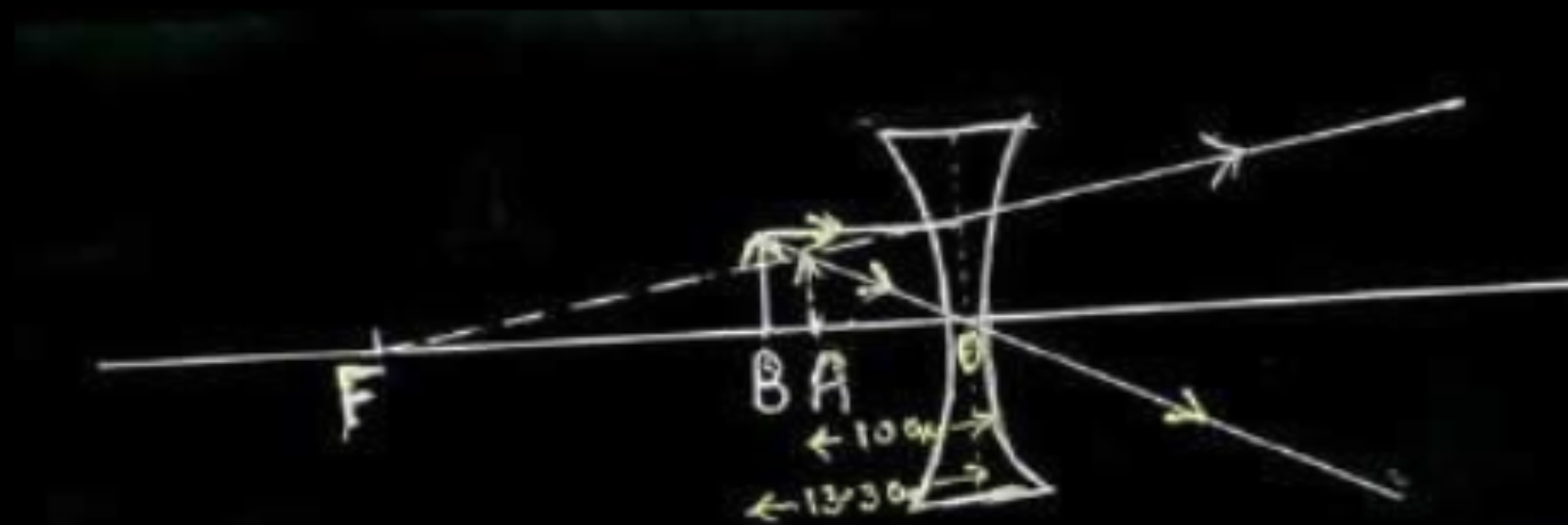
$$\frac{1}{-10 \text{ cm}} - \frac{1}{u} = \frac{1}{-40 \text{ cm}}$$

$$-\frac{1}{u} = \frac{1}{-40} + \frac{1}{10}$$

$$= \frac{-1 + 4}{40} = \frac{3}{40}$$

$$\therefore u = -\frac{40}{3} = -13.3 \text{ cm}$$

(b) Since the power is -ve, the lens used is concave / diverging



$$OA = v = -10 \text{ cm} ; OB = u = -13.3 \text{ cm} ; OF = f = -40 \text{ cm}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$1$$

- 23.** A person may suffer from both myopia and hypermetropia defects.
- (a) What is this condition called ?
 - (b) When does it happen ?
 - (c) Name the type of lens often required by the persons suffering from this defect. Draw labelled diagram of such lenses.

- 23.** A person may suffer from both myopia and hypermetropia defects.
- (a) What is this condition called ?
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3

(a) Presbyopia

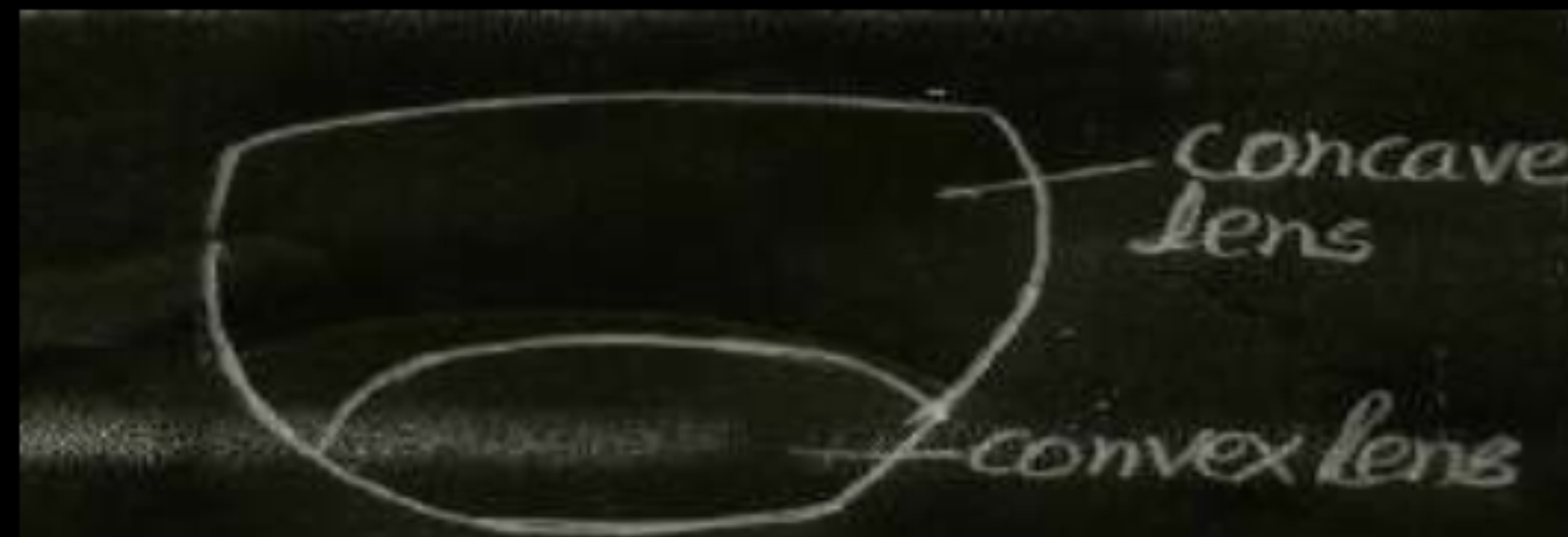
$\frac{1}{2}$

(b) Gradual weakening of the ciliary muscles of the eye/ diminishing flexibility of the eye lens.

1

(c) Bifocal lens

$\frac{1}{2}$



1

24. How will you use two identical glass prisms so that a narrow beam of white light incident on one prism emerges out of the second prism as white light ? Draw and label the ray diagram.

3

By placing second (identical) prism in an inverted position with respect to the first prism.

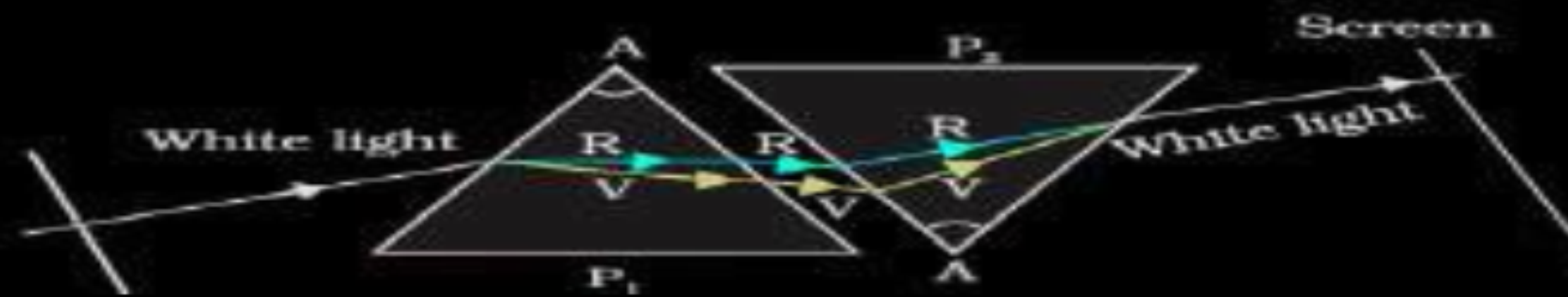


Diagram
Labelling

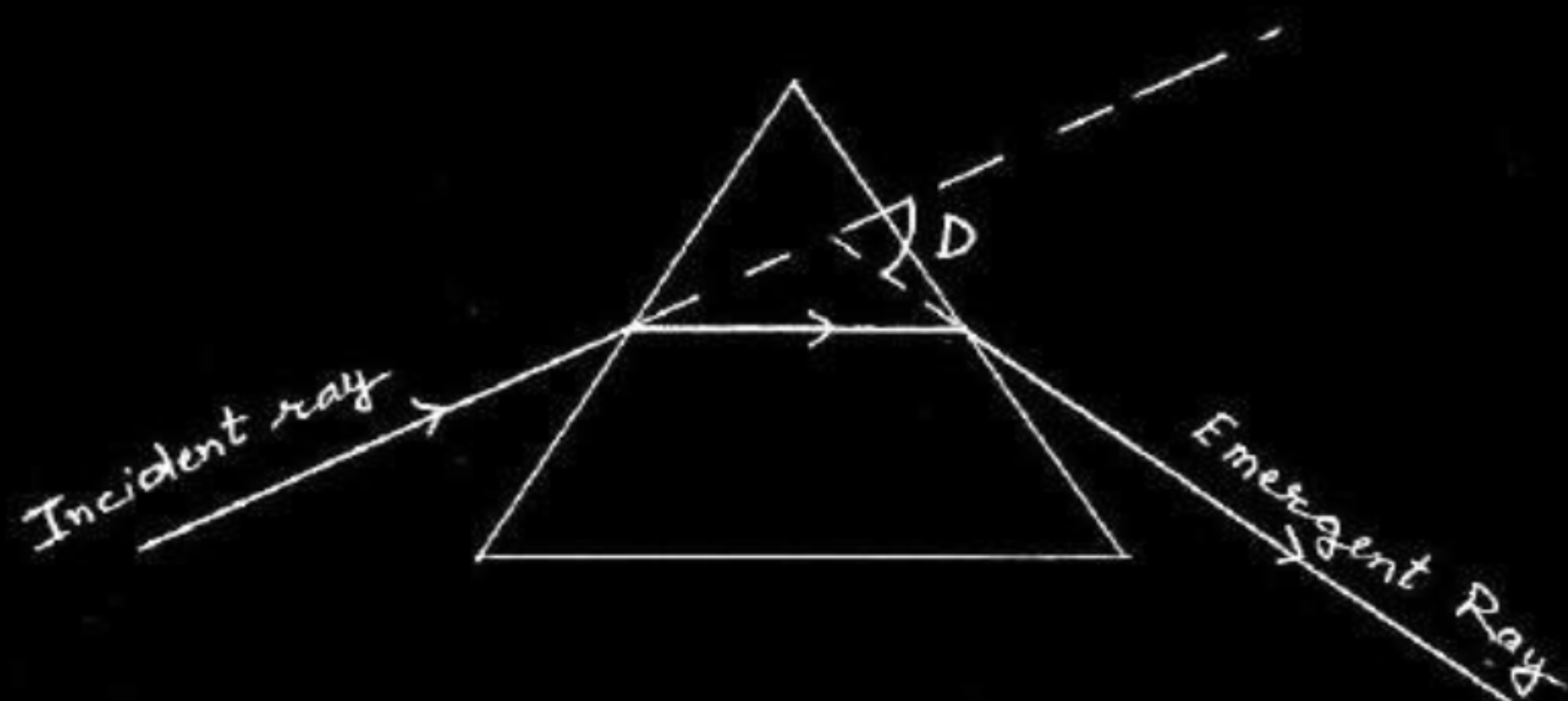
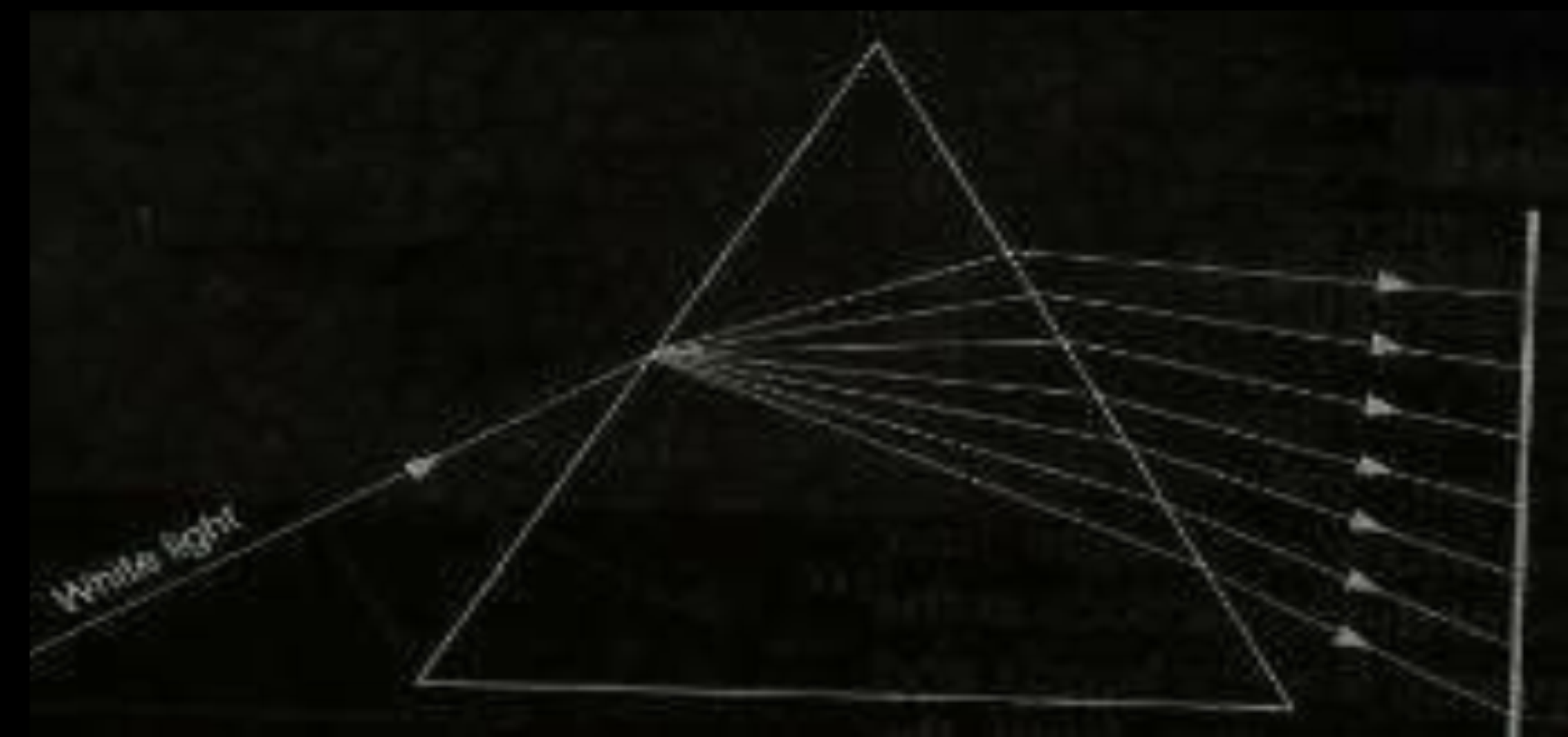
1

$1\frac{1}{2}$
 $\frac{1}{2}$

24. (a) With the help of labelled ray diagram show the path followed by a narrow beam of monochromatic light when it passes through a glass prism.
- (b) What would happen if this beam is replaced by a narrow beam of white light ?

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3

(a)  Incident ray Emergent Ray D Path of the ray Labelling	1 1	(b) Splitting into seven colours / Dispersion / VIBGYOR /  <u>Note :</u> Marks may also be awarded if answer is given in the form of a figure.	1
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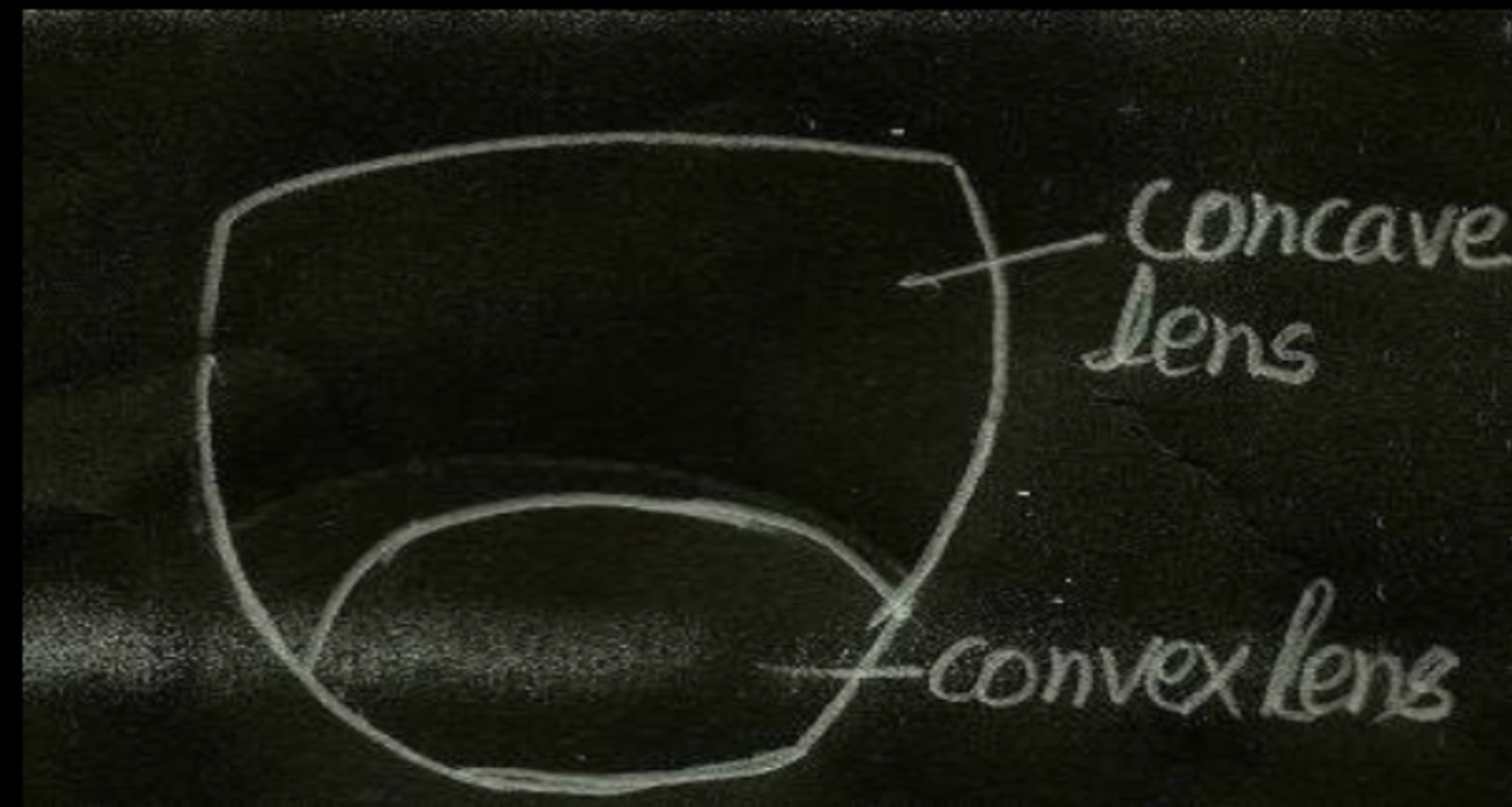
- (a) A person is suffering from both myopia and hypermetropia.
 - (i) What kind of lenses can correct this defect ?
 - (ii) How are these lenses prepared ?
- (b) A person needs a lens of power $+ 3\text{D}$ for correcting his near vision and -3D for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects.

- (a) A person is suffering from both myopia and hypermetropia.
- What kind of lenses can correct this defect ?
 - How are these lenses prepared ?
- (b) A person needs a lens of power + 3D for correcting his near vision and –3D for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects.

3

(a) (i) Bifocal Lens

(ii) Upper part of lens is concave and lower part of the lens is convex. /



(b) $P = +3D$

$$f = \frac{1}{P}$$

$$= \frac{1}{3} \text{ m} = \frac{+100}{3} \text{ cm} = +33.3 \text{ cm}$$

$P = -3D$

$$f = \frac{-100}{3} = -33.3 \text{ cm}$$

$$\frac{1}{2}$$

$$\frac{1}{2}, \frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

Person suffering from cataract has

- (a) elongated eyeball
- (b) excessive curvature of eye lens
- (c) weakened ciliary muscles
- (d) opaque eye lens

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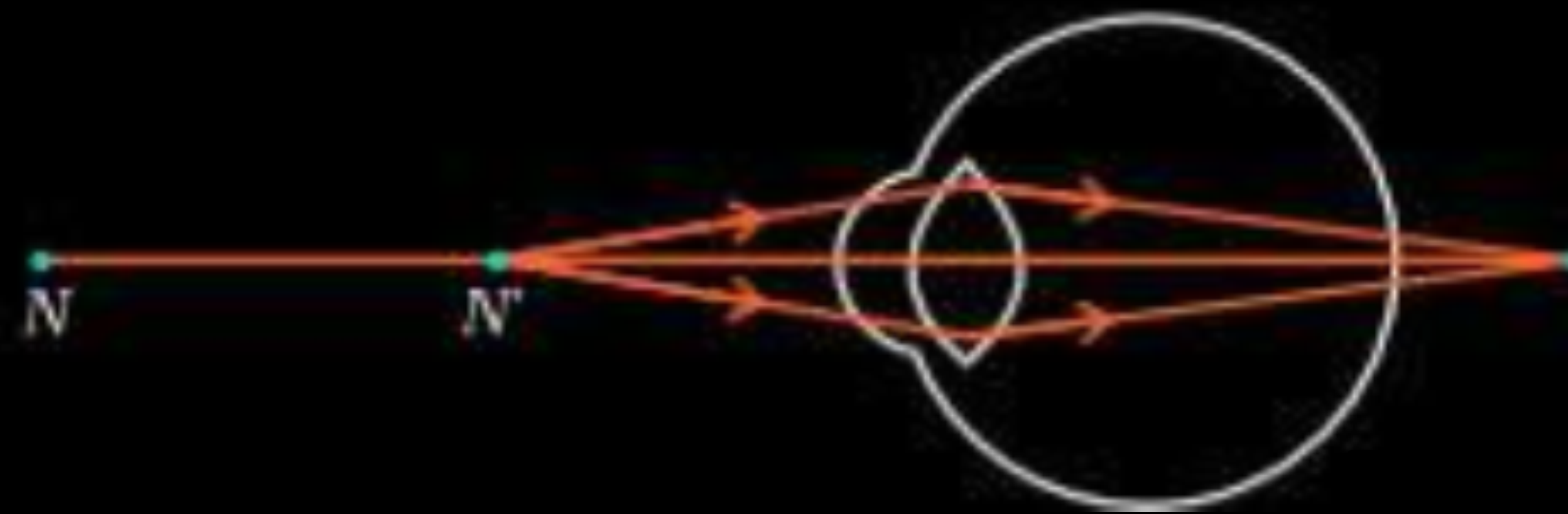
(d) opaque eye lens

24. (a) List two causes of hypermetropia.
- (b) Draw ray diagrams showing (i) a hypermetropic eye and (ii) its correction using suitable optical device.

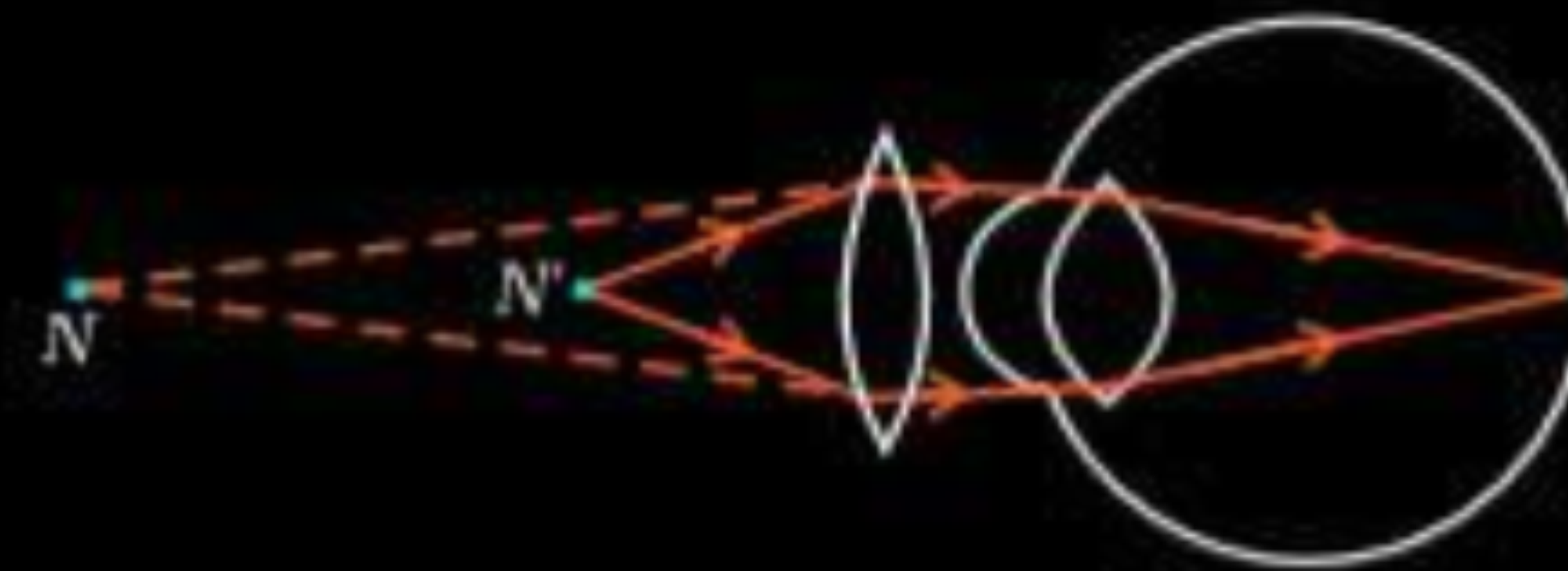
24. (a) List two causes of hypermetropia.
 (b) Draw ray diagrams showing (i) a hypermetropic eye and (ii) its correction using suitable optical device.

3

- (a) (i) Size of eyeball decreases
 (ii) Focal length of eye lens is too long / Power of eye lens decreases.
 (b) Diagrams :



Hypermetropic Eye



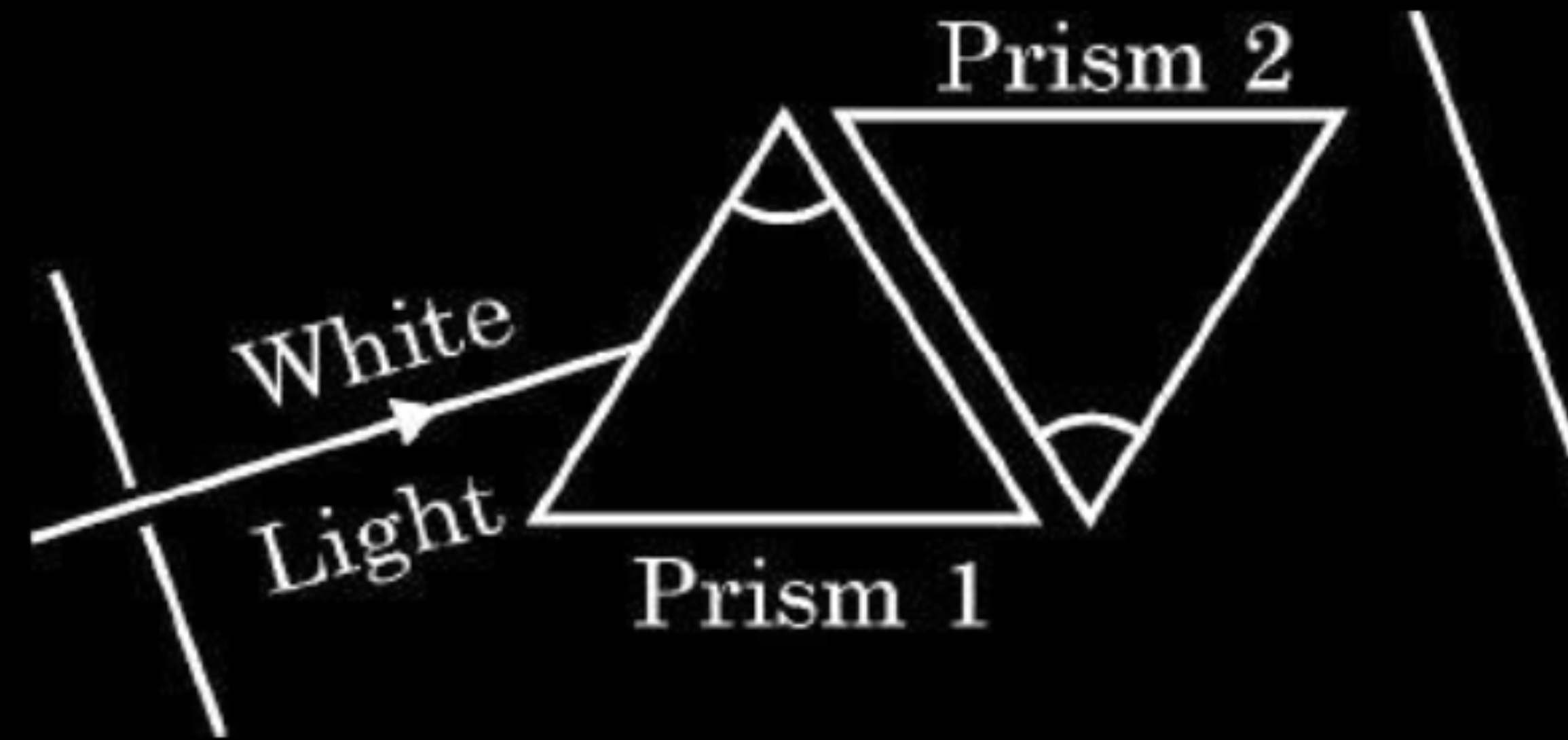
Corrected Eye

$$\frac{1}{2} + \frac{1}{2}$$

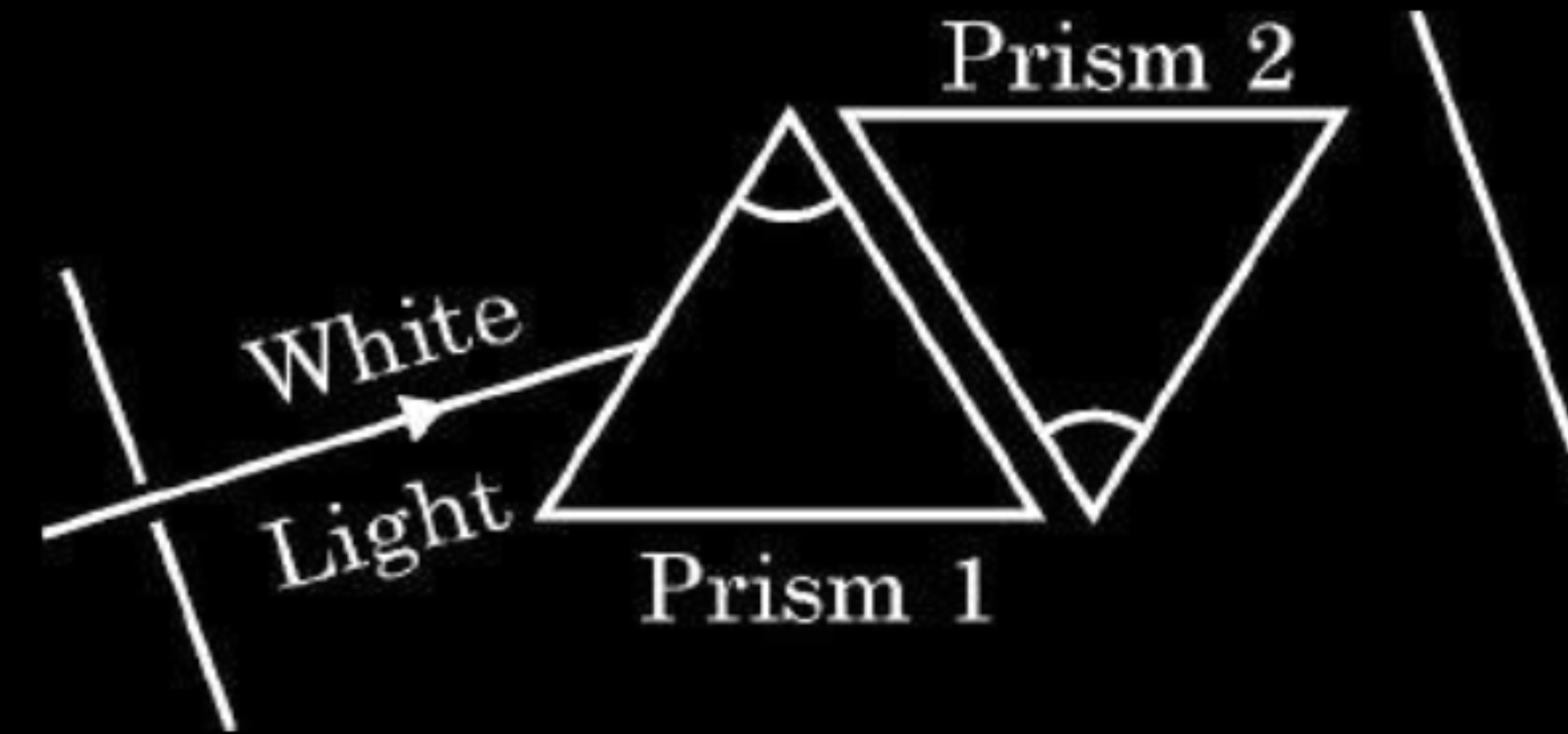
1

1

- (a) State the relation between colour of scattered light and size of the scattering particle.
- (b) The apparent position of an object, when seen through the hot air, fluctuates or wavers. State the basic cause of this observation.
- (c) Complete the path of white light when it passes through two identical prisms placed as shown :

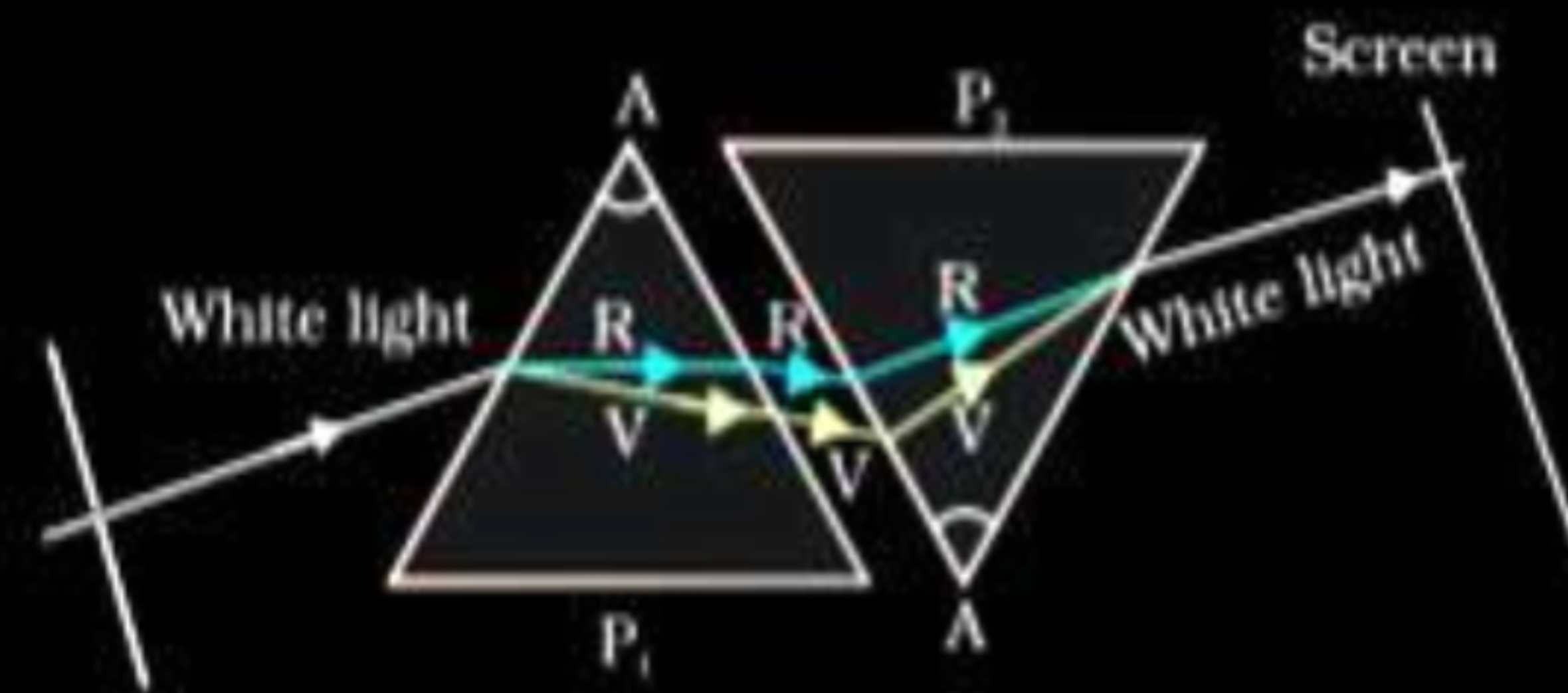


- State the relation between colour of scattered light and size of the scattering particle.
- The apparent position of an object, when seen through the hot air, fluctuates or wavers. State the basic cause of this observation.
- Complete the path of white light when it passes through two identical prisms placed as shown :



3

- Small size particles scatter shorter wavelength (violet) or large sized particles scatter larger wavelength (Red).
- Due to variation in physical condition of hot air.
- Diagram



1

(Splitting of white light is essential)

21. Why is Tyndall effect shown by colloidal particles? State four instances of observing the Tyndall effect

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3

Because of scattering of light.

Instances:

- When a fine beam of light enters a smoke-filled dark room through a small hole.
- When sunlight passes through a canopy of dense forest in foggy/ misty conditions.
- Blue colour of sky.
- Red colour of the sun during sunrise or sunset.

(or any other)

1

$\frac{1}{2} \times 4$

Differentiate between a glass slab and a glass prism. What happens when a narrow beam of (i) a monochromatic light, and (ii) white light passes through (a) glass slab and (b) glass prism?

Differentiate between a glass slab and a glass prism. What happens when a narrow beam of (i) a monochromatic light, and (ii) white light passes through (a) glass slab and (b) glass prism?

• Prism has 2 inclined refracting surfaces whereas a glass slab has 2 parallel refracting surfaces.	1
i) When monochromatic light passes through a glass slab it gets displaced laterally whereas in prism it gets angularly displaced.	$\frac{1}{2}$
ii) When white light passes through a glass slab, it gets laterally displaced whereas in prism, dispersion takes place.	$\frac{1}{2}$

22. Draw a labelled diagram to show (i) reddish appearance of the sun at the sunrise or the sunset and (ii) white appearance of the sun at noon when it is overhead.

22. Draw a labelled diagram to show (i) reddish appearance of the sun at the sunrise or the sunset and (ii) white appearance of the sun at noon when it is overhead.

3

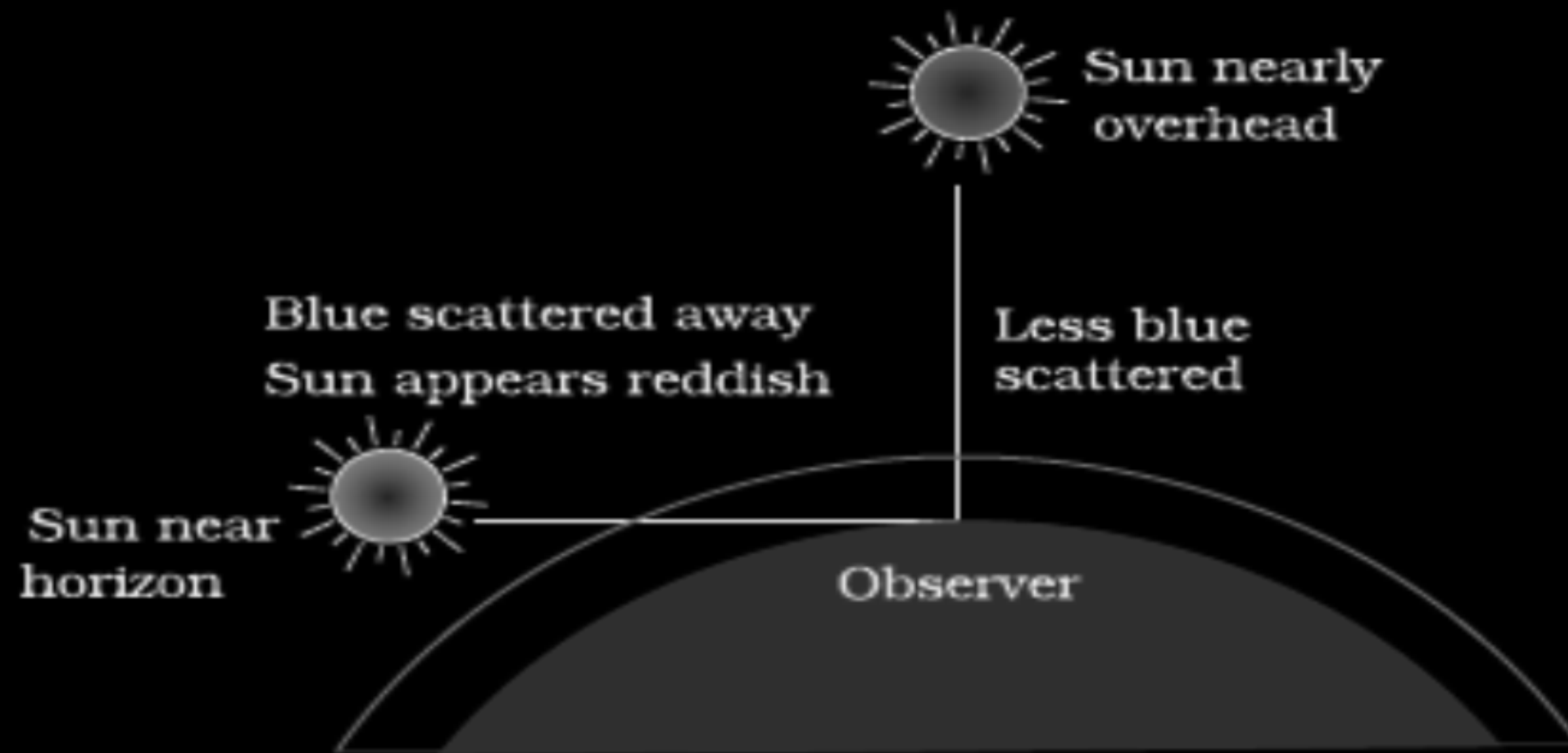
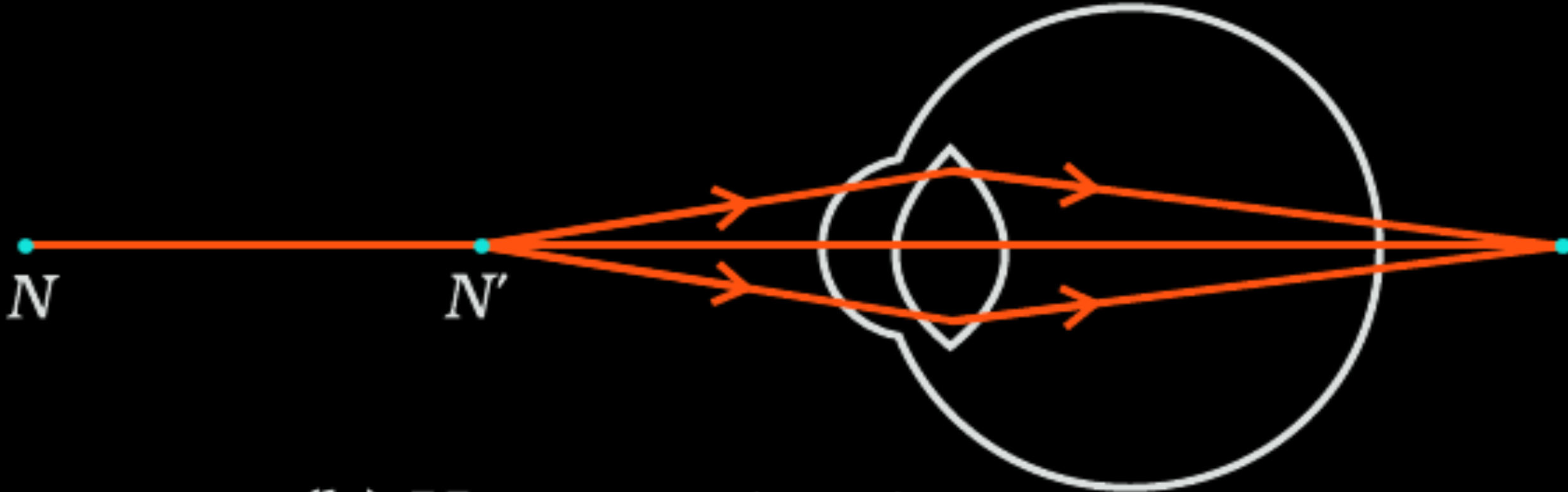
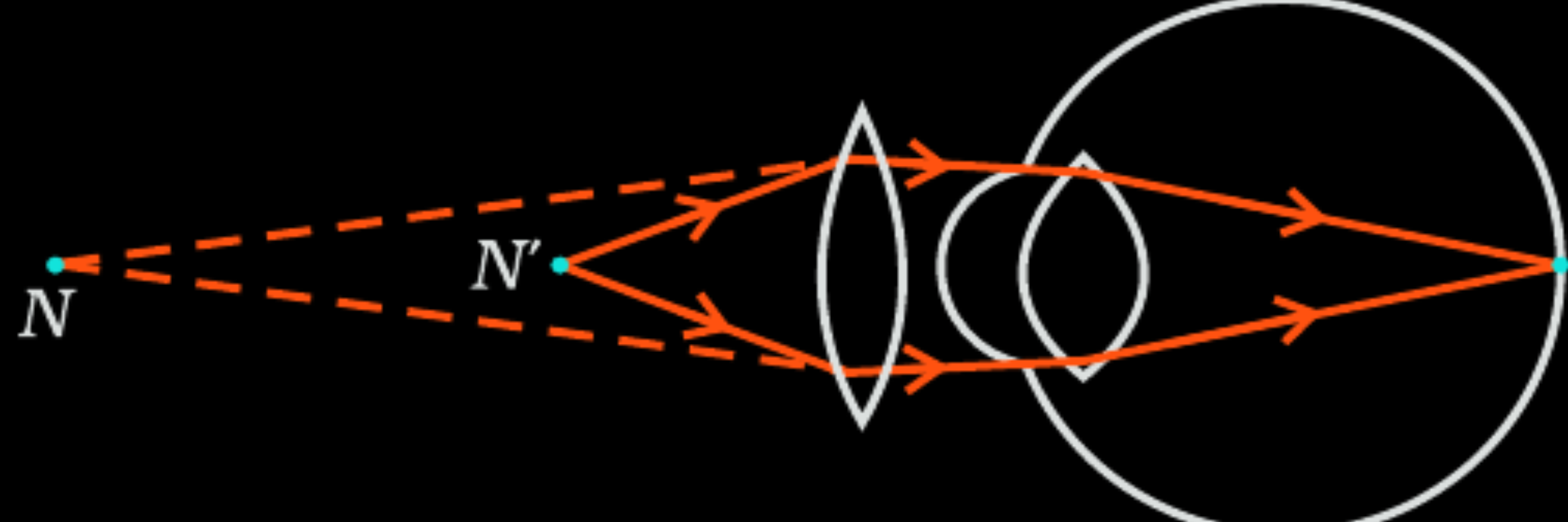


Diagram
Labelling

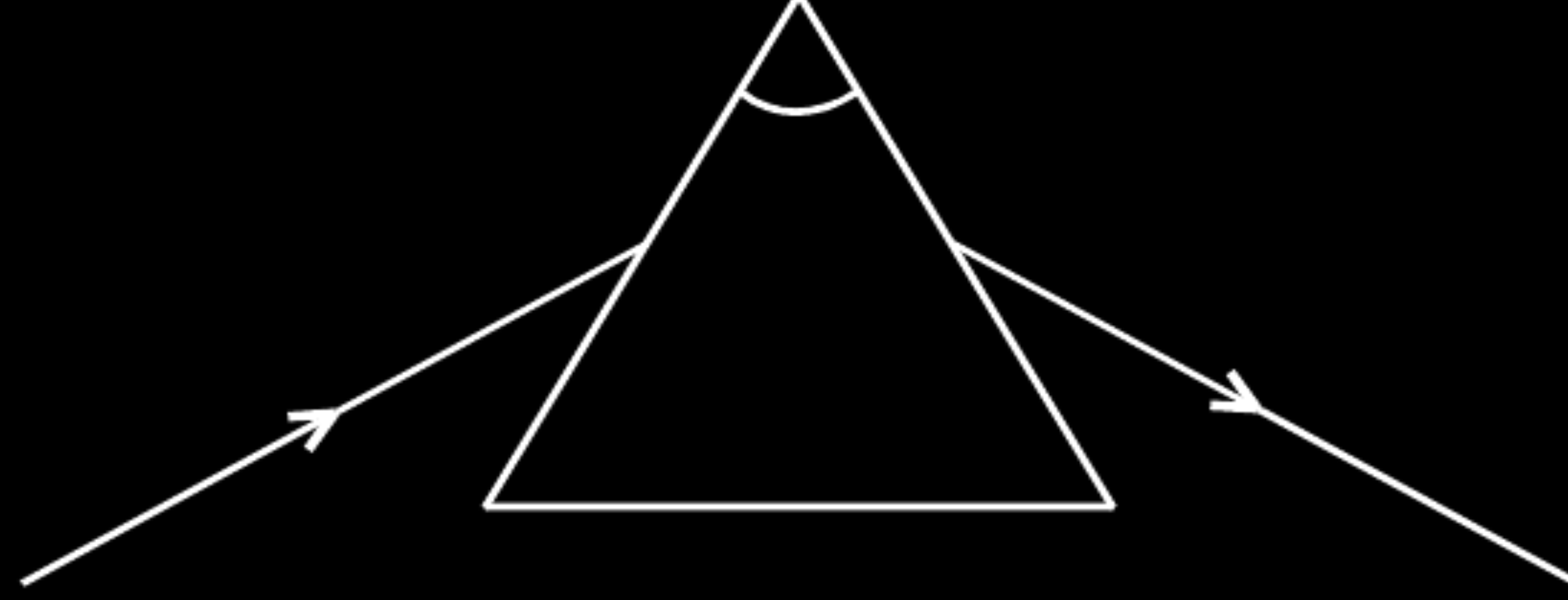
1
 $\frac{1}{2} \times 4$

When do we consider a person to be myopic or hypermetropic ? List two causes of hypermetropia. Explain using ray diagrams how the defect associated with hypermetropic eye can be corrected.

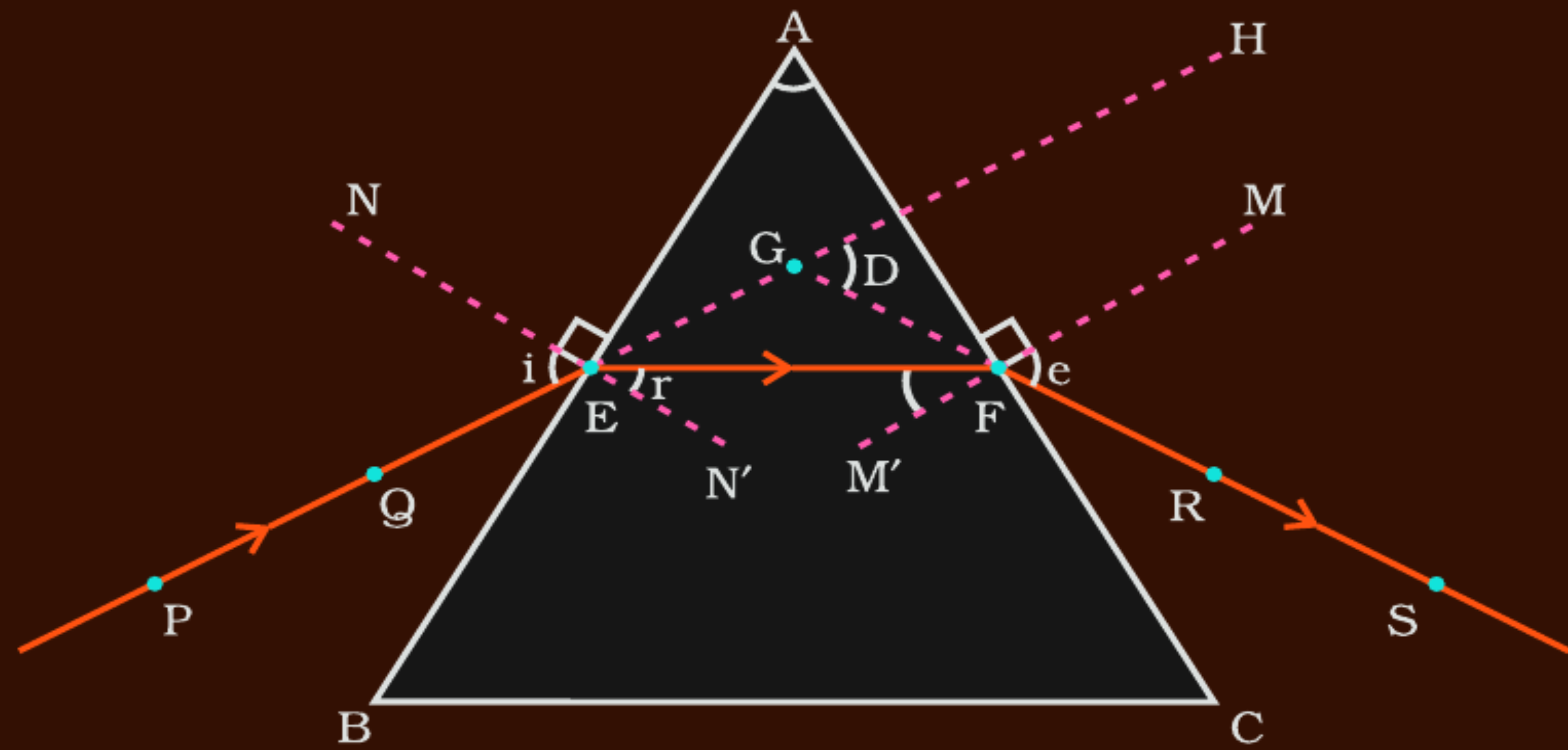
When do we consider a person to be myopic or hypermetropic ? List two causes of hypermetropia. Explain using ray diagrams how the defect associated with hypermetropic eye can be corrected.

Myopia:- Difficult to see the objects placed far away / Hypermtropia: Difficult to see very close or nearby objects.	1
Causes of hypermetropia – (i) The focal length of the eye lens is too long (ii) eye ball has become too small	$\frac{1}{2} + \frac{1}{2}$
 (b) Hypermetropic eye  (c) Correction for Hypermetropic eye	
Note: Diagram with brief description -03; only correct diagram with labelling -2 or only explanation 01	3

A student traces the path of a ray of light through a glass prism as shown in the diagram, but leaves it incomplete and unlabelled. Redraw and complete the diagram. Also label on it $\angle i$, $\angle e$, $\angle r$ and $\angle D$.



A student traces the path of a ray of light through a glass prism as shown in the diagram, but leaves it incomplete and unlabelled. Redraw and complete the diagram. Also label on it $\angle i$, $\angle e$, $\angle r$ and $\angle D$.



PE – Incident ray

EF – Refracted ray

FS – Emergent ray

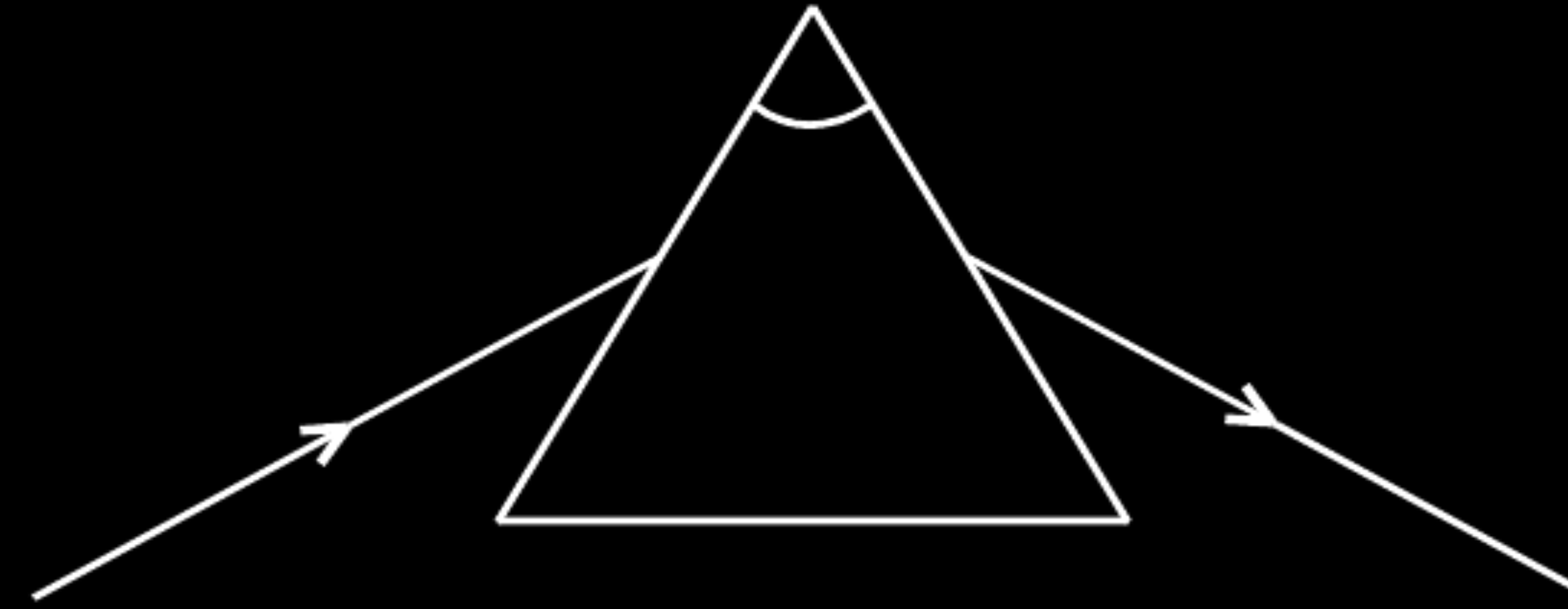
$\angle A$ – Angle of the prism

$\angle i$ – Angle of incidence

$\angle r$ – Angle of refraction

$\angle e$ – Angle of emergence

$\angle D$ – Angle of deviation



Labelling of $\angle i + \angle e + \angle r$ & $\angle D$

$\frac{1}{2} \times 4$

- (a) What is scattering of light ? Explain how the colour of the scattered light depends on the size of the scattering particles.
- (b) Explain the reddish appearance of the Sun at sunrise or sunset. Why does it not appear red at noon ?

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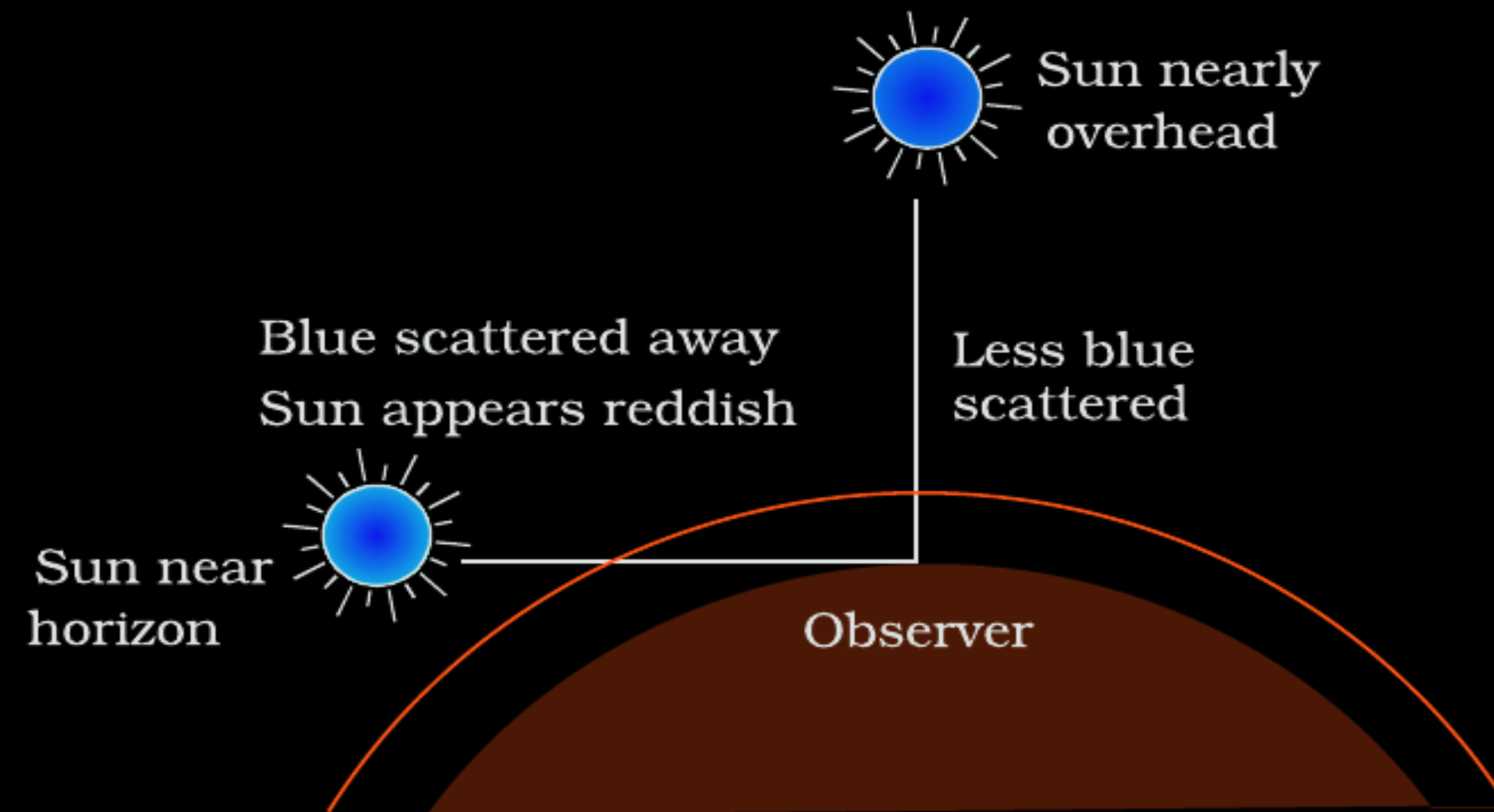


Figure 11.12
Reddening of the Sun at sunrise and sunset

(a) Scattering: Direction of ray of light changes when it collides with particles of comparable size.

- Fine particle scatter shorter wavelengths like blue light.
- Particles of larger size scatter longer wavelengths like red light.
- If particle size is large enough scattered light may appear white.

(b)

- During sunrise and sunset, sun is near horizon. Sunlight travels longer distance. Most of blue light scatters away by particles. Light of longer wavelength reaches our eye so sun appears reddish.
- At noon, sun is directly overhead so sunlight travels lesser distance. Less amount of blue light is scattered giving white appearance to sun

Or can be explained with the help of a diagram given below

What is atmospheric refraction ? List two phenomena which can be explained on the basis of atmospheric refraction.

What is atmospheric refraction ? List two phenomena which can be explained on the basis of atmospheric refraction.

Refraction of light due to variation in optical density of earth.	
Two phenomenon:- (i) Twinkling of stars (ii) Advance sunrise & delayed sunset	$\frac{1}{2} \times 2$

What happens to a beam of white light when it gets refracted through a glass prism ? Which colour deviates the most and the least after refraction through a prism ? What is likely to happen if a second identical prism is placed in an inverted position with respect to the first prism ? Justify your answer.

What happens to a beam of white light when it gets refracted through a glass prism ? Which colour deviates the most and the least after refraction through a prism ? What is likely to happen if a second identical prism is placed in an inverted position with respect to the first prism ? Justify your answer.

The white light splits into seven colours when it gets refracted through the glass prism (VIBGYOR)	1
The colour deviates most – Violet	$\frac{1}{2}$
The colour deviates least – Red	$\frac{1}{2}$
Colours disappear and again white light obtained.	1

A student needs spectacles of power -0.5 D for the correction of his vision.

- (i) Name the defect in vision the student is suffering from.
- (ii) Find the nature and focal length of the corrective lens.
- (iii) List two causes of this defect.

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- (iii) List two causes of this defect.

(i)	Myopia	1 1
(ii)	Concave / diverging lens and focal Length = 200cm	$\frac{1}{2}$
(iii)	(a) excessive curvature of the eye lens (b) elongation of eye ball	$\frac{1}{2}$

Why is the colour of the clear sky blue ?

Why is the colour of the clear sky blue ?

When sunlight passes through the atmosphere, the fine particles in the air scatter the blue colour (short wavelength) more strongly than red. The scattered blue colour enters our eyes.	2
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What is atmospheric refraction ? Explain with the help of a labelled diagram that the position of a star as seen by us is not its true position.

What is atmospheric refraction ? Explain with the help of a labelled diagram that the position of a star as seen by us is not its true position.

- Bending of light due to the variation in optical density of the medium.
- The starlight, on entering into earth's atmosphere undergoes continuous refraction before it reaches the earth.
- The since the atmosphere bends starlight towards the normal, the apparent position to the star is slightly different from its actual position.

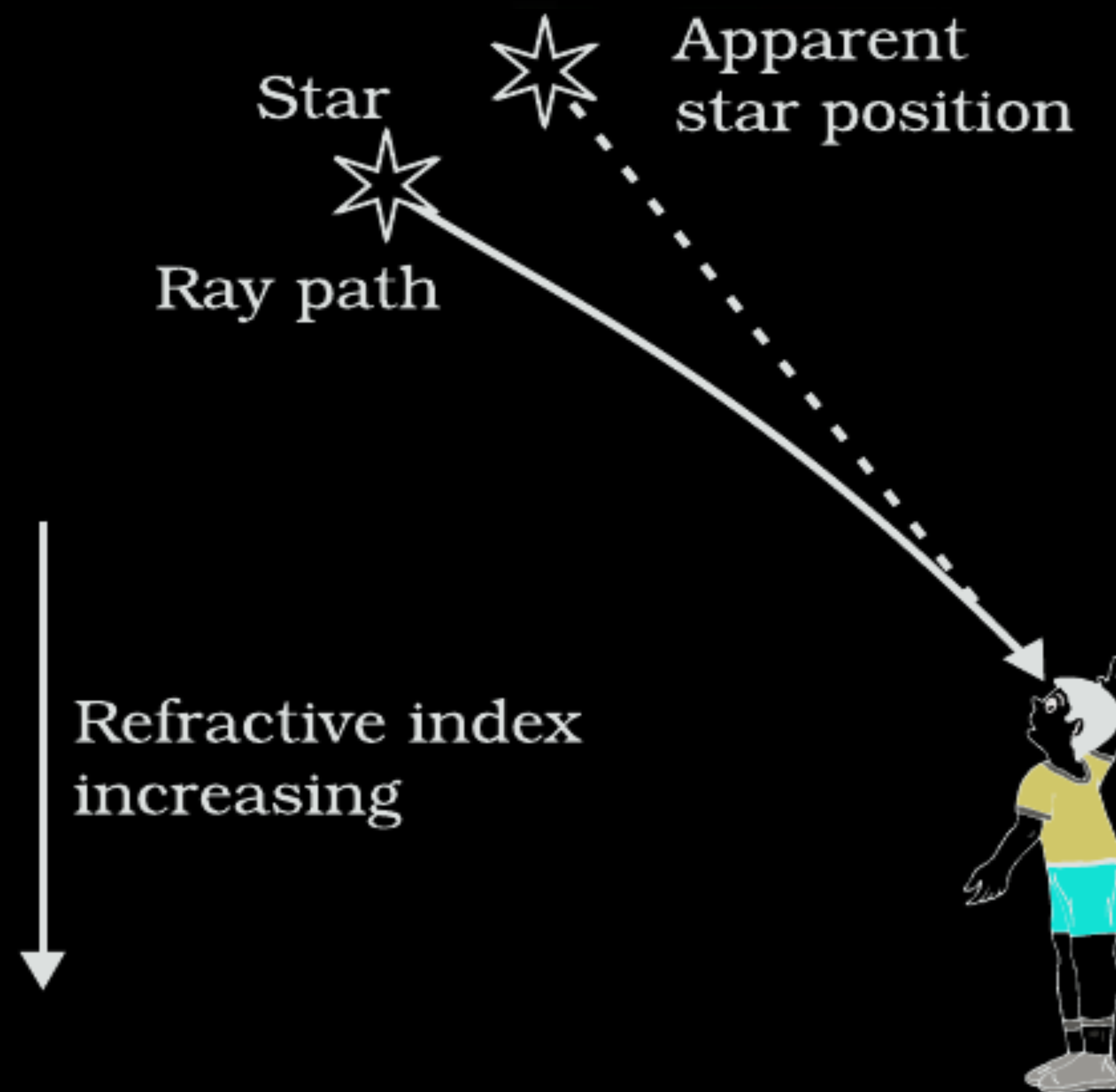


Diagram with Correct labeling

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$1\frac{1}{2}$

When do we consider a student sitting in the class to be myopic ? List two causes of this defect. Explain using a ray diagram how this defect of eye can be corrected.

When do we consider a student sitting in the class to be myopic ? List two causes of this defect. Explain using a ray diagram how this defect of eye can be corrected.

(i) If the student cannot see the words written on the black board then he is considered myopic.

(ii) The defect may arise due to

- 1) Excessive curvature of the eyeball
- 2) Elongation of the eyeball

(iii)



(c) Correction for myopia

1

$\frac{1}{2} \times 2$

1

A glass prism is able to produce a spectrum when white light passes through it but a rectangular block of same transparent glass does not produce any spectrum. Why ?

A glass prism is able to produce a spectrum when white light passes through it but a rectangular block of same transparent glass does not produce any spectrum. Why ?

The extent of bending of a light ray at core opposite parallel faces of the rectangular glass slab is equal. Therefore, the emergent ray is parallel to the incident ray. So it cannot produce any spectrum.
--

List two causes of presbyopia. Draw labelled diagram of a lens used for the correction of this defect of vision.

List two causes of presbyopia. Draw labelled diagram of a lens used for the correction of this defect of vision.

Causes	
i) gradual weakening of the ciliary muscles	1
ii) diminishing flexibility of the eye lens.	1
(No diagram given in t.b. or reference books. Diagram for lens used, use a bifocal lens)	

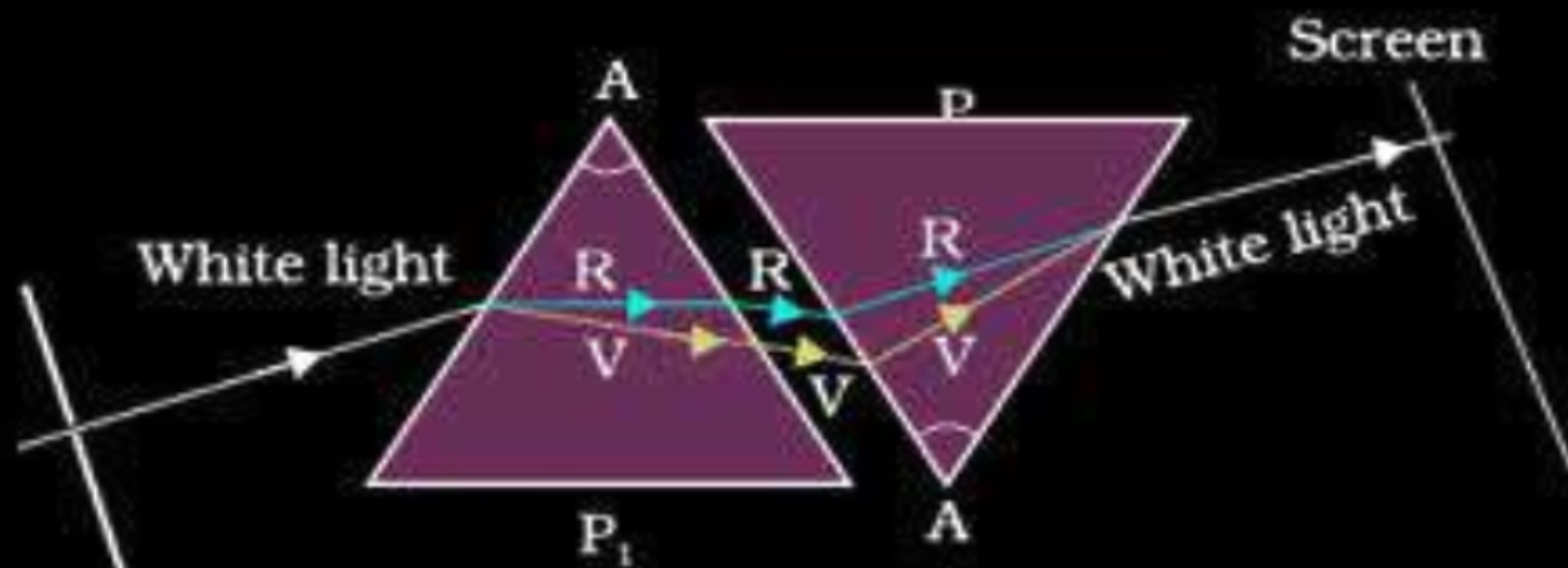
What is the cause of dispersion of white light through a glass prism ? Draw a ray diagram to show the path of light when two identical glass prisms are arranged together in inverted position with respect to each other and a narrow beam of white light is allowed to fall obliquely on one of the faces of the prisms.

What is the cause of dispersion of white light through a glass prism ? Draw a ray diagram to show the path of light when two identical glass prisms are arranged together in inverted position with respect to each other and a narrow beam of white light is allowed to fall obliquely on one of the faces of the prisms.

- Cause of dispersion:
 - (i) Shape of prism
 - (ii) Different colours bend with different angles (different colour has different refractive index or different speed)

diagram

Arrow& labelling



1

1

1

What is scattering of light ? Use this phenomenon to explain why (i) the Sun appears reddish at sun-rise, and (ii) the clear sky appears blue.

What is scattering of light ? Use this phenomenon to explain why (i) the Sun appears reddish at sun-rise, and (ii) the clear sky appears blue.

Scattering of light means to throw light in all possible direction when light interact with particles of medium.	1
(i)The Sun appears reddish at sun-rise : the Sun rays have to travel through a large atmospheric distance near the horizon .As the wave length of red light is maximum in the visible range , hence the scattering is least . The blue light and shorter wavelengths are scattered away by the particles .This gives rise to the reddish appearance of the sun./diagrammatic answers may be given fig:11.12	1
(ii)The sky appears blue: Blue colour has shorter wavelength than red. When sunlight passes through the atmosphere, the fine particles in the air scatter the blue light more strongly than red. Hence the clear sky appears blue.	1

A person is unable to see objects distinctly placed within 50 cm from his eyes.

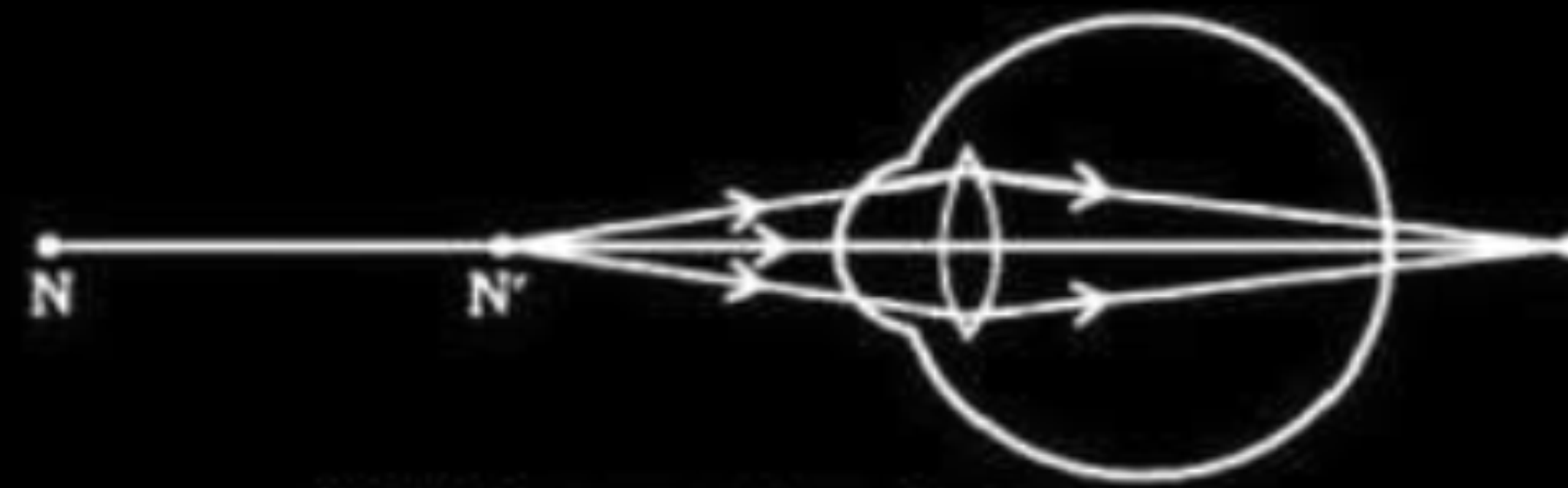
- (a) Name the defect of vision the person is suffering from and list its two possible causes.
- (b) Draw a ray diagram to show the defect in the above case.
- (c) Mention the type of lens used by him for the correction of the defect and calculate its power. Assume that the near point for the normal eye is 25 cm.
- (d) Draw a labelled diagram for the correction of the defect in the above case.

(a) Hypermetropia / farsightedness

Causes:

- i. Shortening of eyeball
- ii. Curvature of eye lens decreases / focal length of eye lens increases.

b)



(b) Hypermetropic eye

(c) Convex lens

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

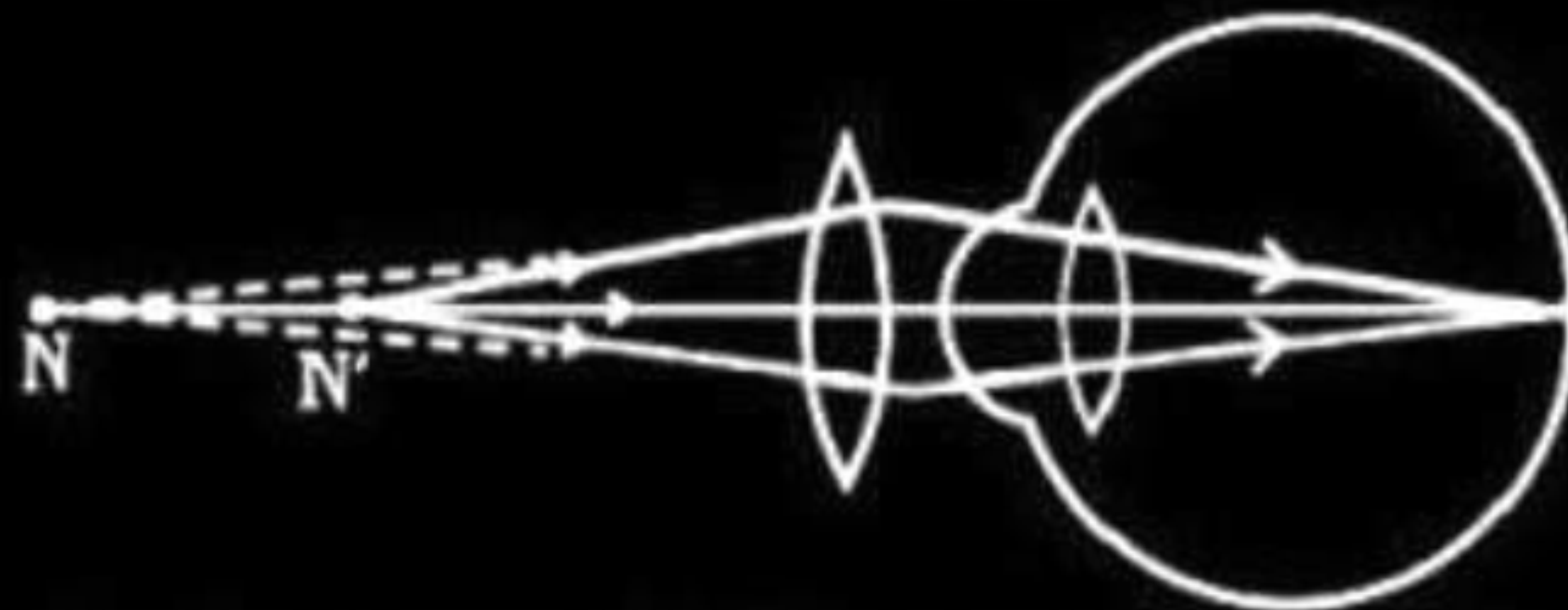
$$= \frac{1}{(-50\text{cm})} - \frac{1}{(-25\text{cm})}$$

$$= \frac{1}{50\text{cm}}$$

Hence ,f=50 cm=0.5m

There fore power =(1/0.5)D=2D

(d)Correction of Hypermetropia



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

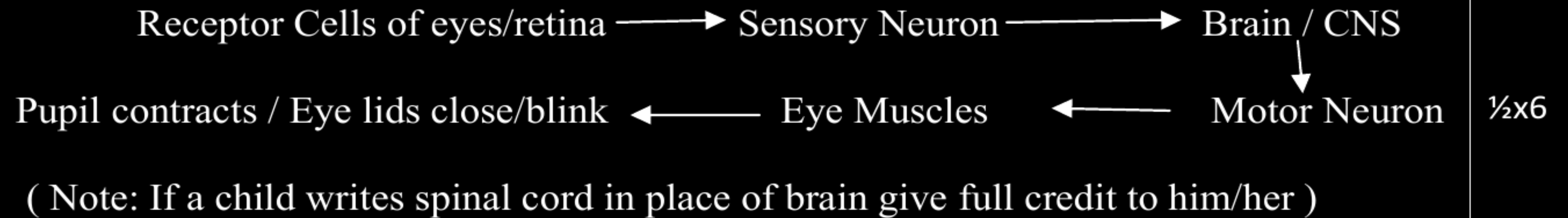
Define the term power of accommodation. Write the modification in the curvature of the eye lens which enables us to see the nearby objects clearly ?

Define the term power of accommodation. Write the modification in the curvature of the eye lens which enables us to see the nearby objects clearly ?

• Power of accommodation – Ability of eye lens to adjust its focal length.	1
• Curvature increases/lens becomes thick	1

Trace the sequence of events which occur when a bright light is focused on your eyes.

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What happens to the image distance in the normal human eye when we decrease the distance of an object, say 10 m to 1 m ? Justify your answer.

What happens to the image distance in the normal human eye when we decrease the distance of an object, say 10 m to 1 m ? Justify your answer.

- Image distance remains the same.
- It is the distance between the eye lens and retina, which remains the same.

1

1

Write the structure of eye lens and state the role of ciliary muscles in the human eye.

Write the structure of eye lens and state the role of ciliary muscles in the human eye.

Structure – Fibrous, jelly like structure	1
Role – To change the curvature of eye lens / to change the focal length of eye lens.	1

- (a) A student is unable to see clearly the words written on the black board placed at a distance of approximately 3 m from him. Name the defect of vision the boy is suffering from. State the possible causes of this defect and explain the method of correcting it.
- (b) Why do stars twinkle ? Explain.

- (a) A student is unable to see clearly the words written on the black board placed at a distance of approximately 3 m from him. Name the defect of vision the boy is suffering from. State the possible causes of this defect and explain the method of correcting it.
- (b) Why do stars twinkle ? Explain.

a.	<u>Defect of vision</u> – Myopia or short sightedness or near sightedness Causes of myopia: i) Excessive curvature of eye lens/eye lens becomes more converging ii) Elongation of eye ball Methods of correction: By the use of concave lens of suitable power or focal length the defect is corrected. / suitable diagrammatic representation.	1 $\frac{1}{2} + \frac{1}{2}$ 1
b.	<u>Due to atmospheric refraction</u> The density of different layers of air keeps on changing due to which the apparent image of the stars keeps on changing. This changing position of stars appears as twinkling of stars.	1 1

- (a) Write the function of each of the following parts of human eye :
- (i) Cornea (ii) Iris (iii) Crystalline lens (iv) Ciliary muscles
- (b) Why does the sun appear reddish early in the morning ? Will this phenomenon be observed by an astronaut on the Moon ? Give reason to justify your answer.

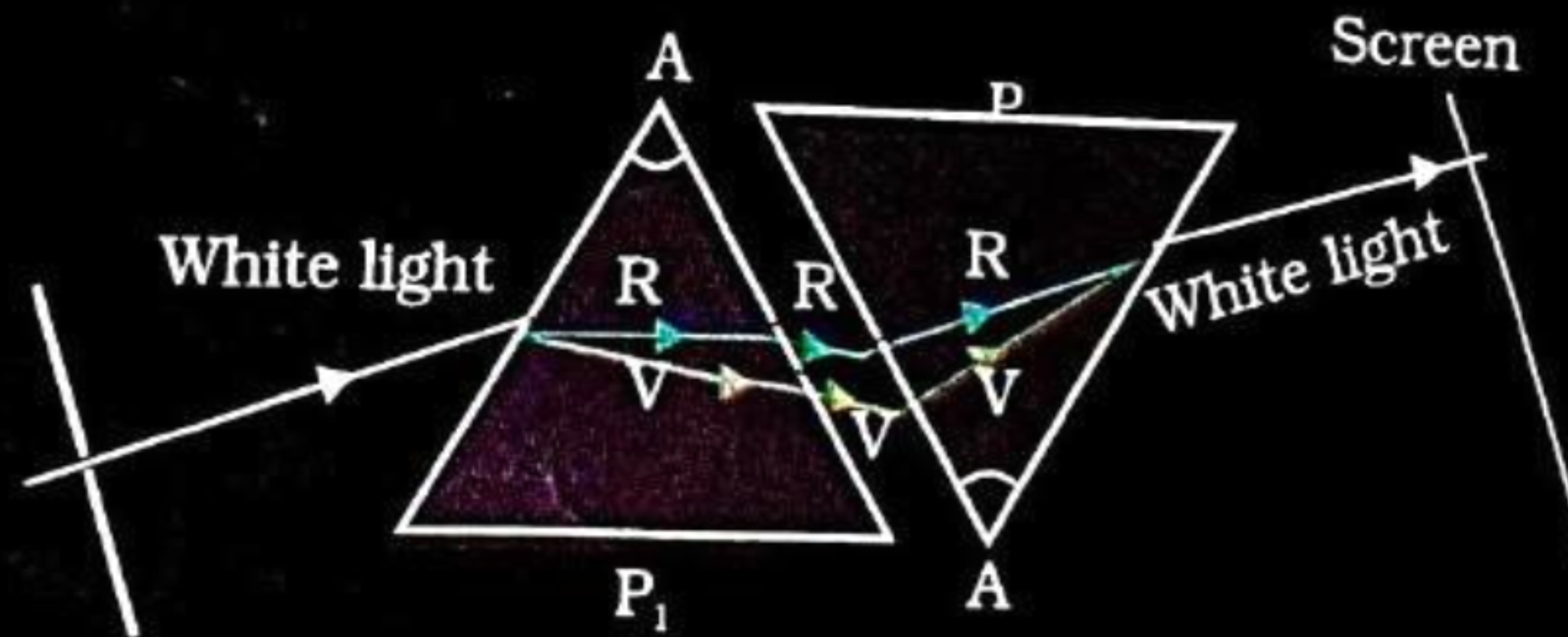
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- (b) Why does the sun appear reddish early in the morning ? Will this phenomenon be observed by an astronaut on the Moon ? Give reason to justify your answer.

a. Function of: • <u>Cornea</u>: focuses light rays / permits the light to enter the eye.. • <u>Iris</u>: Controls amount of light entering the eye. / controls the size of pupil. • <u>Crystalline Lens</u>: Converges light rays onto retina. • <u>Ciliary Muscles</u>: Adjusts focal length of eye lens by contraction and relaxation so that sharp image can be obtained on the retina. / helps in accommodation	$\frac{1}{2} \times 4$
b. In early morning, sun light has to cover larger distance in the atmosphere. So, the shorter wavelengths scatter out. Only the longer wavelengths like red reach our eye. On moon – No Cause: Moon has no atmosphere	$1 \frac{1}{2}$ $\frac{1}{2}$ 1

State the cause of dispersion of white light by a glass prism. How did Newton, using two identical glass prisms, show that white light is made of seven colours ? Draw a ray diagram to show the path of a narrow beam of white light, through a combination of two identical prisms arranged together in inverted position with respect to each other, when it is allowed to fall obliquely on one of the faces of the first prism of the combination.

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Different colours of light bend through different angles with respect to the incident ray / different speed of different colours of light in glass / different values of refractive index of glass for different colours of light.



Direction of ray & labelling

1

1

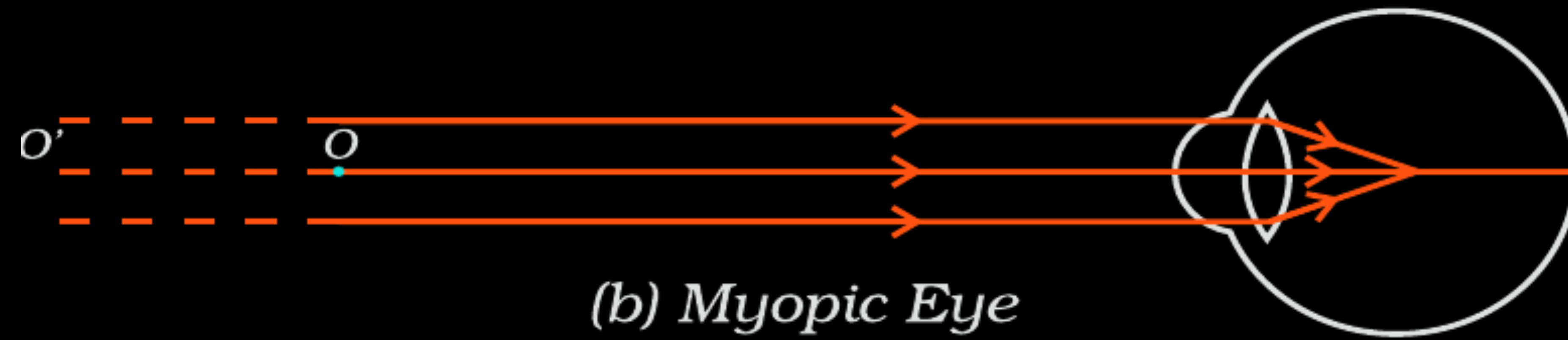
$\frac{1}{2} + \frac{1}{2}$

- (a) A student suffering from myopia is not able to see distinctly the objects placed beyond 5 m. List two possible reasons due to which this defect of vision may have arisen. With the help of ray diagrams, explain
- (i) why the student is unable to see distinctly the objects placed beyond 5 m from his eyes.
 - (ii) the type of the corrective lens used to restore proper vision and how this defect is corrected by the use of this lens.
- (b) If, in this case, the numerical value of the focal length of the corrective lens is 5 m, find the power of the lens as per the new Cartesian sign convention.

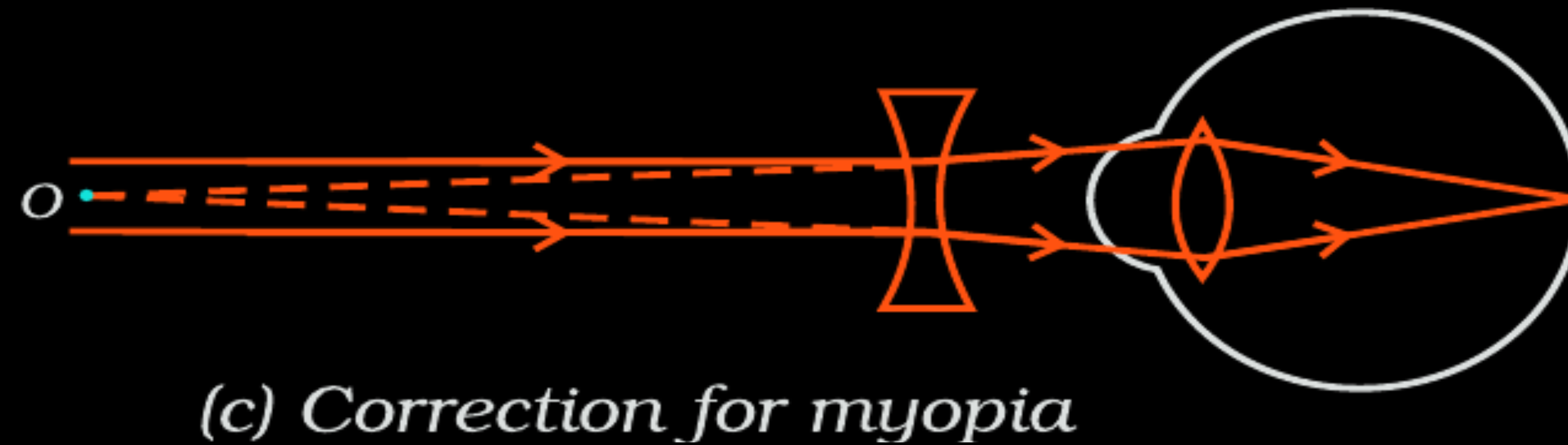
a)

- excessive curvature of the eye lens
- elongation of the eyeball

i)



ii) Concave / diverging lens



b) $f = -5 \text{ m}$ (since lens is concave)

$$P = \frac{1}{f(\text{metre})}$$

$$P = -0.2 \text{ D}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

Describe an activity to show that the colours of white light splitted by a glass prism can be recombined to get white light by another identical glass prism. Also draw ray diagram to show the recombination of the spectrum of white light.

Describe an activity to show that the colours of white light splitted by a glass prism can be recombined to get white light by another identical glass prism. Also draw ray diagram to show the recombination of the spectrum of white light.

Description of activity- When a glass prism is used to obtain a spectrum of sunlight, a second identical prism in an inverted position with respect to the first position will allow all the colours of spectrum to recombine .Thus a beam of white light will emerge from the other side of the second prism.

1 ½



1 ½

What is atmospheric refraction ? Use this phenomenon to explain the following natural events.

- (a) Twinkling of stars
- (b) Advanced sun-rise and delayed sun-set.

Draw diagrams to illustrate your answers.

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(a) Twinkling of stars

(b) Advanced sun-rise and delayed sun-set.

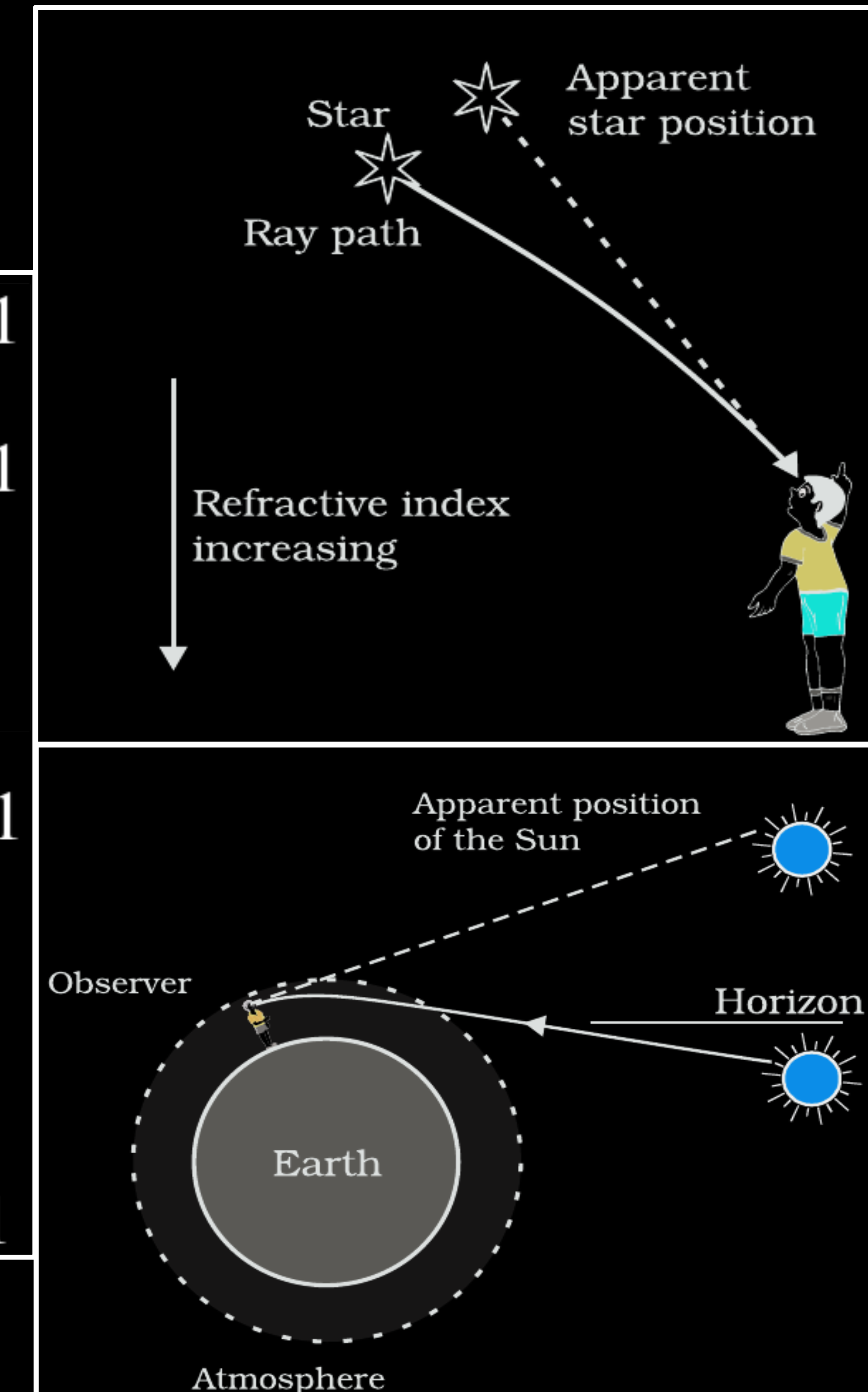
Draw diagrams to illustrate your answers.

Atmospheric refraction- refraction of light caused by the earth's atmosphere due to change in the refractive indices of different layers

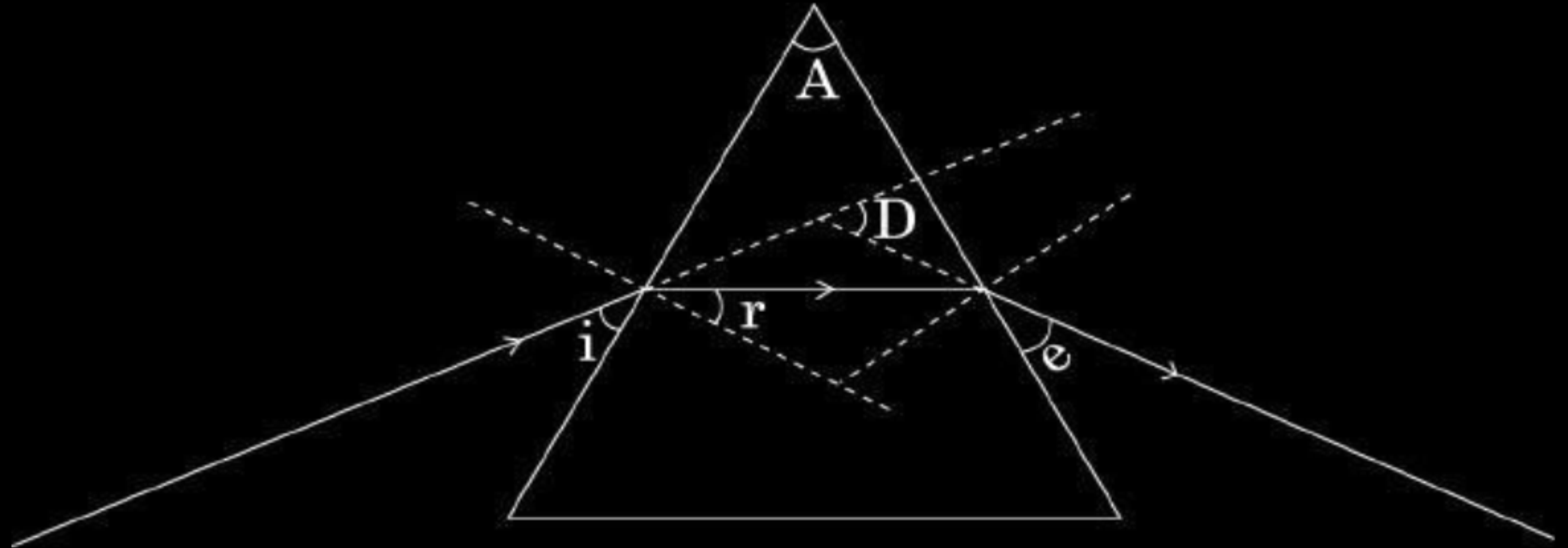
Twinkling of stars- stars are distant point sized source of light. The path of the rays of light coming from the star goes on varying due to atmospheric refraction slightly. Thus apparent position of the stars fluctuates and the amount of star light entering the eye flickers giving the twinkling effect

Advanced sunrise – when the sun is slightly below the horizon, light rays coming from the sun travel from the rarer to denser layers of air. Because of atmospheric refraction of light, light appears to come from a higher position above the horizon. Thus sun appears earlier than actual sunrise.

Delayed sunset- Same reason as similar refraction occurs at the sunset.



In the following ray diagram the correctly marked angle are :



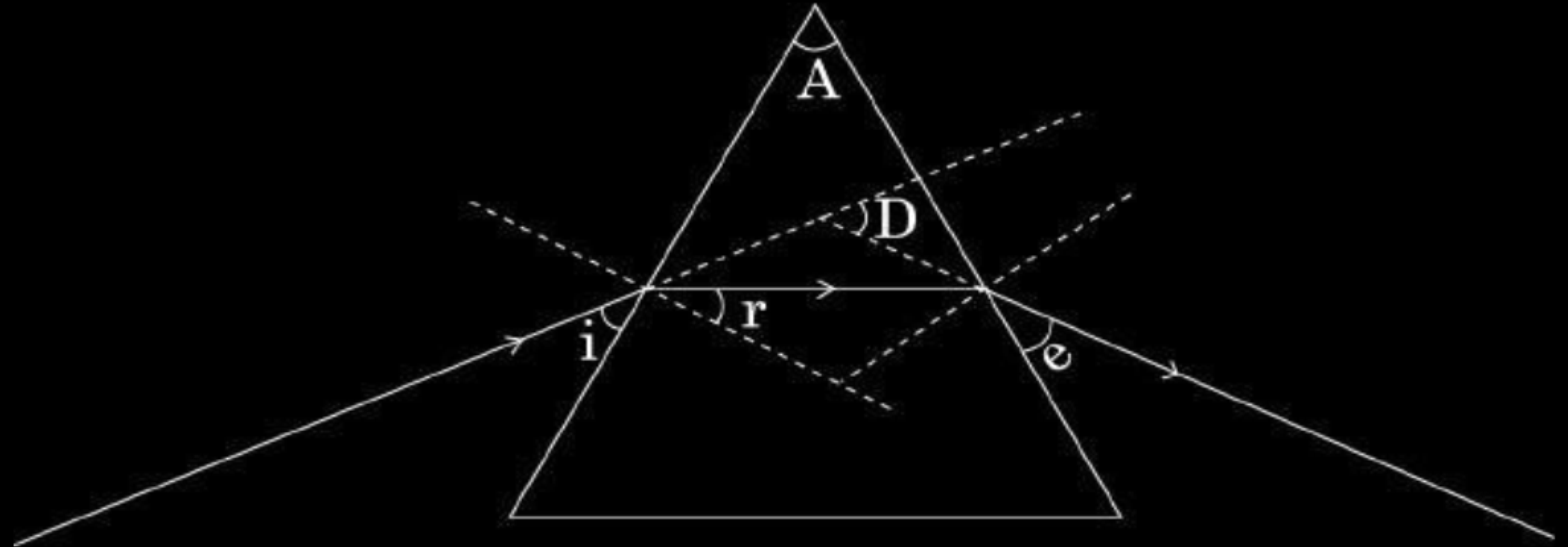
(a) $\angle i$ and $\angle e$

(b) $\angle A$ and $\angle D$

(c) $\angle i$, $\angle e$ and $\angle D$

(d) $\angle r$, $\angle A$ and $\angle D$

In the following ray diagram the correctly marked angle are :



(a) $\angle i$ and $\angle e$

(b) $\angle A$ and $\angle D$

(c) $\angle i$, $\angle e$ and $\angle D$

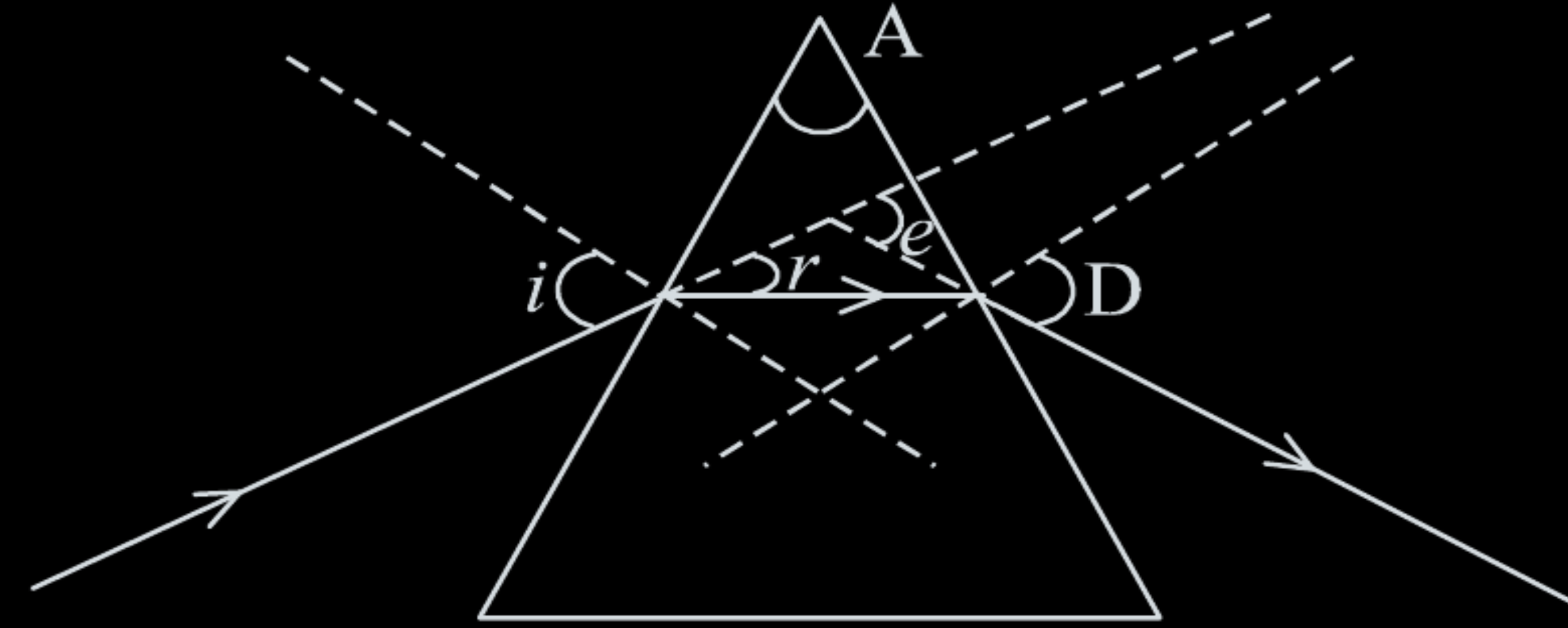
(d) $\angle r$, $\angle A$ and $\angle D$

- (a) Write the function of each of the following parts of human eye :
cornea; iris; crystalline lens; ciliary muscles

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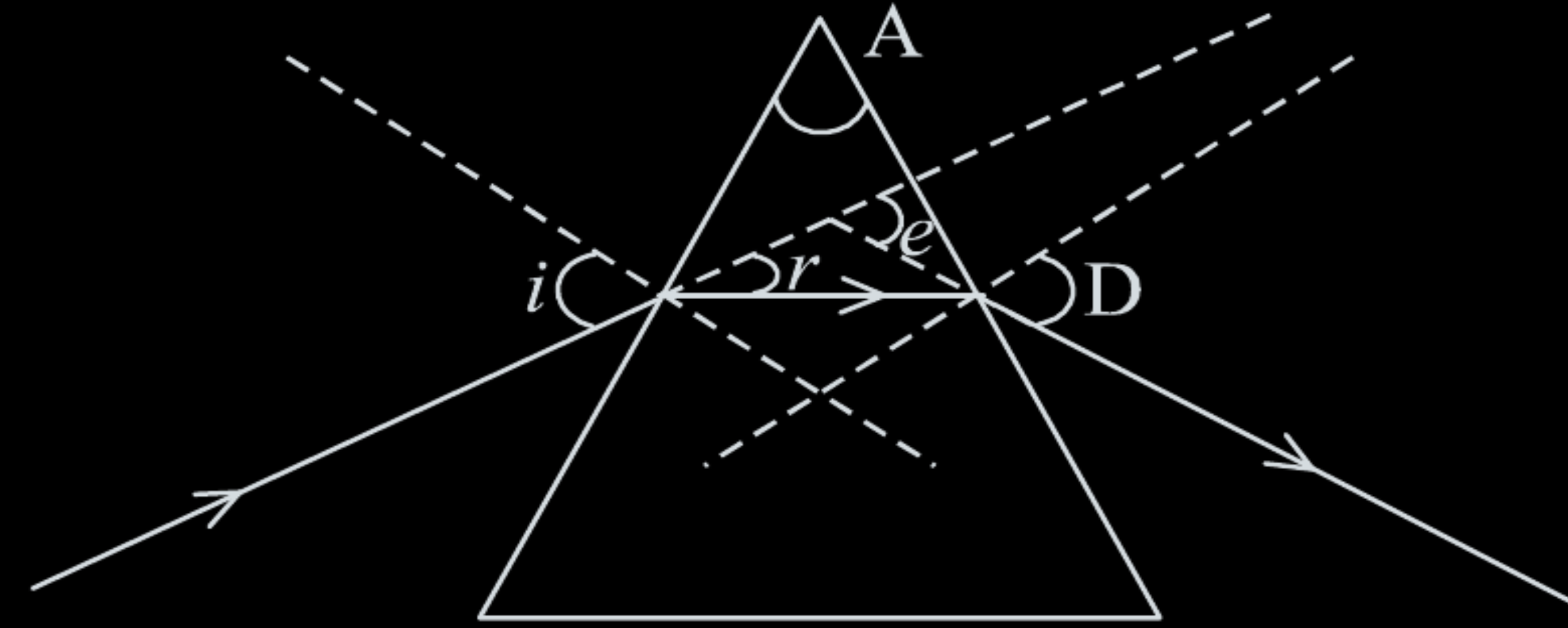
- | | |
|---|---------------|
| • Cornea – Refracts the rays of light falling on the eye | $\frac{1}{2}$ |
| • Iris – Controls the size of the pupil | $\frac{1}{2}$ |
| • Crystalline lens – Focuses the image of the object on the retina | $\frac{1}{2}$ |
| • Ciliary muscles – Holds the eye lens and adjusts its focal length | $\frac{1}{2}$ |

Study the following figure in which a student has marked the angle of incidence ($\angle i$), angle of refraction ($\angle r$), angle of emergence ($\angle e$), angle of prism ($\angle A$) and the angle of deviation ($\angle D$). The correctly marked angles are :



- (a) $\angle A$ and $\angle i$
- (b) $\angle A$, $\angle i$ and $\angle r$
- (c) $\angle A$, $\angle i$, $\angle e$ and $\angle D$
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Study the following figure in which a student has marked the angle of incidence ($\angle i$), angle of refraction ($\angle r$), angle of emergence ($\angle e$), angle of prism ($\angle A$) and the angle of deviation ($\angle D$). The correctly marked angles are :



- (a) $\angle A$ and $\angle i$
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What is meant by scattering of light? The sky appears blue and the sun appears reddish at sunrise and sunset. Explain these phenomena with reason.

What is meant by scattering of light? The sky appears blue and the sun appears reddish at sunrise and sunset. Explain these phenomena with reason.

- | | |
|---|---|
| • Scattering of light – Phenomenon of spreading of light (diffused reflection) by minute particles in a medium. | 1 |
| • The sky appears blue because the blue colour of sunlight scatters much more strongly than the red colour by particles in atmosphere/ air due to its shorter wave length. | 1 |
| • At sun-rise and sun-set most of the blue light and shorter wavelengths are scattered away by the particles in the atmosphere as the light from the sun near the horizon passes through thick layers of air and larger distance. The light that reaches us is of longer wavelength (red colour) giving a reddish appearance. | 1 |

Why does the sun appear reddish early in the morning? Will this phenomenon be observed by an observer on the moon? Justify your answer with a reason.

Why does the sun appear reddish early in the morning? Will this phenomenon be observed by an observer on the moon? Justify your answer with a reason.

Early in the morning, the sun is near the horizon, sunlight reaches us after travelling a longer distance through thick layers of atmosphere. Thus most of the blue light and shorter wavelengths are scattered away by the particles in the atmosphere. The light that reaches us is of longer wavelengths giving a reddish appearance.

2

This phenomenon will not be observed by an observer on the moon.

$\frac{1}{2}$

Because of the absence of atmosphere on the moon.

$\frac{1}{2}$

A student is unable to see clearly the words written on the blackboard placed at a distance of approximately 4 m from him. Name the defect of vision the boy is suffering from. Explain the method of correcting this defect. Draw ray diagram for the :

- (i) defect of vision and also
- (ii) for its correction.

Write the importance of ciliary muscles in the human eye. Name the defect of vision that arises due to gradual weakening of the ciliary muscles in old age. What type of lenses are required by the persons suffering from this defect to see the objects clearly ?

Akshay, sitting in the last row in his class, could not see clearly the words written on the blackboard. When the teacher noticed it, he announced if any student sitting in the front row could volunteer to exchange his seat with Akshay. Salman immediately agreed to exchange his seat with Akshay. He could now see the words written on the blackboard clearly. The teacher thought it fit to send the message to Akshay's parents advising them to get his eyesight checked.

In the context of the above event, answer the following questions :

- (a) Which defect of vision is Akshay suffering from ? Which type of lens is used to correct this defect ?

- Ciliary muscles modify the curvature of the eye lens to enable the eye to focus objects at varying distances/ help in adjusting the focal length of the eye lens

1

- Presbyopia

$\frac{1}{2}$

- Bifocal lens

$\frac{1}{2}$

(a) Defect – Myopia/ Nearsightedness

$\frac{1}{2}$

Corrective lens – Concave/ Diverging lens

$\frac{1}{2}$